AOS-CX 10.14 MPLS Guide

6400 (R0X44C, R0X45C), 8360 Switch Series

Hewlett Packard Enterprise

Published: November 2023

Version: 1

Notices

Contents

About this document 7

Latest version available online 7
Command syntax notation conventions 7
About the examples 8
Identifying switch ports and interfaces 8
Identifying modular switch components 9

Overview 10

MPLS L3 VPN 10

Protocol and feature details 12

LDP and downstream label allocation 12
Label-based forwarding 12
MPLS L3 VPN forwarding 13
MPLS L3 VPN requirements and recommendations 14
MPLS L3 VPN QoS 14
MPLS L3 VPN traffic load balancing 15
Same CE AS 16
LDP packet capture 16
VPNv4 packet capture 17

Supported features and standards 19

Default behaviors 19
Supported features 19
Scale 19
Best practices 20
Supported RFCs and standards 21

Configuration task list 22

Configuring MPLS and LDP globally 22
Configuring MPLS and LDP on an interface 22
Configuring IBGP VPNv4 peering 22
Configuring VPN 23
Configuring EBGP peering between PE and CE 24

Use cases 25

Use case 1: MPLS L3 VPN with dual homed VSX CE 25
- MPLS L3 VPN with dual homed VSX CE diagram 25
- CE1 configuration 25
- CE1 verification 28
- CE2 configuration 31
- CE2 verification 34
- CE3 configuration 38
- CE3 verification 39
- P1 configuration 42
- P1 verification 44
- P2 configuration 53
- P2 verification 54

PE1 configuration 65
PE1 verification 67
PE2 configuration 76
PE3 configuration 87
PE3 verification 89
Use case 2: MPLS L3 VPN with hub and spoke VRFs 98

MPLS commands 141
bind ipv4 (lsp label imposition) 141
bind ipv4 input (static lsp binding) 143
clear mpls statistics 144
crossconnect input (static lsp binding label swap) 145
enable (mpls globally) 147
enable (mpls ldp) 148
enable (mpls static lsp) 149
enable mpls (interface) 149
graceful-restart (mpls ldp) 150
graceful-restart-timers (mpls ldp) 151
label-range (static lsp) 154
label-protocol ldp 155
mpls 156
mpls ldp enable 157
mpls ldp entropy label insertion 158
mpls ldp discovery hello hold time (global) 159
mpls ldp discovery hello hold time (interface) 160
mpls ldp session holdtime (interface) 161
ping mpls 162
router-id (mpls ldp) 163
session hold time (mpls ldp globally) 166
show bgp vpnv4 unicast 167
show capacities mpls 172
show mpls forwarding 173
show mpls label-range static-lsp 175
show mpls ldp bindings 175
show mpls ldp discovery 176
show mpls ldp graceful-restart 178
show mpls ldp neighbor 178
static-lsp 179
traceroute mpls 180

Debugging and troubleshooting 182
Troubleshooting steps 183
Step 1: Verify that MPLS configurations are correct 183
Step 2: Check unicast routing between P/RR and PE 183
Step 3: Check LDP peering and MPLS bindings 184
Step 4: Check MPLS forwarding table 184
Step 5: Check MP-BGP VPNv4 peering between PE/RR 185
Step 6: Check peering between PE and CE 185
Step 7: Check connectivity between PEs and between CEs 186
Step 8: Check VRF tables at ingress PE for EBGP routes 187
Step 9: Check VPNv4 routes at egress PE for EBGP routes 188
Step 10: Check VRF tables at egress PE for EBGP routes 188
Step 11: Check CE routing table 190
Frequently asked questions 190

Support and Other Resources 192

Accessing HPE Aruba Networking Support 192
Accessing Updates 193
Aruba Support Portal 193
My Networking 193
Warranty Information 193
Regulatory Information 193
Documentation Feedback 194

Chapter 1

About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in Support and Other Resources.

Command syntax notation conventions

Convention	Usage
`example-text`	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets (`[ ]`).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ *`<example-text>`* ■ `<example-text>` ■ *`example-text`* ■ *example-text*	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (`< >`). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
`|`	Vertical bar. A logical `OR` that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
`{ }`	Braces. Indicates that at least one of the enclosed items is required.
`[ ]`	Brackets. Indicates that the enclosed item or items are optional.
… or `...`	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

`switch>`

Indicates the operator command context.

`switch#`

Indicates the manager command context.

`switch(`*`CONTEXT-NAME`*`)#`

Indicates the configuration context for a feature. For example:

`switch(config-if)#`

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

`switch(config-vlan-`*`<VLAN-ID>`*`)#`

Where *`<VLAN-ID>`* is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

`member/slot/port`

On the 6400 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
  - Management modules are on the front of the switch in slots 1/1 and 1/2.
  - Line modules are on the front of the switch starting in slot 1/3.
- *port*: Physical number of a port on a line module.

For example, the logical interface `1/3/4` in software is associated with physical port 4 in slot 3 on member 1.

On the 83xx, 9300, and 10000 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface `1/1/4` in software is associated with physical port 4 on the switch.

> If using breakout cables, the port designation changes to x:y, where x is the physical port and y is the lane when split to 4 x 10G or 4 x 25G. For example, the logical interface 1/1/4:2 in software is associated with lane 2 on physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: *member/power supply*:
  - *member*: 1.
  - *power supply*: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: *member/tray/fan*:
  - *member*: 1.
  - *tray*: 1 to 4.
  - *fan*: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: *member/module*:
  - *member*: 1.
  - *member*: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Chapter 2

Overview

Multiprotocol Label Switching (MPLS) is a routing technique in telecommunications networks that directs data from one node to the next based on short path labels rather than long network addresses. The labels identify virtual links (paths) between distant nodes rather than endpoints. MPLS can encapsulate packets of various network protocols, hence the "multiprotocol" reference on its name.

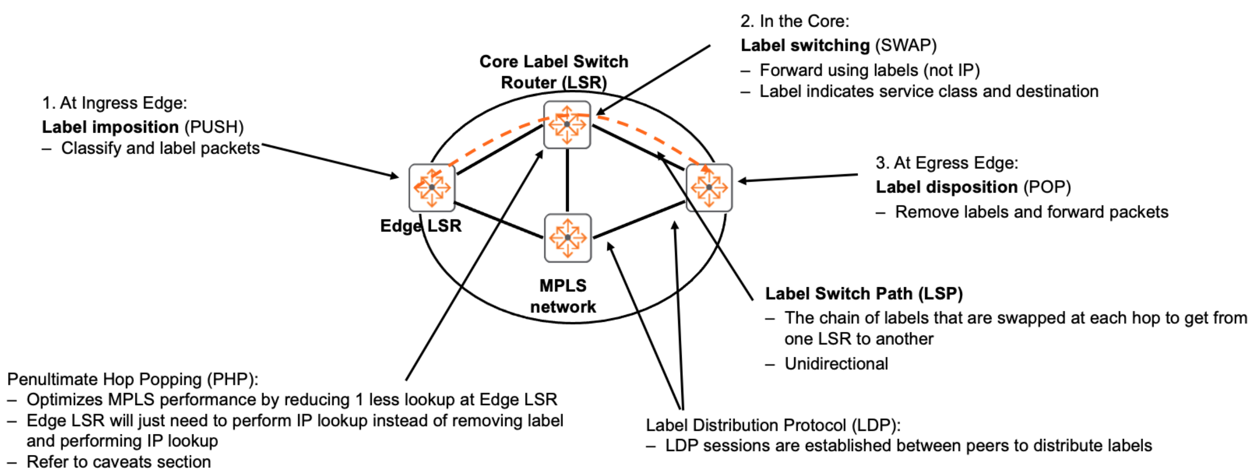
Any features not explicitly mentioned in this document are not supported as of the 10.09 release.

MPLS switches are termed Label Switch Routers (LSRs).

1. At the ingress edge LSR, label imposition or label push is used to classify and label packets.
2. At the transit core LSR, label switching or label swapping is used to forward traffic based on labels instead of IP.
3. At the egress edge LSR, label disposition or label pop is used to remove labels and forward IP packets.

A unidirectional label switched path (LSP) is a chain of labels that are swapped at each hop to get from one LSR to another. Label Distribution Protocol (LDP) sessions are established between LSRs to distribute labels.

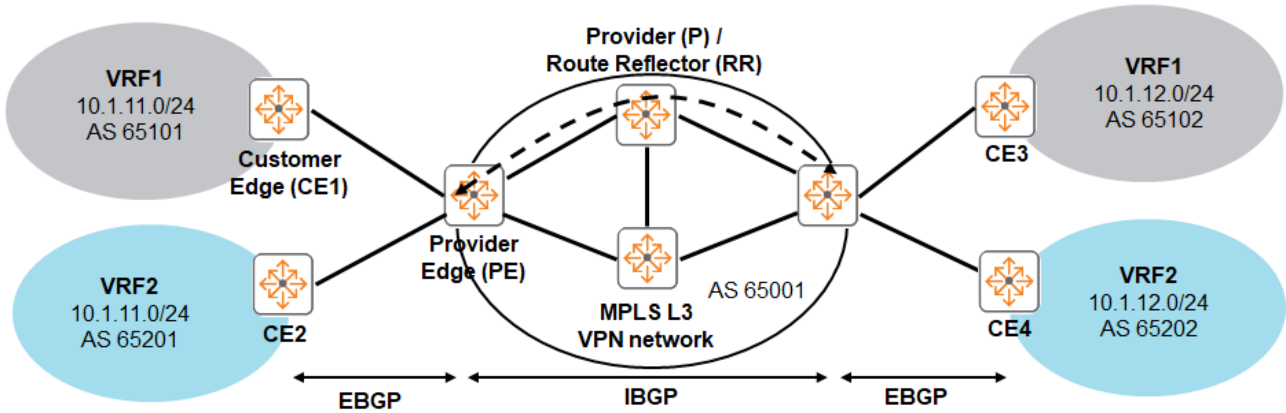
Penultimate Hop Popping (PHP) optimizes MPLS performance by reducing 1 less lookup at the egress edge LSR. With PHP, the egress edge LSR will just need to perform IP lookup instead of removing label and performing IP lookup. The figure below depicts the main MPLS concepts:



MPLS L3 VPN

MPLS L3 VPNs are deployed by service providers to provide L3 network connectivity and multi-tenant traffic isolation using an MPLS network.

In the figure below, the MPLS L3 VPN uses BGP Autonomous System (AS) 65001, CEs in Virtual Routing and Forwarding (VRF1) utilize AS 65101/65102, CEs in VRF2 utilize AS 65201/65202. VRF1 and VRF2 may utilize overlapping IP subnets if desired.



In this example, VRF1 consists of CE1 (LAN subnet 10.1.11.0/24) and CE3 (LAN subnet 10.1.12.0/24). VRF2 consists of CE2 (LAN subnet 10.1.11.0/24) and CE4 (LAN subnet 10.1.12.0/24). The Core LSR now functions as Provider (P) or Route Reflector (RR). The Edge LSR now functions as Provider Edge (PE). Customer Edge (CE) routers are not part of the MPLS L3VPN network but are attached to it. VRFs (without MPLS or VPNv4) could be enabled on CEs to provide locally significant L3 network isolation.

Multi Protocol BGP (MP-BGP) is used between PE routers to exchange VPNv4 addresses, extended communities and labels. Full mesh IBGP peering is avoided within the MPLS L3 VPN network when P routers function as VPNv4 RR. PE would peer to dual RRs for redundancy.

The following details provide additional information regarding the configuration of MPLS features.

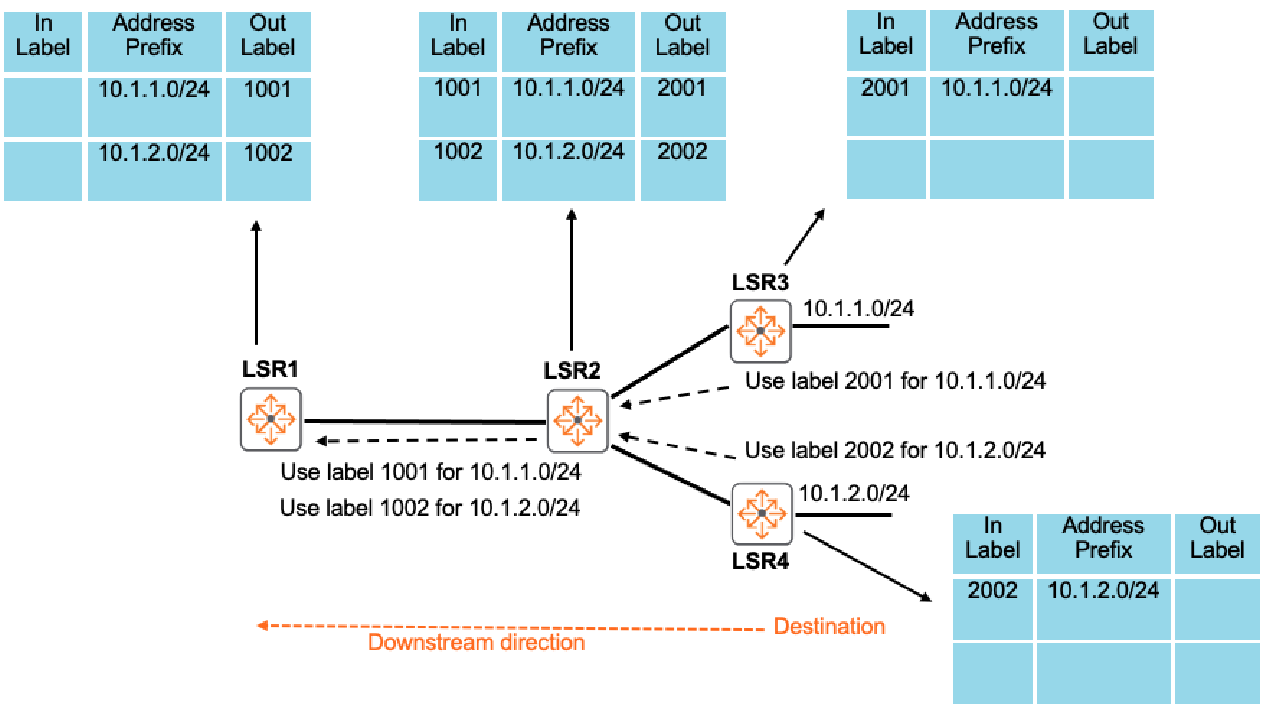
LDP and downstream label allocation

LDP is used for downstream label allocation, Figure 3 describes this MPLS concept. Assuming 10.1.1.0/24 and 10.1.2.0/24 on the right are destination subnets.

From a right to left direction:

1. LSR3 informs LSR2 to use label 2001 for 10.1.1.0/24.
2. LSR4 informs LSR2 to use label 2002 for 10.1.2.0/24.
3. LSR2 will in turn inform LSR1 to use label 1001 for 10.1.1.0/24 and label 1002 for 10.1.2.0/24.

The figure below shows LDP and downstream label allocation:

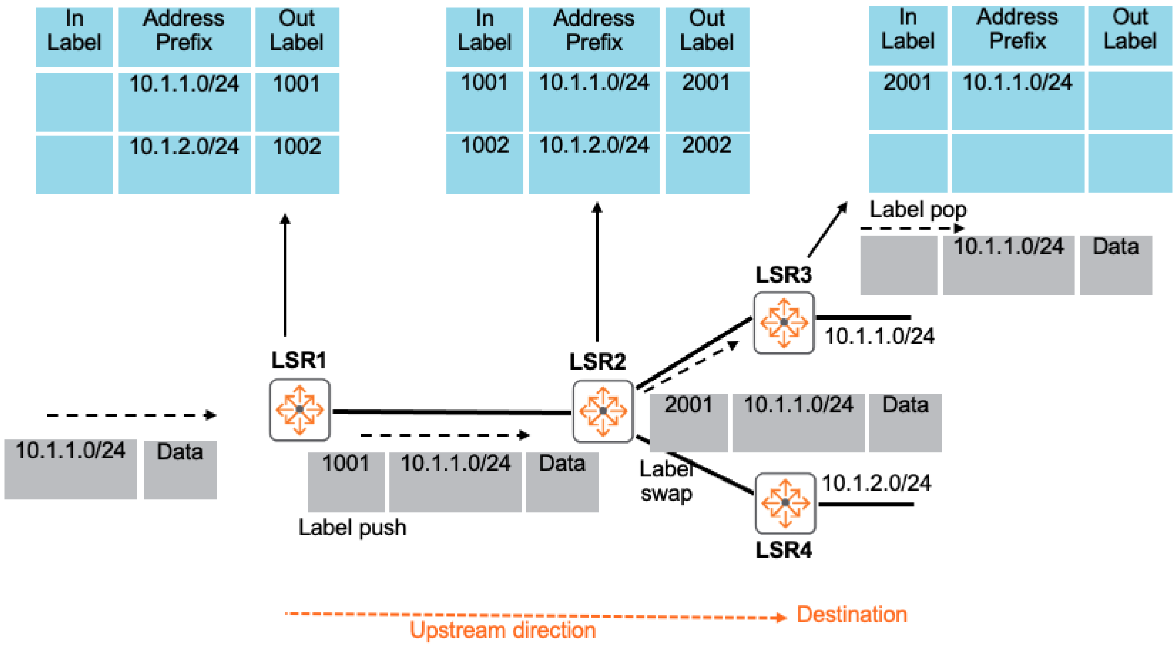


Label-based forwarding

Once the labels are allocated in the downstream direction, label based forwarding in the upstream direction is used next.

1. Data destined for 10.1.1.0/24 arrives at LSR1, LSR1 will refer to the MPLS label table and push 1001 into the packet towards LSR2.
2. LSR2 will receive the packet, see inbound label 1001, swap it with label 2001 and send it towards LSR3.
3. LSR3 will receive the packet, see inbound label 2001, pop it and send the packet using its IP routing table to the destination.

The figure shows label-based forwarding, from the left to right direction:

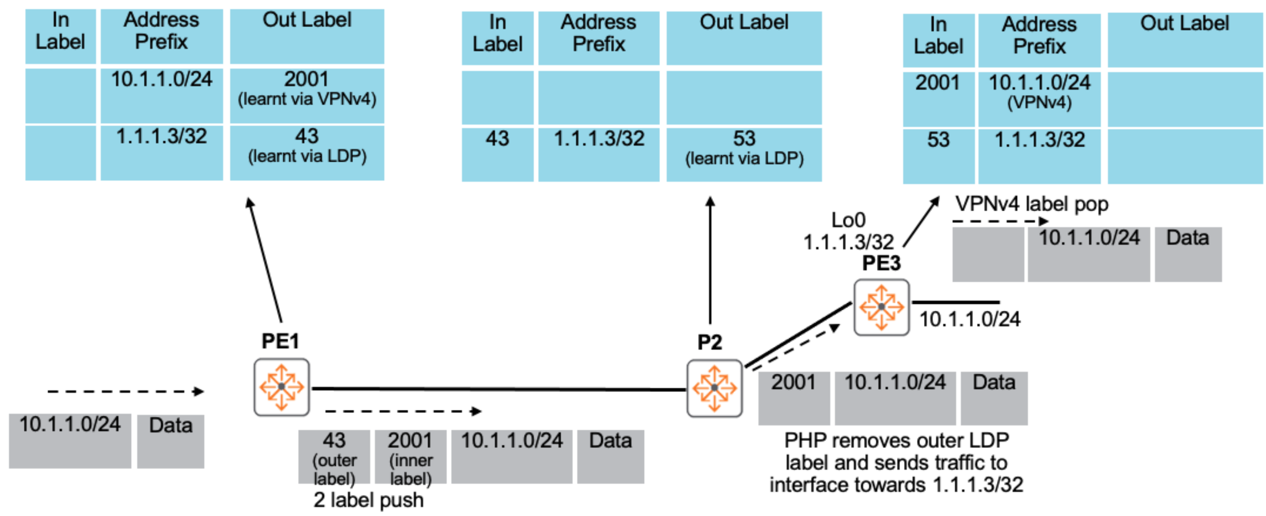


MPLS L3 VPN forwarding

MPLS L3 VPN requires a 2 label stack in order to forward traffic. The outer label (based on LDP) is used to forward traffic towards remote PE loopbacks, while the inner label (based on VPNv4) are used to forward traffic towards CE subnets.

1. Data destined for 10.1.1.0/24 arrives at PE1, PE1 will refer to the LDP, MPLS BGP VPNv4 tables and push both 43 (outer) and 2001 (inner) labels into the packet towards P2.
2. Because of PHP from PE3 to P2, P2 will receive the packet, see inbound label 43, remove outer LDP label and send it towards PE3.
3. PE3 will receive the packet with VPNv4 label, see inbound label 2001, pop it and send the packet using it's IP routing table to the destination.

The figure below shows MPLS L3 VPN forwarding from the left to right direction:



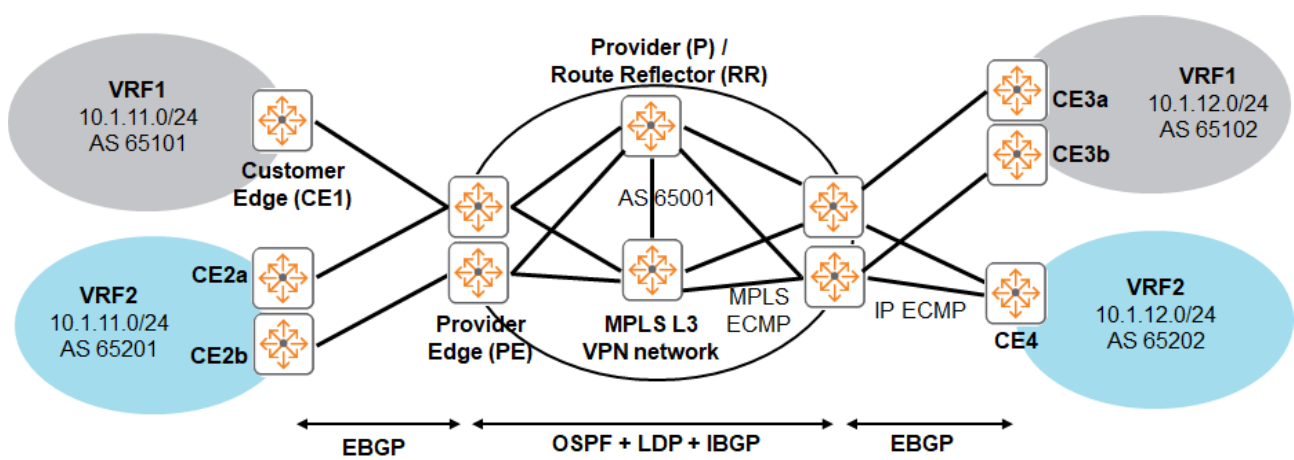
MPLS L3 VPN requirements and recommendations

MPLS L3 VPN has the following requirements:

- OSPF is required to exchange loopback IPs between PE/P
- LDP is required to exchange outer labels
- IBGP MP-BGP VPNv4 peering between PE/P(RR) loopbacks are used to exchange inner VPNv4 labels
- EBGP IPv4 peering is recommended between CE and PE

High availability recommendations:

- Dual standalone Ps
- Dual standalone RRs
- Dual standalone PEs for a site
- Dual CEs (could be VSX or 2 standalone switches)
- Redundant Equal Cost Multi Pathing (ECMP) links



MPLS L3 VPN QoS

MPLS packets interact with Quality of Service features, such as queueing and scheduling. AOS-CX supports static Uniform mode queueing and remarking behavior as described below.

At tunnel entry, packets are prioritized and enqueued before entering the MPLS tunnel. Therefore, a device performing tunnel entry assigns a local priority and queue number to packets identically to non-MPLS packets: local priority is derived from both the packet's priority codepoints and the ingress port's QoS trust mode. Refer to the *Quality of Service Guide* for more information about trust mode and local-priority-to-queue mapping. When an IP packet enters an MPLS tunnel, the uppermost 3 bits of the DSCP field will be written into the egress packet's EXP value in the newly pushed MPLS header.

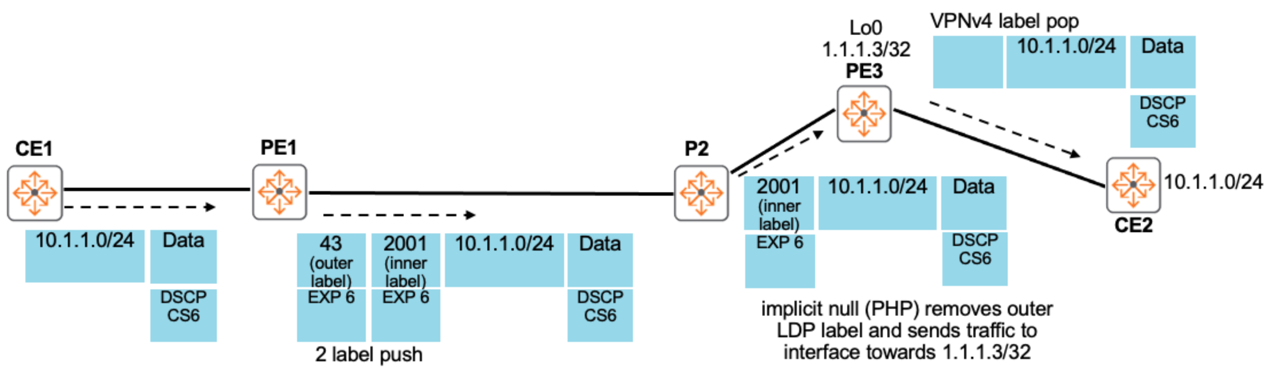
At MPLS transit and tunnel exit, packets are assigned a local priority equal to the EXP value of the incoming packet's outermost MPLS label. The COS and DSCP values of an incoming MPLS packet, and the trust mode of the ingress port, are ignored. EXP values are NOT subject to the CoS map or DSCP map applied on the system, but the local priority derived from the EXP value is still translated to a queue number through the queue profile applied on the system.

At tunnel exit, the device operates in Uniform mode. For IP packets, the EXP value from the popped MPLS label is written to the upper 3 bits of the DSCP field of the egress IP packet's header. The remaining bits of the DSCP field will be set to 0.

For MPLS Layer 3 Virtual Private Network (L3VPN), the protocols that are currently supported for communication between Customer Edge (CE) and Provider Edge (PE) are Open Shortest Path First (OSPF) and Border Gateway Protocol (BGP). Routing Information Protocol(RIP) is not currently supported for communication between Customer Edge and Provider Edge.

MPLS L3 VPN supports QoS for congestion management. In the figure below implicit null/PHP is enabled on egress PE3 towards P2.

1. If CE1 sends traffic to PE1 with CS6 DSCP marking PE1 will automatically copy the CS6 DSCP marking to MPLS EXP 6 for both inner and outer labels
2. P2 will remove the outer label and forward traffic to interface towards PE3 with EXP 6 marking
3. PE3 will receive, pop the VPNv4 label and forward traffic with DSCP CS6



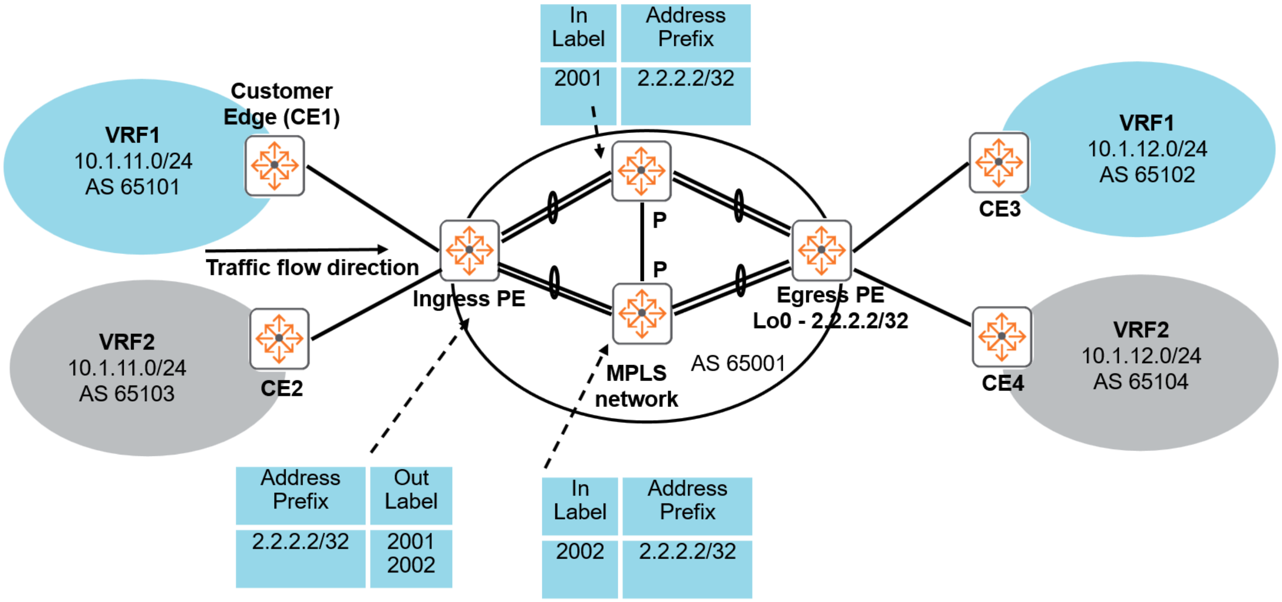
MPLS L3 VPN traffic load balancing

MPLS L3 VPN supports traffic load balancing via MPLS Equal Cost Multi Pathing (ECMP) across Link Aggregation Group (LAG) and links over multiple paths that have the same cost.

In the figure below, when ingress PE has 2 way ECMP towards egress PE loopback (2.2.2.2/32) via 2 intermediate P routers:

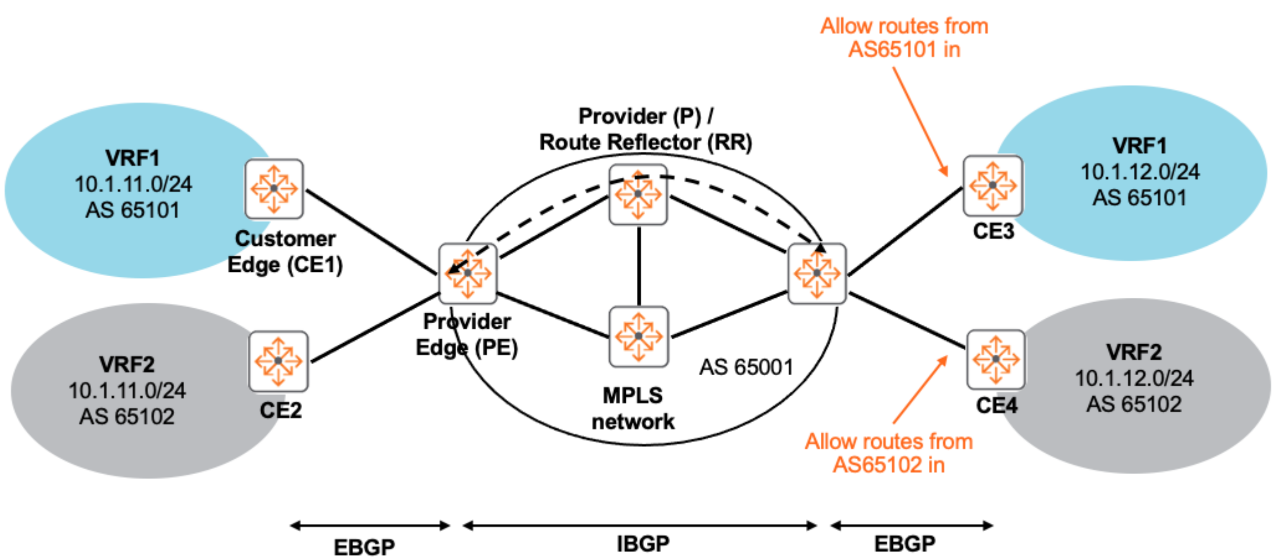
1. Traffic from ingress PE to egress PE will be load balanced across the links via the 2 P routers on a per flow basis

2. On ingress PE, some flows will use outbound label (2001) via top P router, while other flows will use outbound label (2002) via bottom P router



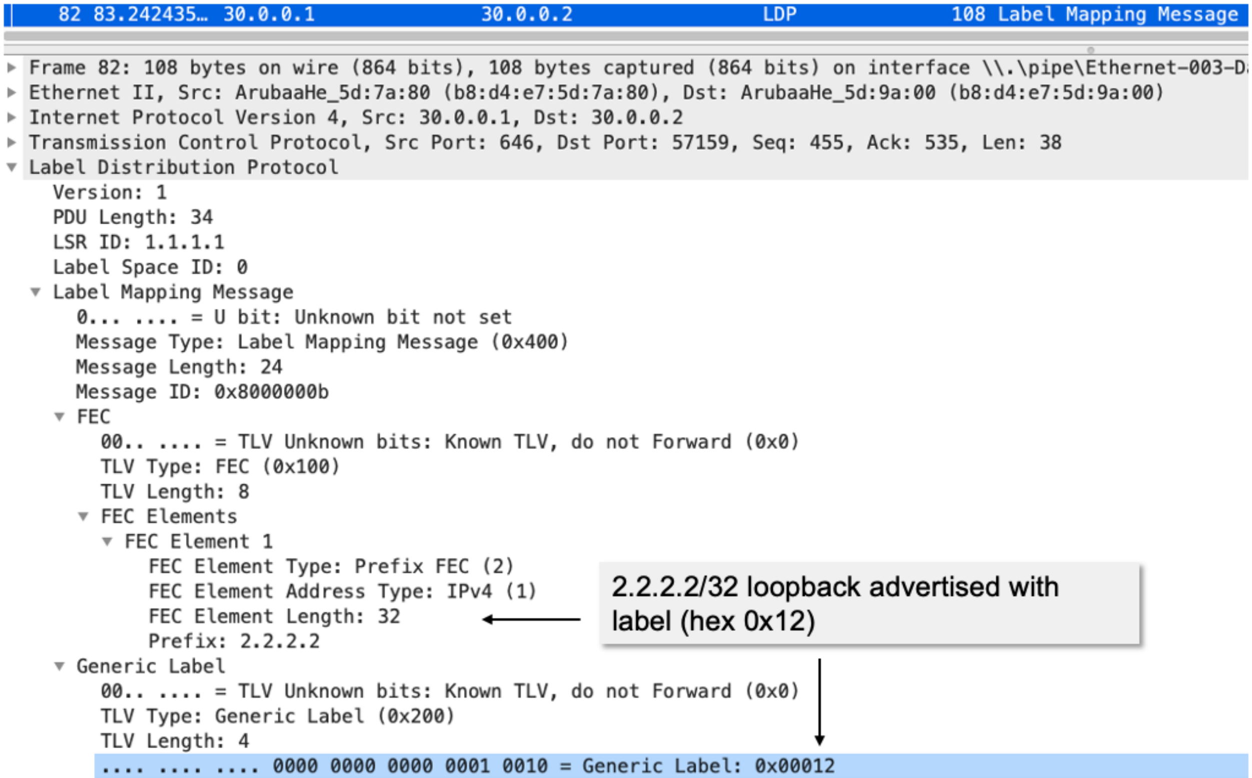
Same CE AS

CEs should be deployed using unique AS numbers to help identify which AS a route originates from, however it is possible to deploy the same CE AS as shown in the figure below, where VRF1 uses AS 65101 and VRF2 uses AS 65102. This is achieved by allowing the same inbound on the CE as a BGP router will (by default) drop routes from the same AS it is configured.



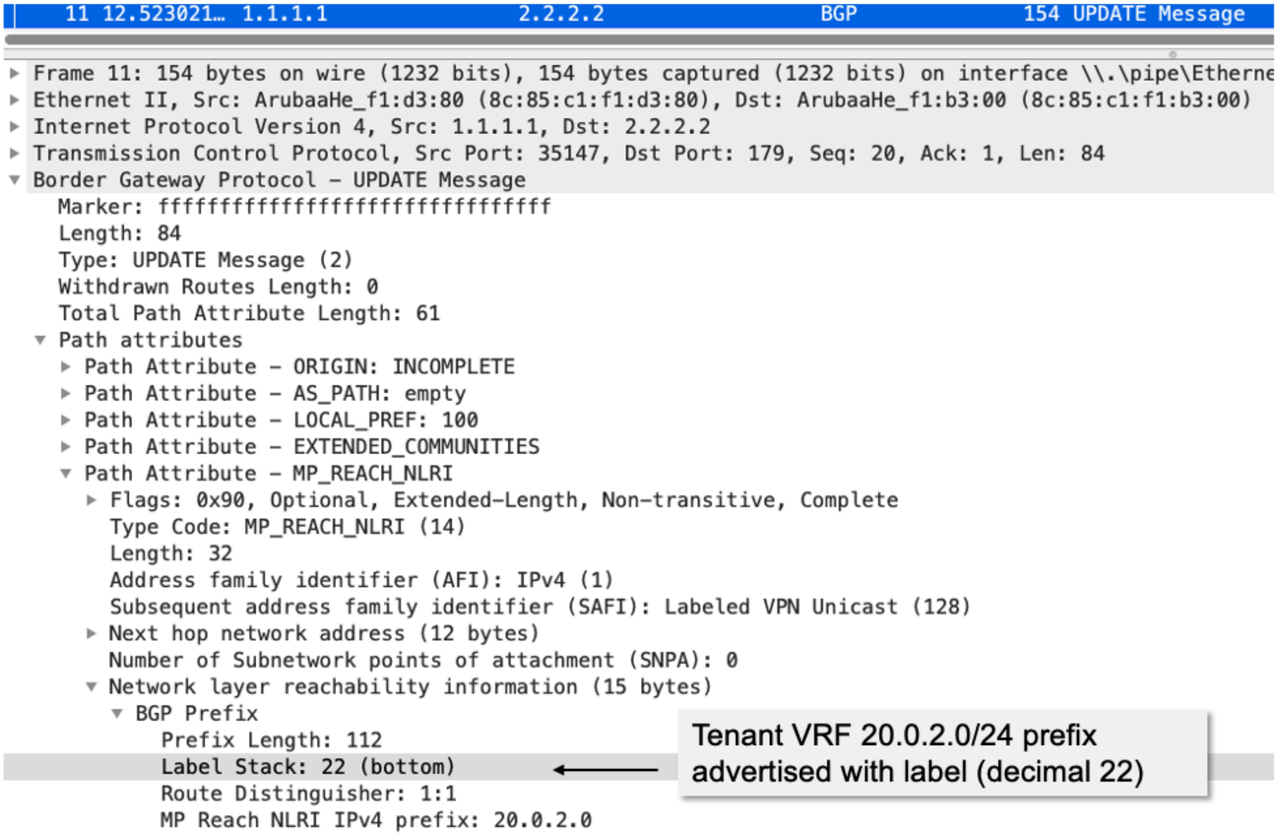
LDP packet capture

As shown in the figure below, a packet capture of LDP traffic between 2 LSRs is provided for reference. This example shows how a PE loopback IP of 2.2.2.2/32 is advertised with outer label of hex 0x12. Packet captures might be required if there is a need to troubleshoot LDP between 2 LSRs.



VPNv4 packet capture

As shown in the figure below, a packet capture of MP-BGP VPNv4 traffic between 2 PEs is provided for reference. This example shows how a VRF tenant subnet of 20.0.2.0/24 is advertised with inner label of decimal 22. Packet captures might be required if there is a need to troubleshoot VPNv4 labels between 2 PEs. CEs should be deployed using unique AS numbers to help identify which AS a route originates from, however it is possible to deploy duplicate CE AS as shown in Figure 10, where VRF1 uses AS 65101 and VRF2 uses AS 65102. This is achieved by allowing the same inbound on the CE, as a BGP router will (by default) drop routes from the same AS it is configured.


11 12.523021… 1.1.1.1 2.2.2.2 BGP 154 UPDATE Message
Frame 11: 154 bytes on wire (1232 bits), 154 bytes captured (1232 bits) on interface \\.\pipe\Etherne
Ethernet II, Src: ArubaaHe_f1:d3:80 (8c:85:c1:f1:d3:80), Dst: ArubaaHe_f1:b3:00 (8c:85:c1:f1:b3:00)
Internet Protocol Version 4, Src: 1.1.1.1, Dst: 2.2.2.2
Transmission Control Protocol, Src Port: 35147, Dst Port: 179, Seq: 20, Ack: 1, Len: 84
Border Gateway Protocol – UPDATE Message
Marker: ffffffffffffffffffffffffffffffff
Length: 84
Type: UPDATE Message (2)
Withdrawn Routes Length: 0
Total Path Attribute Length: 61
Path attributes
Path Attribute – ORIGIN: INCOMPLETE
Path Attribute – AS_PATH: empty
Path Attribute – LOCAL_PREF: 100
Path Attribute – EXTENDED_COMMUNITIES
Path Attribute – MP_REACH_NLRI
Flags: 0x90, Optional, Extended-Length, Non-transitive, Complete
Type Code: MP_REACH_NLRI (14)
Length: 32
Address family identifier (AFI): IPv4 (1)
Subsequent address family identifier (SAFI): Labeled VPN Unicast (128)
Next hop network address (12 bytes)
Number of Subnetwork points of attachment (SNPA): 0
Network layer reachability information (15 bytes)
BGP Prefix
Prefix Length: 112
Label Stack: 22 (bottom)
Route Distinguisher: 1:1
MP Reach NLRI IPv4 prefix: 20.0.2.0
Tenant VRF 20.0.2.0/24 prefix advertised with label (decimal 22)

Chapter 4
Supported features and standards

Default behaviors

- Default QoS mode: Uniform
- Default TTL propagation mode: Uniform
- Default egress label: Explicit null
- Per-VPN/VRF label

Supported features

- MPLS L3 VPN
  - IPv4 unicast
- Static LSP
- OSPF between P/RR and PE/P
- LDP between P/RR and PE/P
- MP-BGP VPNv4 RR on PE
- MP-BGP VPNv4 RR client on PE
- QoS uniform mode only
- Label bindings are supported with /32 address
  - o E.g. if the network between P and PE is X.0.0.0/24 with an interface address of P as X.0.0.1/24 and address of PE as X.0.0.2/24, the LDP binding on PE for the X.0.0.0/24 network will be shown as X.0.0.1/32
- VRFs (intranet and extranet VRF with route leaking)
- Multi-VRF (VRF-Lite) CE
- PE-CE connectivity:
  - EBGP
  - Static route
  - Default route
  - Connected route (PE acting as an L3 first hop)
- Ping over MPLS networks
- Traceroute over MPLS networks

Scale

8360 Switch Series	Profile: Aggregation-Leaf (Default)	Profile: Core-Spine
Max VPNv4 VRFs/EBGP as PE-CE protocol	256	256
Max VPNv4 neighbors/Max dynamic LSPs	256	256
Max VPNv4 VRF routes	65,000	600,000
Max VPNv4 neighbors/Max routes	256 neighbors/32,000 routes	256 neighbors/600,000 routes
Max LDP neighbors	128	128
Max static LSP	4096	4096

6400v2T Switch Series	Profile: v2-Aggregation-High-Bandwidth (non-default)	Profile: v2-Leaf-Extended-High-Bandwidth(non-default)	Profile: v2-Core-High-Bandwidth(non-default)
Max VPNv4 VRFs/EBGP as PE-CE protocol	256	256	256
Max VPNv4 neighbors/Max dynamic LSPs	256	256	256
Max VPNv4 VRF routes	65,000	65,000	600,000
Max VPNv4 neighbors/Max routes	256 neighbors/32,000 routes	256 neighbors/32,000 routes	256 neighbors/600,000 routes
Max LDP neighbors	128	128	128
Max static LSP	4096	4096	4096

Best practices

Recommended best practices for MPLS configuration are as follows:

- Use single area OSPF within the MPLS network to advertise and learn LSR loopbacks
- Use a single AS for the MPLS network
- Utilize EBGP between CE and PE for dynamic routing, use unique CE AS# if possible
- Use redundant VPNv4 RRs to avoid full mesh IBGP peers
- Use peer groups to simplify configs on both P/RR and PE

```
router bgp 65001
neighbor RR-client peer-group
neighbor RR-client remote-as 65001
neighbor RR-client description all RR clients
neighbor RR-client update-source loopback 0
neighbor 1.1.1.1 peer-group RR-client
neighbor 2.2.2.2 peer-group RR-client
address-family vpnv4 unicast
        neighbor RR-client route-reflector-client
        neighbor RR-client send-community both
        neighbor 1.1.1.1 activate
        neighbor 2.2.2.2 activate
```

- The `l3vpn-only` command should only be configured under the VRF context
- Implicit null is not supported, explicit null has to be configured on 3rd party egress PE towards AOS-CX P
- If egress PE and P are AOS-CX switches, changing it to explicit null is not required as explicit null is automatically sent

Supported RFCs and standards

The MPLS and MPLS L3 VPN implementation uses standards based protocols as described in the following RFCs:

- RFC 2547, BGP/MPLS VPNs
- RFC 3031, Multiprotocol Label Switching Architecture
- RFC 3032, MPLS Label Stack Encoding
- RFC 3036, LDP Specification
- RFC 3443, Time To Live (TTL) Processing in Multi-Protocol Label Switching (MPLS) Networks
- RFC 4182, Removing a Restriction on the use of MPLS Explicit NULL
- RFC 4364, BGP/MPLS IP Virtual Private Networks (VPNs)
- RFC 6388, Label Distribution Protocol Extensions for Point-to-Multipoint and Multipoint-to-Multipoint Label Switched Paths

Configuration task list

The following examples provide configuration steps to follow for basic MPLS configuration. For commands related to BGP configuration refer to the Border Gateway Protocol (BGP) chapter of the *IP Routing Guide*.

Configuring MPLS and LDP globally

Step	Command	Comments
1. Enter MPLS context	`mpls`	
2. Enable MPLS globally	`enable`	
3. Enter LDP context	`label-protocol ldp`	
4. Enable LDP globally	`enable`	
5. Specify loopback as LDP router-ID	`router-id <IP-ADDR>`	Refer to the *IP Routing Guide* for command description.

Configuring MPLS and LDP on an interface

Step	Command	Comments
1. Enter desired interface	`interface <IFNAME>`	Desired IP address and OSPF will also need to be added.
2. Enable MPLS	`mpls enable`	
3. Enable LDP	`mpls ldp enable`	

Configuring IBGP VPNv4 peering

Step	Command	Comments
1. Enter BGP context	`router bgp <AS-NUMBER>`	
2. Specify remote BGP neighbor and remote AS	`neighbor <IP-ADDRESS> remote-as <AS-NUMBER>`	The same AS number should

Step	Command	Comments
		be used between P/PE. OSPF should be enabled before BGP to advertise loopback IPs between LSRs.
3. Specify source IP/interface for IBGP peering	`neighbor <IP-ADDRESS> update-source loopback <NUMBER>`	
4. Enter VPNv4 address family	`address-family vpnv4 unicast`	
5. Activate IBGP neighbor	`neighbor <IP-ADDRESS> activate`	
6. Send extended communities	`neighbor <IP-ADDRESS> send-community extended`	

The descriptions, syntax, and examples for the commands listed above can be found in the IP Routing Guide.

Configuring VPN

Step	Command	Comments
1. Create desired VRF	`vrf <VRF-NAME>`	This command can be found in the IP Routing Guide.
2. Specify a unique RD per PE, per VRF	`rd number:number`	This command can be found in the VXLAN Guide.
3. Enable VPNv4 address family	`l3vpn-only`	`l3vpn-only` under VRF context is required for VPNv4 address family to be active.
4. Enter IPv4 unicast address family	`address-family ipv4 unicast`	This command can be found in the IP Routing Guide.
5. Specify desired route-target for export	`route-target export <AS-NUMBER:ID> | <IP-ADDRESS:ID`	The **route target both number:number** command may also be used as an alternative.This command can be found in the VXLAN Guide.
6. Specify desired route-target for import	`route-target import <AS-NUMBER:ID> | <IP-ADDRESS:ID`	

Configuring EBGP peering between PE and CE

Step	Command	Comments
1. Enter interface facing CE	`interface <IFNAME>`	Desired IP address will also need to be added.
2. Attach VRF to interface	`vrf attach <VRF-NAME>`	
3. Enter BGP context	`router bgp <AS-NUMBER>`	
4. Enter desired VRF context	`vrf <VRF-NAME>`	
5. Specify remote CE IP	`neighbor <IP-ADDRESS> remote-as <AS-NUMBER>`	EBGP should be used between CE/PE with different AS .
6. Enter IPv4 unicast context for VRF	`address-family ipv4 unicast`	
7. Active the EBGP neighbor	`neighbor IP-address activate`	
8. Redistribute connected route to remote PEs and CEs	`redistribute connected`	Optional. Only required if connected routes need to be advertised.

The descriptions, syntax, and examples for the commands listed above can be found in the IP Routing Guide.

Chapter 6
Use cases

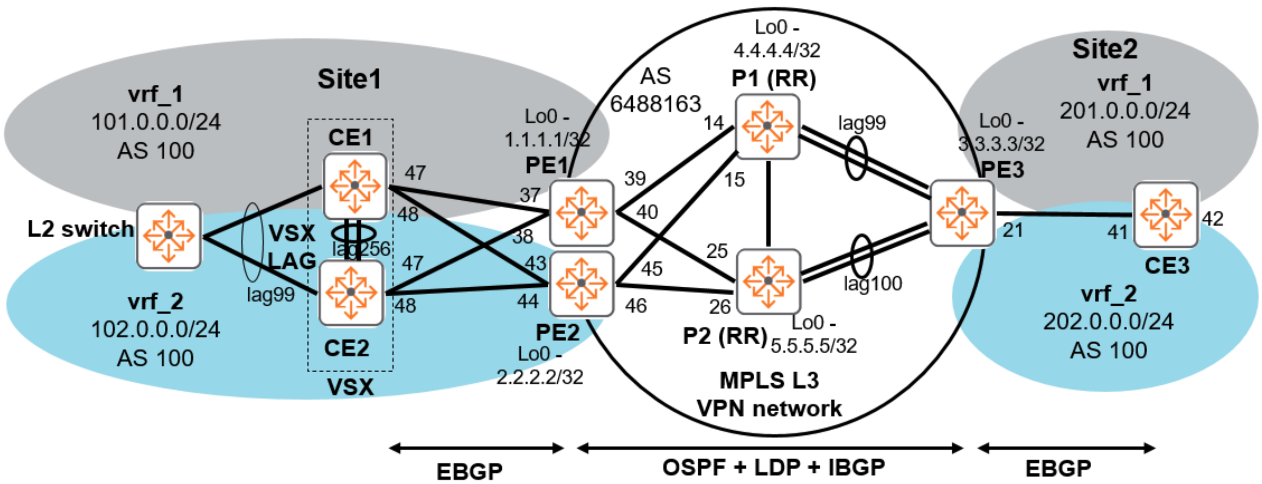
The following use cases provide examples of networks using MPLS VPNs.

Use case 1: MPLS L3 VPN with dual homed VSX CE

This use case covers:

- An MPLS L3 VPN network with redundant PEs, Ps and VRFs across the MPLS network
- Redundant VSX CEs at Site1 and standalone CE at Site2 with locally significant VRFs
- BGP community is used inbound/outbound on PE1 and PE2 to prevent loops from Site1
- Sample configurations
- Relevant verification commands

MPLS L3 VPN with dual homed VSX CE diagram



CE1 configuration

```
CE1# show running
Current configuration:
!
!Version ArubaOS-CX GL.10.09.0001BF
!export-password: default
hostname CE1
no ip icmp redirect
profile l3-core
logging console severity crit
vrf vrf_1
vrf vrf_2
cli-session
    timeout 0
!
```

```
!
!
!
!
ssh server vrf mgmt
vlan 1,2001-2002,2011-2012,2021-2022
interface mgmt
    no shutdown
    ip dhcp
system interface-group 1 speed 10g
    !interface group 1 contains ports 1/1/1-1/1/12
system interface-group 4 speed 10g
    !interface group 4 contains ports 1/1/37-1/1/48
interface lag 99 multi-chassis
    no shutdown
    description Towards L2SW
    no routing
    vlan trunk native 1
    vlan trunk allowed 2001-2002
    lacp mode active
interface lag 256
    no shutdown
    description Towards ISL
    no routing
    vlan trunk native 1 tag
    vlan trunk allowed all
    lacp mode active
interface 1/1/1
    no shutdown
    mtu 9000
    description Towards L2SW
    lag 99
interface 1/1/2
    no shutdown
    mtu 9000
    description Towards ISL
    lag 256
interface 1/1/3
    no shutdown
    mtu 9000
    description Towards ISL
    lag 256
interface 1/1/46
    no shutdown
    mtu 9000
    description Towards KA
    ip mtu 9000
    ip address 99.0.0.1/24
interface 1/1/47
    no shutdown
    mtu 9000
    description Towards PE1
    no routing
    vlan trunk native 1
    vlan trunk allowed 2011-2012
interface 1/1/48
    no shutdown
    mtu 9000
    description Towards PE2
    no routing
    vlan trunk native 1
    vlan trunk allowed 2021-2022
```

```
interface vlan 2001
    vrf attach vrf_1
    description Towards LAN
    ip mtu 9000
    vsx active-forwarding
    ip address 20.0.1.2/24
interface vlan 2002
    vrf attach vrf_2
    description Towards LAN
    ip mtu 9000
    vsx active-forwarding
    ip address 20.0.1.2/24
interface vlan 2011
    vrf attach vrf_1
    description Towards PE1
    ip mtu 9000
    ip address 20.1.1.1/24
interface vlan 2012
    vrf attach vrf_2
    description Towards PE1
    ip mtu 9000
    ip address 20.1.1.1/24
interface vlan 2021
    vrf attach vrf_1
    description Towards PE2
    ip mtu 9000
    ip address 20.2.1.1/24
interface vlan 2022
    vrf attach vrf_2
    description Towards PE2
    ip mtu 9000
    ip address 20.2.1.1/24
vsx
    inter-switch-link lag 256
    role primary
    keepalive peer 99.0.0.2 source 99.0.0.1
!
!
!
!
!
router bgp 100
!
    vrf vrf_1
        neighbor 20.0.1.1 remote-as 100
        neighbor 20.0.1.1 description Towards LAN
        neighbor 20.0.1.3 remote-as 100
        neighbor 20.0.1.3 description Towards CE2
        neighbor 20.1.1.2 remote-as 6488163
        neighbor 20.1.1.2 description Towards PE1
        neighbor 20.2.1.2 remote-as 6488163
        neighbor 20.2.1.2 description Towards PE2
        address-family ipv4 unicast
            neighbor 20.0.1.1 activate
            neighbor 20.0.1.1 next-hop-self
            neighbor 20.0.1.1 send-community both
            neighbor 20.0.1.1 soft-reconfiguration inbound
            neighbor 20.0.1.3 activate
            neighbor 20.0.1.3 allowas-in 1
            neighbor 20.0.1.3 next-hop-self
            neighbor 20.0.1.3 send-community both
            neighbor 20.0.1.3 soft-reconfiguration inbound
```

```
            neighbor 20.1.1.2 activate
            neighbor 20.1.1.2 allowas-in 1
            neighbor 20.1.1.2 send-community both
            neighbor 20.1.1.2 soft-reconfiguration inbound
            neighbor 20.2.1.2 activate
            neighbor 20.2.1.2 allowas-in 1
            neighbor 20.2.1.2 send-community both
            neighbor 20.2.1.2 soft-reconfiguration inbound
        exit-address-family
!
    vrf vrf_2
        neighbor 20.0.1.1 remote-as 100
        neighbor 20.0.1.1 description Towards LAN
        neighbor 20.0.1.3 remote-as 100
        neighbor 20.0.1.3 description Towards CE2
        neighbor 20.1.1.2 remote-as 6488163
        neighbor 20.1.1.2 description Towards PE1
        neighbor 20.2.1.2 remote-as 6488163
        neighbor 20.2.1.2 description Towards PE2
        address-family ipv4 unicast
            neighbor 20.0.1.1 activate
            neighbor 20.0.1.1 next-hop-self
            neighbor 20.0.1.1 send-community both
            neighbor 20.0.1.1 soft-reconfiguration inbound
            neighbor 20.0.1.3 activate
            neighbor 20.0.1.3 allowas-in 1
            neighbor 20.0.1.3 next-hop-self
            neighbor 20.0.1.3 send-community both
            neighbor 20.0.1.3 soft-reconfiguration inbound
            neighbor 20.1.1.2 activate
            neighbor 20.1.1.2 allowas-in 1
            neighbor 20.1.1.2 send-community both
            neighbor 20.1.1.2 soft-reconfiguration inbound
            neighbor 20.2.1.2 activate
            neighbor 20.2.1.2 allowas-in 1
            neighbor 20.2.1.2 send-community both
            neighbor 20.2.1.2 soft-reconfiguration inbound
        exit-address-family
!
https-server vrf mgmt
```

CE1 verification

```
CE1# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS               : 100          BGP Router Identifier  : 99.0.0.1
 Peers                  : 0            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------

Address-family : IPv6 Unicast
-----------------------------

Address-family : L2VPN EVPN
-----------------------------
```

```
VRF : vrf_1
BGP Summary
-----------
 Local AS               : 100          BGP Router Identifier  : 20.2.1.1
 Peers                  : 4            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.0.1.1        100         56      63      00h:26m:24s  Established   Up
 20.0.1.3        100         37      36      00h:29m:13s  Established   Up
 20.1.1.2        6488163     37      36      00h:28m:34s  Established   Up
 20.2.1.2        6488163     35      35      00h:28m:01s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

VRF : vrf_2
BGP Summary
-----------
 Local AS               : 100          BGP Router Identifier  : 20.2.1.1
 Peers                  : 4            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.0.1.1        100         58      65      00h:26m:25s  Established   Up
 20.0.1.3        100         38      38      00h:29m:18s  Established   Up
 20.1.1.2        6488163     37      40      00h:28m:33s  Established   Up
 20.2.1.2        6488163     38      38      00h:27m:57s  Established   Up

Address-family : IPv6 Unicast
-----------------------------
```

```
CE1#  show bgp all-vrf all
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 99.0.0.1

Address-family : IPv4 Unicast
-----------------------------
    Network           Nexthop     Metric    LocPrf   Weight Path
Total number of entries 0

Address-family : IPv6 Unicast
-----------------------------
    Network           Nexthop     Metric    LocPrf   Weight Path
Total number of entries 0

Address-family : L2VPN EVPN
-----------------------------
```

```
    Network           Nexthop     Metric    LocPrf   Weight Path
------------------------------
Total number of entries 0

VRF : vrf_1
Local Router-ID 20.2.1.1

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*>i 101.0.0.0/24       20.0.1.1        0          0          0       i
* i 201.0.0.0/24       20.0.1.3        0          100        0       6488163 100 i
*>e 201.0.0.0/24       20.1.1.2        0          100        0       6488163 100 i
*=e 201.0.0.0/24       20.2.1.2        0          100        0       6488163 100 i
Total number of entries 4

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

VRF : vrf_2
Local Router-ID 20.2.1.1

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*>i 102.0.0.0/24       20.0.1.1        0          0          0       i
* i 202.0.0.0/24       20.0.1.3        0          100        0       6488163 100 i
*=e 202.0.0.0/24       20.1.1.2        0          100        0       6488163 100 i
*>e 202.0.0.0/24       20.2.1.2        0          100        0       6488163 100 i
Total number of entries 4

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0
```

```
CE1# show ip route all-vrf

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix               Nexthop            Interface     VRF(egress)   Origin/
Distance/    Age
                                                                    Type     Metric
--------------------------------------------------------------------------------
------------
99.0.0.0/24         -                1/1/46       -                 C
[0/0]        -
99.0.0.1/32         -                1/1/46       -                 L
[0/0]        -

VRF: vrf_1
```

```
Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
-----------------------------------------------------------------------------------
------------
20.0.1.0/24         -                vlan2001      -                 C
[0/0]        -
20.0.1.2/32         -                vlan2001      -                 L
[0/0]        -
20.1.1.0/24         -                vlan2011      -                 C
[0/0]        -
20.1.1.1/32         -                vlan2011      -                 L
[0/0]        -
20.2.1.0/24         -                vlan2021      -                 C
[0/0]        -
20.2.1.1/32         -                vlan2021      -                 L
[0/0]        -
101.0.0.0/24        20.0.1.1         vlan2001      -                 B/I
[200/0]      01h:11m:54s
201.0.0.0/24        20.1.1.2         vlan2011      -                 B/E
[20/0]       01h:10m:33s
                    20.2.1.2         vlan2021      -
[20/0]       01h:10m:33s

VRF: vrf_2

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
-----------------------------------------------------------------------------------
------------
20.0.1.0/24         -                vlan2002      -                 C
[0/0]        -
20.0.1.2/32         -                vlan2002      -                 L
[0/0]        -
20.1.1.0/24         -                vlan2012      -                 C
[0/0]        -
20.1.1.1/32         -                vlan2012      -                 L
[0/0]        -
20.2.1.0/24         -                vlan2022      -                 C
[0/0]        -
20.2.1.1/32         -                vlan2022      -                 L
[0/0]        -
102.0.0.0/24        20.0.1.1         vlan2002      -                 B/I
[200/0]      00h:54m:05s
202.0.0.0/24        20.2.1.2         vlan2022      -                 B/E
[20/0]       00h:54m:05s
                    20.1.1.2         vlan2012      -
[20/0]       00h:54m:05s

Total Route Count : 18
```

CE2 configuration

```
CE2# show running-config
Current configuration:
!
!Version ArubaOS-CX GL.10.09.0001BF
!export-password: default
```

```
hostname CE2
no ip icmp redirect
profile l3-core
logging console severity crit
vrf vrf_1
vrf vrf_2
cli-session
    timeout 0
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2001-2002,2031-2032,2041-2042
interface mgmt
    no shutdown
    ip dhcp
system interface-group 1 speed 10g
    !interface group 1 contains ports 1/1/1-1/1/12
system interface-group 4 speed 10g
    !interface group 4 contains ports 1/1/37-1/1/48
interface lag 99 multi-chassis
    no shutdown
    description Towards L2SW
    no routing
    vlan trunk native 1
    vlan trunk allowed 2001-2002
    lacp mode active
interface lag 256
    no shutdown
    description Towards ISL
    no routing
    vlan trunk native 1 tag
    vlan trunk allowed all
    lacp mode active
interface 1/1/1
    no shutdown
    mtu 9000
    description Towards L2SW
    lag 99
interface 1/1/2
    no shutdown
    mtu 9000
    description Towards ISL
    lag 256
interface 1/1/3
    no shutdown
    mtu 9000
    description Towards ISL
    lag 256
interface 1/1/46
    no shutdown
    mtu 9000
    description Towards KA
    ip mtu 9000
    ip address 99.0.0.2/24
interface 1/1/47
    no shutdown
    mtu 9000
    description Towards PE1
    no routing
```

```
    vlan trunk native 1
    vlan trunk allowed 2031-2032
interface 1/1/48
    no shutdown
    mtu 9000
    description Towards PE2
    no routing
    vlan trunk native 1
    vlan trunk allowed 2041-2042
interface vlan 2001
    vrf attach vrf_1
    description Towards LAN
    ip mtu 9000
    vsx active-forwarding
    ip address 20.0.1.3/24
interface vlan 2002
    vrf attach vrf_2
    description Towards LAN
    ip mtu 9000
    vsx active-forwarding
    ip address 20.0.1.3/24
interface vlan 2031
    vrf attach vrf_1
    description Towards PE1
    ip mtu 9000
    ip address 20.3.1.1/24
interface vlan 2032
    vrf attach vrf_2
    description Towards PE1
    ip mtu 9000
    ip address 20.3.1.1/24
interface vlan 2041
    vrf attach vrf_1
    description Towards PE2
    ip mtu 9000
    ip address 20.4.1.1/24
interface vlan 2042
    vrf attach vrf_2
    description Towards PE2
    ip mtu 9000
    ip address 20.4.1.1/24
vsx
    inter-switch-link lag 256
    role secondary
    keepalive peer 99.0.0.1 source 99.0.0.2
!
!
!
!
!
router bgp 100
!
    vrf vrf_1
        neighbor 20.0.1.1 remote-as 100
        neighbor 20.0.1.1 description Towards LAN
        neighbor 20.0.1.2 remote-as 100
        neighbor 20.0.1.2 description Towards CE1
        neighbor 20.3.1.2 remote-as 6488163
        neighbor 20.3.1.2 description Towards PE1
        neighbor 20.4.1.2 remote-as 6488163
        neighbor 20.4.1.2 description Towards PE2
        address-family ipv4 unicast
```

```
            neighbor 20.0.1.1 activate
            neighbor 20.0.1.1 next-hop-self
            neighbor 20.0.1.1 send-community both
            neighbor 20.0.1.1 soft-reconfiguration inbound
            neighbor 20.0.1.2 activate
            neighbor 20.0.1.2 allowas-in 1
            neighbor 20.0.1.2 next-hop-self
            neighbor 20.0.1.2 send-community both
            neighbor 20.0.1.2 soft-reconfiguration inbound
            neighbor 20.3.1.2 activate
            neighbor 20.3.1.2 allowas-in 1
            neighbor 20.3.1.2 send-community both
            neighbor 20.3.1.2 soft-reconfiguration inbound
            neighbor 20.4.1.2 activate
            neighbor 20.4.1.2 allowas-in 1
            neighbor 20.4.1.2 send-community both
            neighbor 20.4.1.2 soft-reconfiguration inbound
        exit-address-family
!
    vrf vrf_2
        neighbor 20.0.1.1 remote-as 100
        neighbor 20.0.1.1 description Towards LAN
        neighbor 20.0.1.2 remote-as 100
        neighbor 20.0.1.2 description Towards CE1
        neighbor 20.3.1.2 remote-as 6488163
        neighbor 20.3.1.2 description Towards PE1
        neighbor 20.4.1.2 remote-as 6488163
        neighbor 20.4.1.2 description Towards PE2
        address-family ipv4 unicast
            neighbor 20.0.1.1 activate
            neighbor 20.0.1.1 next-hop-self
            neighbor 20.0.1.1 send-community both
            neighbor 20.0.1.1 soft-reconfiguration inbound
            neighbor 20.0.1.2 activate
            neighbor 20.0.1.2 allowas-in 1
            neighbor 20.0.1.2 next-hop-self
            neighbor 20.0.1.2 send-community both
            neighbor 20.0.1.2 soft-reconfiguration inbound
            neighbor 20.3.1.2 activate
            neighbor 20.3.1.2 allowas-in 1
            neighbor 20.3.1.2 send-community both
            neighbor 20.3.1.2 soft-reconfiguration inbound
            neighbor 20.4.1.2 activate
            neighbor 20.4.1.2 allowas-in 1
            neighbor 20.4.1.2 send-community both
            neighbor 20.4.1.2 soft-reconfiguration inbound
        exit-address-family
!
https-server vrf mgmt
```

CE2 verification

```
CE2#  show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS               : 100          BGP Router Identifier  : 99.0.0.2
 Peers                  : 0            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0
```

```
Address-family : IPv4 Unicast
-----------------------------

Address-family : IPv6 Unicast
-----------------------------

Address-family : L2VPN EVPN
-----------------------------

VRF : vrf_1
BGP Summary
-----------
 Local AS               : 100          BGP Router Identifier  : 20.4.1.1
 Peers                  : 4            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.0.1.1        100         56      65      00h:26m:24s  Established   Up
 20.0.1.2        100         36      37      00h:29m:13s  Established   Up
 20.3.1.2        6488163     36      36      00h:28m:34s  Established   Up
 20.4.1.2        6488163     35      35      00h:27m:58s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

VRF : vrf_2
BGP Summary
-----------
 Local AS               : 100          BGP Router Identifier  : 20.4.1.1
 Peers                  : 4            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.0.1.1        100         58      65      00h:26m:25s  Established   Up
 20.0.1.2        100         38      38      00h:29m:18s  Established   Up
 20.3.1.2        6488163     38      40      00h:28m:43s  Established   Up
 20.4.1.2        6488163     37      38      00h:27m:56s  Established   Up

Address-family : IPv6 Unicast
-----------------------------
```

```
CE2#  show bgp all-vrf all
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 99.0.0.2

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop          Metric     LocPrf     Weight Path
```

```
Total number of entries 0

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : L2VPN EVPN
-----------------------------
     Network                                             Nexthop         Metric
   LocPrf    Weight   Path
----------------------------------------------------------------------------------
--------------------------
Total number of entries 0

VRF : vrf_1
Local Router-ID 20.4.1.1

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*>i 101.0.0.0/24       20.0.1.1        0          0          0       i
* i 201.0.0.0/24       20.0.1.2        0          100        0       6488163 100 i
*>e 201.0.0.0/24       20.3.1.2        0          100        0       6488163 100 i
*=e 201.0.0.0/24       20.4.1.2        0          100        0       6488163 100 i
Total number of entries 4

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

VRF : vrf_2
Local Router-ID 20.4.1.1

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*>i 102.0.0.0/24       20.0.1.1        0          0          0       i
* i 202.0.0.0/24       20.0.1.2        0          100        0       6488163 100 i
*=e 202.0.0.0/24       20.3.1.2        0          100        0       6488163 100 i
*>e 202.0.0.0/24       20.4.1.2        0          100        0       6488163 100 i
Total number of entries 4

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0
```

```
CE2# show ip route all-vrf

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default
```

```
Prefix                  Nexthop      Interface      VRF(egress)    Origin/   Distance/
Age
                                                                   Type      Metric
--------------------------------------------------------------------------------
-------
99.0.0.0/24         -                1/1/46        -                 C
[0/0]         -
99.0.0.2/32         -                1/1/46        -                 L
[0/0]         -

VRF: vrf_1

Prefix                  Nexthop            Interface      VRF(egress)    Origin/
Distance/     Age
                                                                   Type      Metric
--------------------------------------------------------------------------------
------------
20.0.1.0/24         -                vlan2001      -                 C
[0/0]         -
20.0.1.3/32         -                vlan2001      -                 L
[0/0]         -
20.3.1.0/24         -                vlan2031      -                 C
[0/0]         -
20.3.1.1/32         -                vlan2031      -                 L
[0/0]         -
20.4.1.0/24         -                vlan2041      -                 C
[0/0]         -
20.4.1.1/32         -                vlan2041      -                 L
[0/0]         -
101.0.0.0/24        20.0.1.1         vlan2001      -                 B/I
[200/0]       01h:11m:55s
201.0.0.0/24        20.4.1.2         vlan2041      -                 B/E
[20/0]        01h:10m:34s
                    20.3.1.2         vlan2031      -
[20/0]        01h:10m:34s

VRF: vrf_2

Prefix                  Nexthop            Interface      VRF(egress)    Origin/
Distance/     Age
                                                                   Type      Metric
--------------------------------------------------------------------------------
------------
20.0.1.0/24         -                vlan2002      -                 C
[0/0]         -
20.0.1.3/32         -                vlan2002      -                 L
[0/0]         -
20.3.1.0/24         -                vlan2032      -                 C
[0/0]         -
20.3.1.1/32         -                vlan2032      -                 L
[0/0]         -
20.4.1.0/24         -                vlan2042      -                 C
[0/0]         -
20.4.1.1/32         -                vlan2042      -                 L
[0/0]         -
102.0.0.0/24        20.0.1.1         vlan2002      -                 B/I
[200/0]       00h:54m:06s
202.0.0.0/24        20.4.1.2         vlan2042      -                 B/E
[20/0]        00h:54m:06s
                    20.3.1.2         vlan2032      -
[20/0]        00h:54m:06s
```

```
Total Route Count : 18
```

CE3 configuration

```
CE3# show running-config
Current configuration:
!
!Version ArubaOS-CX GL.10.09.0001BF
!export-password: default
hostname CE3
profile l3-core
logging console severity crit
vrf vrf_1
vrf vrf_2
cli-session
    timeout 0
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2051-2052,2061-2062
interface mgmt
    no shutdown
    ip dhcp
system interface-group 4 speed 10g
    !interface group 4 contains ports 1/1/37-1/1/48
interface 1/1/41
    no shutdown
    mtu 9000
    description Towards PE3
    no routing
    vlan trunk native 1
    vlan trunk allowed 2051-2052
interface 1/1/42
    no shutdown
    mtu 9000
    description Towards LAN
    no routing
    vlan trunk native 1
    vlan trunk allowed 2061-2062
interface vlan 2051
    vrf attach vrf_1
    description Towards PE3
    ip mtu 9000
    ip address 20.5.1.2/24
interface vlan 2052
    vrf attach vrf_2
    description Towards PE3
    ip mtu 9000
    ip address 20.5.1.2/24
interface vlan 2061
    vrf attach vrf_1
    description Towards LAN
    ip mtu 9000
    ip address 20.6.1.1/24
interface vlan 2062
    vrf attach vrf_2
    description Towards LAN
```

```
    ip mtu 9000
    ip address 20.6.1.1/24
!
!
!
!
!
router bgp 100
!
    vrf vrf_1
        neighbor 20.5.1.1 remote-as 6488163
        neighbor 20.5.1.1 description Towards PE3
        neighbor 20.6.1.2 remote-as 100
        neighbor 20.6.1.2 description Towards LAN
        address-family ipv4 unicast
            neighbor 20.5.1.1 activate
            neighbor 20.5.1.1 allowas-in 1
            neighbor 20.5.1.1 send-community both
            neighbor 20.5.1.1 soft-reconfiguration inbound
            neighbor 20.6.1.2 activate
            neighbor 20.6.1.2 next-hop-self
            neighbor 20.6.1.2 send-community both
            neighbor 20.6.1.2 soft-reconfiguration inbound
        exit-address-family
!
    vrf vrf_2
        neighbor 20.5.1.1 remote-as 6488163
        neighbor 20.5.1.1 description Towards PE3
        neighbor 20.6.1.2 remote-as 100
        neighbor 20.6.1.2 description Towards LAN
        address-family ipv4 unicast
            neighbor 20.5.1.1 activate
            neighbor 20.5.1.1 allowas-in 1
            neighbor 20.5.1.1 send-community both
            neighbor 20.5.1.1 soft-reconfiguration inbound
            neighbor 20.6.1.2 activate
            neighbor 20.6.1.2 next-hop-self
            neighbor 20.6.1.2 send-community both
            neighbor 20.6.1.2 soft-reconfiguration inbound
        exit-address-family
!
https-server vrf mgmt
```

CE3 verification

```
CE3#  show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS                : 100          BGP Router Identifier  : (null)
 Peers                   : 0            Log Neighbor Changes   : No
 Cfg. Hold Time          : 180          Cfg. Keep Alive        : 60
 Confederation Id        : 0

Address-family : IPv4 Unicast
-----------------------------

Address-family : IPv6 Unicast
-----------------------------
```

```
Address-family : L2VPN EVPN
-----------------------------

VRF : vrf_1
BGP Summary
-----------
 Local AS               : 100          BGP Router Identifier  : 20.6.1.1
 Peers                  : 2            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.5.1.1        6488163     34      33      00h:25m:12s  Established   Up
 20.6.1.2        100         54      62      00h:25m:04s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

VRF : vrf_2
BGP Summary
-----------
 Local AS               : 100          BGP Router Identifier  : 20.6.1.1
 Peers                  : 2            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.5.1.1        6488163     34      34      00h:25m:11s  Established   Up
 20.6.1.2        100         56      63      00h:25m:04s  Established   Up

Address-family : IPv6 Unicast
-----------------------------
```

```
CE3#  show bgp all-vrf all
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Router-ID not configured

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : L2VPN EVPN
-----------------------------
     Network                                              Nexthop         Metric
   LocPrf    Weight   Path
```

```
-----------------------------------------------------------------------------------
--------------------------
Total number of entries 0

VRF : vrf_1
Local Router-ID 20.6.1.1

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*>e 101.0.0.0/24       20.5.1.1        0          100        0       6488163 100 i
*>i 201.0.0.0/24       20.6.1.2        0          0          0       i
Total number of entries 2

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

VRF : vrf_2
Local Router-ID 20.6.1.1

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*>e 102.0.0.0/24       20.5.1.1        0          100        0       6488163 100 i
*>i 202.0.0.0/24       20.6.1.2        0          0          0       i
Total number of entries 2

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0
```

```
CE3# show ip route all-vrf

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: vrf_1

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
-----------------------------------------------------------------------------------
------------
20.5.1.0/24        -                vlan2051      -               C
[0/0]        -
20.5.1.2/32        -                vlan2051      -               L
[0/0]        -
20.6.1.0/24        -                vlan2061      -               C
[0/0]        -
20.6.1.1/32        -                vlan2061      -               L
[0/0]        -
101.0.0.0/24       20.5.1.1         vlan2051      -               B/E
[20/0]       01h:10m:42s
```

```
201.0.0.0/24        20.6.1.2         vlan2061      -                 B/I
[200/0]      01h:10m:34s

VRF: vrf_2

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
--------------------------------------------------------------------------------------
------------
20.5.1.0/24         -                vlan2052      -                 C
[0/0]        -
20.5.1.2/32         -                vlan2052      -                 L
[0/0]        -
20.6.1.0/24         -                vlan2062      -                 C
[0/0]        -
20.6.1.1/32         -                vlan2062      -                 L
[0/0]        -
102.0.0.0/24        20.5.1.1         vlan2052      -                 B/E
[20/0]       00h:54m:06s
202.0.0.0/24        20.6.1.2         vlan2062      -                 B/I
[200/0]      00h:54m:06s

Total Route Count : 12
```

P1 configuration

```
P1# show running-config
Current configuration:
!
!Version ArubaOS-CX LL.10.09.0001BF
!export-password: default
hostname P1
profile core-spine
logging console severity crit
cli-session
    timeout 0
!
!
!
!
!
!
ssh server vrf mgmt
vlan 1
interface mgmt
    no shutdown
    ip dhcp
system interface-group 1 speed 10g
    !interface group 1 contains ports 1/1/1-1/1/4
interface lag 99
    no shutdown
    description Towards PE3
    ip mtu 9198
    ip address 30.0.2.1/24
    lacp mode active
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface 1/1/14
```

```
    no shutdown
    mtu 9198
    description Towards PE1
    ip mtu 9198
    ip address 30.0.0.2/24
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface 1/1/15
    no shutdown
    mtu 9198
    description Towards PE2
    ip mtu 9198
    ip address 30.0.1.2/24
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface 1/1/16
    no shutdown
    mtu 9198
    description Towards PE3
    lag 99
interface 1/1/17
    no shutdown
    mtu 9198
    description Towards PE3
    lag 99
interface 1/1/27
    description Towards PE3
interface 1/1/28
    description Towards PE3
interface loopback 0
    ip address 4.4.4.4/32
    ip ospf 1 area 0.0.0.0
!
!
!
!
!
router ospf 1
    area 0.0.0.0
router bgp 6488163
    neighbor 1.1.1.1 remote-as 6488163
    neighbor 1.1.1.1 description Towards PE1
    neighbor 1.1.1.1 update-source loopback 0
    neighbor 2.2.2.2 remote-as 6488163
    neighbor 2.2.2.2 description Towards PE2
    neighbor 2.2.2.2 update-source loopback 0
    neighbor 3.3.3.3 remote-as 6488163
    neighbor 3.3.3.3 description Towards PE2
    neighbor 3.3.3.3 update-source loopback 0
    neighbor 5.5.5.5 remote-as 6488163
    neighbor 5.5.5.5 description Towards RR2
    neighbor 5.5.5.5 update-source loopback 0
    address-family vpnv4 unicast
        neighbor 1.1.1.1 activate
        neighbor 1.1.1.1 route-reflector-client
        neighbor 1.1.1.1 send-community both
        neighbor 2.2.2.2 activate
        neighbor 2.2.2.2 route-reflector-client
        neighbor 2.2.2.2 send-community both
        neighbor 3.3.3.3 activate
```

```
        neighbor 3.3.3.3 route-reflector-client
        neighbor 3.3.3.3 send-community both
        neighbor 5.5.5.5 activate
        neighbor 5.5.5.5 send-community both
    exit-address-family
!
https-server vrf mgmt
mpls
    enable
    label-protocol ldp
        enable
        router-id loopback0
```

P1 verification

```
P1# show ip ospf neighbors
VRF : default                          Process : 1
===================================================

Total Number of Neighbors : 3

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
1.1.1.1          1         FULL/DR           30.0.0.1           1/1/14

2.2.2.2          1         FULL/BDR          30.0.1.1           1/1/15

3.3.3.3          1         FULL/BDR          30.0.2.2           lag99
```

```
P1# show mpls ldp neighbor
Local LDP Identifier: 4.4.4.4:0, Peer LDP Identifier: 2.2.2.2:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:13:36
    State: operational
    LDP Discovery Sources: 1/1/15
    Addresses bound to this peer:
         2.2.2.2 30.0.1.1 30.0.4.1

Local LDP Identifier: 4.4.4.4:0, Peer LDP Identifier: 3.3.3.3:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:12:42
    State: operational
    LDP Discovery Sources: lag99
    Addresses bound to this peer:
         3.3.3.3 30.0.2.2 30.0.5.2

Local LDP Identifier: 4.4.4.4:0, Peer LDP Identifier: 1.1.1.1:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:13:36
    State: operational
    LDP Discovery Sources: 1/1/14
    Addresses bound to this peer:
         1.1.1.1 30.0.0.1 30.0.3.1
```

```
P1# show mpls ldp binding
30.0.4.1/32
        No local binding
        remote binding: lsr:2.2.2.2:0 label: exp-null
30.0.2.1/32
        local binding: label: exp-null
        No remote binding
30.0.1.2/32
        local binding: label: exp-null
        No remote binding
30.0.0.1/32
        No local binding
        remote binding: lsr:1.1.1.1:0 label: exp-null
30.0.1.1/32
        No local binding
        remote binding: lsr:2.2.2.2:0 label: exp-null
30.0.5.2/32
        No local binding
        remote binding: lsr:3.3.3.3:0 label: exp-null
30.0.2.2/32
        No local binding
        remote binding: lsr:3.3.3.3:0 label: exp-null
2.2.2.2/32
        local binding: label: 18
        remote binding: lsr:1.1.1.1:0 label:18
        remote binding: lsr:3.3.3.3:0 label:17
        remote binding: lsr:2.2.2.2:0 label: exp-null
5.5.5.5/32
        local binding: label: 17
        remote binding: lsr:1.1.1.1:0 label:17
        remote binding: lsr:3.3.3.3:0 label:19
        remote binding: lsr:2.2.2.2:0 label:18
30.0.0.2/32
        local binding: label: exp-null
        No remote binding
3.3.3.3/32
        local binding: label: 19
        remote binding: lsr:1.1.1.1:0 label:19
        remote binding: lsr:3.3.3.3:0 label: exp-null
        remote binding: lsr:2.2.2.2:0 label:19
4.4.4.4/32
        local binding: label: exp-null
        remote binding: lsr:1.1.1.1:0 label:16
        remote binding: lsr:3.3.3.3:0 label:18
        remote binding: lsr:2.2.2.2:0 label:17
30.0.3.1/32
        No local binding
        remote binding: lsr:1.1.1.1:0 label: exp-null
1.1.1.1/32
        local binding: label: 16
        remote binding: lsr:1.1.1.1:0 label: exp-null
        remote binding: lsr:3.3.3.3:0 label:16
        remote binding: lsr:2.2.2.2:0 label:16
```

```
P1# show mpls forwarding

MPLS Bindings

Entry Bindings   : 4
Exit Bindings    : 2
Transit Bindings : 4
```

```
PHP Mode          : Explicit-Null
QoS Mode          : Uniform
TTL Propagation   : Uniform

Entry Bindings:

Origin Prefix       Ingress  Nexthop    Outgoing  Egress     Egress   Status
                    VRF      Address    Label     Interface  VRF
--------------------------------------------------------------------------------
LDP    5.5.5.5/32  default  30.0.2.2   19        lag99      default  operational


Exit Bindings:

Origin   Prefix              Incoming  Service   Egress            Status
                             Label     Label     VRF
--------------------------------------------------------------------------------
static   n/a                 exp-null  -         default           operational
static   n/a                 7         -         default           operational

Transit Bindings:

Origin  Prefix       Incoming  Egress     Egress   Nexthop    Outgoing  Status

                     Label     Interface  VRF      Address    Label

-----------------------------------------------------------------------------
LDP     1.1.1.1/32   16        1/1/14     default  30.0.0.1  exp-null  Operational

LDP     5.5.5.5/32   17        lag99      default  30.0.2.2  19        Operational

LDP     2.2.2.2/32   18        1/1/15     default  30.0.1.1  exp-null  Operational

LDP     3.3.3.3/32   19        lag99      default  30.0.2.2  exp-null  Operational
```

```
P1# show bgp vpnv4 un neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 1.1.1.1 (Internal)
    Description         : Towards PE1
    Peer-group          :

    Remote Router Id    : 1.1.1.1           Local Router Id    : 4.4.4.4
    Remote AS           : 6488163           Local AS           : 6488163
    Remote Port         : 42093             Local Port         : 179
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      : loopback0
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:15m:55s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error
```

```
    Graceful-Restart    : Enabled           Gr. Restart Time    : 120
    Gr. Stalepath Time  : 300               Remove Private-AS   : No
    TTL                 : 255               Local Cluster-ID    :
    Weight              : 0                 Fall-over           : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      7        3
    Keepalives                  18       18
    Route Refresh                0        0
    Total                       26       22

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : Yes                Send Community    : both
    Allow-AS in         : 0                  Advt. Interval    : 30
    Max. Prefix         : 300000             Soft Reconfig In  :
    Nexthop-Self        :                    Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 2.2.2.2 (Internal)
    Description         : Towards PE2
    Peer-group          :

    Remote Router Id    : 2.2.2.2           Local Router Id     : 4.4.4.4
    Remote AS           : 6488163           Local AS            : 6488163
    Remote Port         : 44703             Local Port          : 179
    State               : Established       Admin Status        : Up
    Conn. Established   : 1                 Conn. Dropped       : 0
    Passive             : No                Update-Source       : loopback0
    Cfg. Hold Time      : 180               Cfg. Keep Alive     : 60
    Neg. Hold Time      : 180               Neg. Keep Alive     : 60
    Up/Down Time        : 00h:15m:21s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS       : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error
```

```
    Graceful-Restart    : Enabled           Gr. Restart Time    : 120
    Gr. Stalepath Time  : 300               Remove Private-AS   : No
    TTL                 : 255               Local Cluster-ID    :
    Weight              : 0                 Fall-over           : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      5        3
    Keepalives                  17       18
    Route Refresh                0        0
    Total                       23       22

    Capability                              Advertised      Received
    ----------------------------            -----------     ----------
    Route Refresh                           Yes             Yes
    Graceful Restart                        Yes             Yes
    Add-Path                                No              No
    Four Octet ASN                          Yes             Yes
    Address family IPv4 Unicast             No              No
    Address family IPv6 Unicast             No              No
    Address family VPNv4 Unicast            Yes             Yes
    Address family L2VPN EVPN               No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : Yes                Send Community    : both
    Allow-AS in         : 0                  Advt. Interval    : 30
    Max. Prefix         : 300000             Soft Reconfig In  :
    Nexthop-Self        :                    Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 3.3.3.3 (Internal)
    Description         : Towards PE2
    Peer-group          :

    Remote Router Id    : 3.3.3.3           Local Router Id    : 4.4.4.4
    Remote AS           : 6488163           Local AS           : 6488163
    Remote Port         : 179               Local Port         : 33231
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      : loopback0
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:14m:48s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error
```

```
    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 255               Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      3        3
    Keepalives                  18       17
    Route Refresh                0        0
    Total                       22       21

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : Yes                Send Community    : both
    Allow-AS in         : 0                  Advt. Interval    : 30
    Max. Prefix         : 300000             Soft Reconfig In  :
    Nexthop-Self        :                    Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 5.5.5.5 (Internal)
    Description         : Towards RR2
    Peer-group          :

    Remote Router Id    : 5.5.5.5           Local Router Id    : 4.4.4.4
    Remote AS           : 6488163           Local AS           : 6488163
    Remote Port         : 179               Local Port         : 40545
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      : loopback0
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:15m:33s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error
```

```
    Graceful-Restart    : Enabled           Gr. Restart Time    : 120
    Gr. Stalepath Time  : 300               Remove Private-AS   : No
    TTL                 : 255               Local Cluster-ID    :
    Weight              : 0                 Fall-over           : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      7        6
    Keepalives                  17       17
    Route Refresh                0        0
    Total                       25       24

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                 Send Community    : both
    Allow-AS in         : 0                  Advt. Interval    : 30
    Max. Prefix         : 300000             Soft Reconfig In  :
    Nexthop-Self        :                    Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :
```

```
P1#  show bgp vpnv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 4.4.4.4

    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 1:1                  (Label 20)
*>i 101.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 101.0.0.0/24       2.2.2.2         0          100        0       100 i
* i 101.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 2:1                  (Label 22)
*>i 102.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 102.0.0.0/24       2.2.2.2         0          100        0       100 i
* i 102.0.0.0/24       1.1.1.1         0          100        0       100 i
```

```
Route Distinguisher: 1:3                 (Label 21)
*>i 201.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 201.0.0.0/24       3.3.3.3         0          100        0       100 i

Route Distinguisher: 2:3                 (Label 22)
*>i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 10
```

```
P1# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 4.4.4.4
 Peers                  : 4            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------

Address-family : IPv6 Unicast
-----------------------------

Address-family : VPNv4 Unicast
------------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 1.1.1.1         6488163     36      43      00h:27m:27s  Established   Up
 2.2.2.2         6488163     37      40      00h:26m:53s  Established   Up
 3.3.3.3         6488163     35      40      00h:26m:20s  Established   Up
 5.5.5.5         6488163     39      40      00h:27m:05s  Established   Up

Address-family : L2VPN EVPN
-----------------------------
```

```
P1#  show bgp all-vrf all
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 4.4.4.4

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : VPNv4 Unicast
------------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 1:1                 (Label 20)
*>i 101.0.0.0/24       1.1.1.1         0          100        0       100 i
```

```
* i 101.0.0.0/24       2.2.2.2         0          100        0       100 i
* i 101.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 2:1                  (Label 22)
*>i 102.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 102.0.0.0/24       2.2.2.2         0          100        0       100 i
* i 102.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 1:3                  (Label 21)
*>i 201.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 201.0.0.0/24       3.3.3.3         0          100        0       100 i

Route Distinguisher: 2:3                  (Label 22)
*>i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 10

Address-family : L2VPN EVPN
-----------------------------
     Network                                              Nexthop         Metric
   LocPrf    Weight   Path
----------------------------------------------------------------------------------
--------------------------
Total number of entries 0
```

```
P1# show ip route all-vrf

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
----------------------------------------------------------------------------------
------------
1.1.1.1/32          30.0.0.1         1/1/14        -                 O
[110/100]    01h:13m:03s
2.2.2.2/32          30.0.1.1         1/1/15        -                 O
[110/100]    01h:12m:33s
3.3.3.3/32          30.0.2.2         lag99         -                 O
[110/50]     01h:11m:57s
4.4.4.4/32          -                loopback0     -                 L
[0/0]        -
5.5.5.5/32          30.0.2.2         lag99         -                 O
[110/100]    01h:11m:52s
30.0.0.0/24         -                1/1/14        -                 C
[0/0]        -
30.0.0.2/32         -                1/1/14        -                 L
[0/0]        -
30.0.1.0/24         -                1/1/15        -                 C
[0/0]        -
30.0.1.2/32         -                1/1/15        -                 L
[0/0]        -
30.0.2.0/24         -                lag99         -                 C
```

```
[0/0]        -
30.0.2.1/32        -                lag99        -                 L
[0/0]        -
30.0.3.0/24        30.0.0.1         1/1/14       -                 O
[110/200]    01h:11m:52s
                   30.0.2.2         lag99        -
[110/200]    01h:11m:52s
30.0.4.0/24        30.0.1.1         1/1/15       -                 O
[110/200]    01h:11m:52s
                   30.0.2.2         lag99        -
[110/200]    01h:11m:52s
30.0.5.0/24        30.0.2.2         lag99        -                 O
[110/100]    01h:11m:57s

Total Route Count : 14
```

P2 configuration

```
P2# show running
Current configuration:
!
!Version ArubaOS-CX LL.10.09.0001BF
!export-password: default
hostname P2
profile core-spine
logging console severity crit
cli-session
    timeout 0
!
!
!
!
!
!
ssh server vrf mgmt
vlan 1
interface mgmt
    no shutdown
    ip dhcp
system interface-group 1 speed 10g
    !interface group 1 contains ports 1/1/1-1/1/4
interface lag 100
    no shutdown
    description Towards PE3
    ip mtu 9198
    ip address 30.0.5.1/24
    lacp mode active
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface 1/1/25
    no shutdown
    mtu 9198
    description Towards PE1
    ip mtu 9198
    ip address 30.0.3.2/24
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface 1/1/26
```

```
    no shutdown
    mtu 9198
    description Towards PE2
    ip mtu 9198
    ip address 30.0.4.2/24
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface 1/1/27
    no shutdown
    mtu 9198
    description Towards PE3
    lag 100
interface 1/1/28
    no shutdown
    mtu 9198
    description Towards PE3
    lag 100
interface loopback 0
    ip address 5.5.5.5/32
    ip ospf 1 area 0.0.0.0
!
!
!
!
!
router ospf 1
    area 0.0.0.0
router bgp 6488163
    neighbor 1.1.1.1 remote-as 6488163
    neighbor 1.1.1.1 description Towards PE1
    neighbor 1.1.1.1 update-source loopback 0
    neighbor 2.2.2.2 remote-as 6488163
    neighbor 2.2.2.2 description Towards PE2
    neighbor 2.2.2.2 update-source loopback 0
    neighbor 3.3.3.3 remote-as 6488163
    neighbor 3.3.3.3 description Towards PE3
    neighbor 3.3.3.3 update-source loopback 0
    neighbor 4.4.4.4 remote-as 6488163
    neighbor 4.4.4.4 description Towards RR1
    neighbor 4.4.4.4 update-source loopback 0
    address-family vpnv4 unicast
        neighbor 1.1.1.1 activate
        neighbor 1.1.1.1 route-reflector-client
        neighbor 1.1.1.1 send-community both
        neighbor 2.2.2.2 activate
        neighbor 2.2.2.2 route-reflector-client
        neighbor 2.2.2.2 send-community both
        neighbor 3.3.3.3 activate
        neighbor 3.3.3.3 route-reflector-client
        neighbor 3.3.3.3 send-community both
        neighbor 4.4.4.4 activate
        neighbor 4.4.4.4 send-community both
    exit-address-family
!
https-server vrf mgmt
mpls
    enable
    label-protocol ldp
        enable
        router-id loopback0
```

P2 verification

```
P2# show ip ospf neighbors
VRF : default                          Process : 1
===================================================

Total Number of Neighbors : 3

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
1.1.1.1          1         FULL/DR           30.0.3.1           1/1/25

2.2.2.2          1         FULL/DR           30.0.4.1           1/1/26

3.3.3.3          1         FULL/BDR          30.0.5.2           lag100
```

```
P2#  show mpls ldp neighbor
Local LDP Identifier: 5.5.5.5:0, Peer LDP Identifier: 3.3.3.3:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:12:42
    State: operational
    LDP Discovery Sources: lag100
    Addresses bound to this peer:
         3.3.3.3 30.0.2.2 30.0.5.2

Local LDP Identifier: 5.5.5.5:0, Peer LDP Identifier: 1.1.1.1:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:13:19
    State: operational
    LDP Discovery Sources: 1/1/25
    Addresses bound to this peer:
         1.1.1.1 30.0.0.1 30.0.3.1

Local LDP Identifier: 5.5.5.5:0, Peer LDP Identifier: 2.2.2.2:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:13:14
    State: operational
    LDP Discovery Sources: 1/1/26
    Addresses bound to this peer:
         2.2.2.2 30.0.1.1 30.0.4.1
```

```
P2#  show mpls ldp binding
30.0.3.2/32
        local binding: label: exp-null
        No remote binding
30.0.1.1/32
        No local binding
        remote binding: lsr:2.2.2.2:0 label: exp-null
2.2.2.2/32
        local binding: label: 18
        remote binding: lsr:1.1.1.1:0 label:18
        remote binding: lsr:3.3.3.3:0 label:17
        remote binding: lsr:2.2.2.2:0 label: exp-null
30.0.5.1/32
        local binding: label: exp-null
        No remote binding
30.0.4.2/32
        local binding: label: exp-null
        No remote binding
```

```
5.5.5.5/32
        local binding: label: exp-null
        remote binding: lsr:1.1.1.1:0 label:17
        remote binding: lsr:3.3.3.3:0 label:19
        remote binding: lsr:2.2.2.2:0 label:18
30.0.0.1/32
        No local binding
        remote binding: lsr:1.1.1.1:0 label: exp-null
30.0.2.2/32
        No local binding
        remote binding: lsr:3.3.3.3:0 label: exp-null
30.0.5.2/32
        No local binding
        remote binding: lsr:3.3.3.3:0 label: exp-null
4.4.4.4/32
        local binding: label: 17
        remote binding: lsr:1.1.1.1:0 label:16
        remote binding: lsr:3.3.3.3:0 label:18
        remote binding: lsr:2.2.2.2:0 label:17
30.0.4.1/32
        No local binding
        remote binding: lsr:2.2.2.2:0 label: exp-null
1.1.1.1/32
        local binding: label: 16
        remote binding: lsr:1.1.1.1:0 label: exp-null
        remote binding: lsr:3.3.3.3:0 label:16
        remote binding: lsr:2.2.2.2:0 label:16
3.3.3.3/32
        local binding: label: 19
        remote binding: lsr:1.1.1.1:0 label:19
        remote binding: lsr:3.3.3.3:0 label: exp-null
        remote binding: lsr:2.2.2.2:0 label:19
30.0.3.1/32
        No local binding
        remote binding: lsr:1.1.1.1:0 label: exp-null
```

```
P2# show mpls forwarding

MPLS Bindings

Entry Bindings   : 4
Exit Bindings    : 2
Transit Bindings : 4
PHP Mode         : Explicit-Null
QoS Mode         : Uniform
TTL Propagation  : Uniform

Entry Bindings:

Origin Prefix      Ingress  Nexthop    Outgoing  Egress     Egress   Status
                   VRF      Address    Label     Interface  VRF
-------------------------------------------------------------------------------
LDP    4.4.4.4/32  default  30.0.5.2   18        lag100     default  operational

Exit Bindings:

Origin   Prefix              Incoming  Service   Egress            Status
                             Label     Label     VRF
-------------------------------------------------------------------------------
static   n/a                 exp-null  -         default           operational
static   n/a                 7         -         default           operational
```

```
Transit Bindings:

Origin  Prefix       Incoming  Egress     Egress   Nexthop   Outgoing  Status

                     Label     Interface  VRF      Address   Label

----------------------------------------------------------------------------
LDP     1.1.1.1/32   16        1/1/25     default  30.0.3.1  exp-null  operational
LDP     4.4.4.4/32   17        lag100     default  30.0.5.2  18        operational
 LDP     2.2.2.2/32   18        1/1/26     default  30.0.4.1  exp-null
operational  LDP     3.3.3.3/32   18        lag100     default  30.0.5.2  exp-null
 operational
```

```
P2# show bgp vpnv4 un neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 1.1.1.1 (Internal)
    Description         : Towards PE1
    Peer-group          :

    Remote Router Id    : 1.1.1.1           Local Router Id    : 5.5.5.5
    Remote AS           : 6488163           Local AS           : 6488163
    Remote Port         : 40029             Local Port         : 179
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      : loopback0
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:15m:40s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 255               Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      5        3
    Keepalives                  16       17
    Route Refresh                0        0
    Total                       22       21

    Capability                          Advertised      Received
    ----------------------------        -----------     ----------
    Route Refresh                       Yes             Yes
    Graceful Restart                    Yes             Yes
    Add-Path                            No              No
    Four Octet ASN                      Yes             Yes
```

```
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : Yes                 Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  :
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 2.2.2.2 (Internal)
    Description         : Towards PE2
    Peer-group          :

    Remote Router Id    : 2.2.2.2           Local Router Id    : 5.5.5.5
    Remote AS           : 6488163           Local AS           : 6488163
    Remote Port         : 179               Local Port         : 33107
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      : loopback0
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:15m:30s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 255               Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      5        3
    Keepalives                  17       17
    Route Refresh                0        0
    Total                       23       21

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
```

```
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : Yes                 Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  :
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 3.3.3.3 (Internal)
    Description         : Towards PE3
    Peer-group          :

    Remote Router Id    : 3.3.3.3           Local Router Id    : 5.5.5.5
    Remote AS           : 6488163           Local AS           : 6488163
    Remote Port         : 179               Local Port         : 39781
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      : loopback0
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:14m:46s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 255               Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      3        3
    Keepalives                  17       17
    Route Refresh                0        0
    Total                       21       21

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
```

```
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : Yes                 Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  :
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 4.4.4.4 (Internal)
    Description         : Towards RR1
    Peer-group          :

    Remote Router Id    : 4.4.4.4           Local Router Id    : 5.5.5.5
    Remote AS           : 6488163           Local AS           : 6488163
    Remote Port         : 40545             Local Port         : 179
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      : loopback0
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:15m:33s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 255               Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      6        7
    Keepalives                  17       17
    Route Refresh                0        0
    Total                       24       25

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
```

```
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                    Send Community    : both
    Allow-AS in         : 0                     Advt. Interval    : 30
    Max. Prefix         : 300000                Soft Reconfig In  :
    Nexthop-Self        :                       Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :
```

```
P2# show bgp vpnv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 5.5.5.5

    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 2:1                  (Label 21)
*>i 101.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 101.0.0.0/24       2.2.2.2         0          100        0       100 i

Route Distinguisher: 1:1                  (Label 20)
* i 101.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 2:3                  (Label 20)
*>i 201.0.0.0/24       3.3.3.3         0          100        0       100 i

Route Distinguisher: 1:3                  (Label 21)
* i 201.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 5
```

```
P2# show mpls forwarding

MPLS Bindings

Entry Bindings   : 4
Exit Bindings    : 2
Transit Bindings : 4
PHP Mode         : Explicit-Null
QoS Mode         : Uniform
TTL Propagation  : Uniform

Entry Bindings:

Origin Prefix      Ingress  Nexthop    Outgoing  Egress     Egress   Status
                   VRF      Address    Label     Interface  VRF
```

```
-------------------------------------------------------------------------------
LDP    4.4.4.4/32  default  30.0.5.2   18        lag100     default  operational


Exit Bindings:

Origin   Prefix              Incoming  Service   Egress           Status
                             Label     Label     VRF
-------------------------------------------------------------------------------
static   n/a                 exp-null  -         default          operational
static   n/a                 7         -         default          operational

Transit Bindings:

Origin  Prefix       Incoming  Egress     Egress   Nexthop   Outgoing  Status

                     Label     Interface  VRF      Address   Label

----------------------------------------------------------------------------------
LDP     1.1.1.1/32   16        1/1/25     default  30.0.3.1  exp-null  operational
LDP     4.4.4.4/32   17        lag100     default  30.0.5.2  18        operational
 LDP     2.2.2.2/32   18        1/1/26     default  30.0.4.1  exp-null
operational  LDP     3.3.3.3/32   18        lag100     default  30.0.5.2  exp-null
 operational
```

```
P2# show bgp vpnv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 5.5.5.5

    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 1:1                  (Label 20)
*>i 101.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 101.0.0.0/24       2.2.2.2         0          100        0       100 i
* i 101.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 2:1                  (Label 22)
*>i 102.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 102.0.0.0/24       2.2.2.2         0          100        0       100 i
* i 102.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 1:3                  (Label 21)
*>i 201.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 201.0.0.0/24       3.3.3.3         0          100        0       100 i

Route Distinguisher: 2:3                  (Label 22)
*>i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 10
```

```
P2# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 5.5.5.5
 Peers                  : 4            Log Neighbor Changes   : No
```

```
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------

Address-family : IPv6 Unicast
-----------------------------

Address-family : VPNv4 Unicast
------------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 1.1.1.1         6488163     34      39      00h:27m:12s  Established   Up
 2.2.2.2         6488163     35      41      00h:27m:02s  Established   Up
 3.3.3.3         6488163     36      39      00h:26m:17s  Established   Up
 4.4.4.4         6488163     40      39      00h:27m:05s  Established   Up

Address-family : L2VPN EVPN
-----------------------------
```

```
P2# show bgp all-vrf all
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 5.5.5.5

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : VPNv4 Unicast
------------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 1:1                  (Label 20)
*>i 101.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 101.0.0.0/24       2.2.2.2         0          100        0       100 i
* i 101.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 2:1                  (Label 22)
*>i 102.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 102.0.0.0/24       2.2.2.2         0          100        0       100 i
* i 102.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 1:3                  (Label 21)
*>i 201.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 201.0.0.0/24       3.3.3.3         0          100        0       100 i

Route Distinguisher: 2:3                  (Label 22)
*>i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 10
```

```
Address-family : L2VPN EVPN
-----------------------------
     Network                                                     Nexthop         Metric
   LocPrf    Weight   Path
-----------------------------------------------------------------------------------
--------------------------
Total number of entries 0
```

```
P2# show ip route all-vrf

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
-----------------------------------------------------------------------------------
------------
1.1.1.1/32          30.0.3.1         1/1/25        -                 O
[110/100]    01h:12m:46s
2.2.2.2/32          30.0.4.1         1/1/26        -                 O
[110/100]    01h:12m:21s
3.3.3.3/32          30.0.5.2         lag100        -                 O
[110/50]     01h:11m:52s
4.4.4.4/32          30.0.5.2         lag100        -                 O
[110/100]    01h:11m:52s
5.5.5.5/32          -                loopback0     -                 L
[0/0]        -
30.0.0.0/24         30.0.5.2         lag100        -                 O
[110/200]    01h:11m:52s
                    30.0.3.1         1/1/25        -
[110/200]    01h:11m:52s
30.0.1.0/24         30.0.4.1         1/1/26        -                 O
[110/200]    01h:11m:52s
                    30.0.5.2         lag100        -
[110/200]    01h:11m:52s
30.0.2.0/24         30.0.5.2         lag100        -                 O
[110/100]    01h:11m:52s
30.0.3.0/24         -                1/1/25        -                 C
[0/0]        -
30.0.3.2/32         -                1/1/25        -                 L
[0/0]        -
30.0.4.0/24         -                1/1/26        -                 C
[0/0]        -
30.0.4.2/32         -                1/1/26        -                 L
[0/0]        -
30.0.5.0/24         -                lag100        -                 C
[0/0]        -
30.0.5.1/32         -                lag100        -                 L
[0/0]        -

Total Route Count : 14
```

PE1 configuration

```
PE1# show running
Current configuration:
!
!Version ArubaOS-CX LL.10.09.0001BF
!export-password: default
hostname PE1
profile core-spine
logging console severity crit
vrf vrf_1
    rd 1:1
    l3vpn-only
    address-family ipv4 unicast
        route-target export 1:1
        route-target import 1:1
    exit-address-family
vrf vrf_2
    rd 2:1
    l3vpn-only
    address-family ipv4 unicast
        route-target export 2:1
        route-target import 2:1
    exit-address-family
cli-session
    timeout 0
!
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2011-2012,2031-2032
interface mgmt
    no shutdown
    ip dhcp
interface 1/1/37
    no shutdown
    mtu 9000
    description Towards CE1
    no routing
    vlan trunk native 1
    vlan trunk allowed 2011-2012
interface 1/1/38
    no shutdown
    mtu 9000
    description Towards CE2
    no routing
    vlan trunk native 1
    vlan trunk allowed 2031-2032
interface 1/1/39
    no shutdown
    mtu 9198
    description Towards P1
    ip mtu 9198
    ip address 30.0.0.1/24
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface 1/1/40
    no shutdown
    mtu 9198
```

```
    description Towards P2
    ip mtu 9198
    ip address 30.0.3.1/24
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface loopback 0
    ip address 1.1.1.1/32
    ip ospf 1 area 0.0.0.0
interface vlan 2011
    vrf attach vrf_1
    description Towards CE1
    ip mtu 9000
    ip address 20.1.1.2/24
interface vlan 2012
    vrf attach vrf_2
    description Towards CE1
    ip mtu 9000
    ip address 20.1.1.2/24
interface vlan 2031
    vrf attach vrf_1
    description Towards CE2
    ip mtu 9000
    ip address 20.3.1.2/24
interface vlan 2032
    vrf attach vrf_2
    description Towards CE2
    ip mtu 9000
    ip address 20.3.1.2/24
!
ip community-list standard site1_community description For_setting_community_From_
Site1
ip community-list standard site1_community seq 10 permit 99:99
!
!
!
route-map rmap1 permit seq 10
     description For_setting_community_For Site1_Received_EBGP_routes
     set community 99:99
route-map rmap2 deny seq 10
     description For_Denying_routes_with_Site1_community_To_EBGP_Peer
     match community-list site1_community
route-map rmap2 permit seq 20
     description For_Allowing_all_other_route
!
router ospf 1
    area 0.0.0.0
router bgp 6488163
    neighbor 4.4.4.4 remote-as 6488163
    neighbor 4.4.4.4 description Towards RR1{P1}
    neighbor 4.4.4.4 update-source loopback 0
    neighbor 5.5.5.5 remote-as 6488163
    neighbor 5.5.5.5 description Towards RR2{P2}
    neighbor 5.5.5.5 update-source loopback 0
    address-family vpnv4 unicast
        neighbor 4.4.4.4 activate
        neighbor 4.4.4.4 send-community both
        neighbor 5.5.5.5 activate
        neighbor 5.5.5.5 send-community both
    exit-address-family
!
    vrf vrf_1
```

```
        neighbor 20.1.1.1 remote-as 100
        neighbor 20.1.1.1 description Towards CE1
        neighbor 20.3.1.1 remote-as 100
        neighbor 20.3.1.1 description Towards CE2
        address-family ipv4 unicast
            neighbor 20.1.1.1 activate
            neighbor 20.1.1.1 route-map rmap1 in
            neighbor 20.1.1.1 route-map rmap2 out
            neighbor 20.1.1.1 send-community both
            neighbor 20.1.1.1 soft-reconfiguration inbound
            neighbor 20.3.1.1 activate
            neighbor 20.3.1.1 route-map rmap1 in
            neighbor 20.3.1.1 route-map rmap2 out
            neighbor 20.3.1.1 send-community both
            neighbor 20.3.1.1 soft-reconfiguration inbound
        exit-address-family
!
    vrf vrf_2
        neighbor 20.1.1.1 remote-as 100
        neighbor 20.1.1.1 description Towards CE1
        neighbor 20.3.1.1 remote-as 100
        neighbor 20.3.1.1 description Towards CE2
        address-family ipv4 unicast
            neighbor 20.1.1.1 activate
            neighbor 20.1.1.1 route-map rmap1 in
            neighbor 20.1.1.1 route-map rmap2 out
            neighbor 20.1.1.1 send-community both
            neighbor 20.1.1.1 soft-reconfiguration inbound
            neighbor 20.3.1.1 activate
            neighbor 20.3.1.1 route-map rmap1 in
            neighbor 20.3.1.1 route-map rmap2 out
            neighbor 20.3.1.1 send-community both
            neighbor 20.3.1.1 soft-reconfiguration inbound
        exit-address-family
!
https-server vrf mgmt
mpls
    enable
    label-protocol ldp
        enable
        router-id loopback0
```

PE1 verification

```
PE1# show ip ospf neighbors
VRF : default                          Process : 1
===================================================

Total Number of Neighbors : 2

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
4.4.4.4          1         FULL/BDR          30.0.0.2           1/1/39

5.5.5.5          1         FULL/BDR          30.0.3.2           1/1/40
```

```
PE1# show mpls ldp neighbor
Local LDP Identifier: 1.1.1.1:0, Peer LDP Identifier: 4.4.4.4:0
    Graceful Restart: No
```

```
    Session Holdtime: 40 sec
    Up time: 00:13:36
    State: operational
    LDP Discovery Sources: 1/1/39
    Addresses bound to this peer:
         30.0.0.2 30.0.1.2 30.0.2.1 4.4.4.4

Local LDP Identifier: 1.1.1.1:0, Peer LDP Identifier: 5.5.5.5:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:13:19
    State: operational
    LDP Discovery Sources: 1/1/40
    Addresses bound to this peer:
         30.0.3.2 30.0.4.2 30.0.5.1 5.5.5.5
```

```
PE1# show mpls ldp binding
5.5.5.5/32
        local binding: label: 17
        remote binding: lsr:5.5.5.5:0 label: exp-null
        remote binding: lsr:4.4.4.4:0 label:17
30.0.1.2/32
        No local binding
        remote binding: lsr:4.4.4.4:0 label: exp-null
30.0.3.2/32
        No local binding
        remote binding: lsr:5.5.5.5:0 label: exp-null
30.0.2.1/32
        No local binding
        remote binding: lsr:4.4.4.4:0 label: exp-null
1.1.1.1/32
        local binding: label: exp-null
        remote binding: lsr:5.5.5.5:0 label:16
        remote binding: lsr:4.4.4.4:0 label:16
4.4.4.4/32
        local binding: label: 16
        remote binding: lsr:5.5.5.5:0 label:17
        remote binding: lsr:4.4.4.4:0 label: exp-null
30.0.0.1/32
        local binding: label: exp-null
        No remote binding
3.3.3.3/32
        local binding: label: 19
        remote binding: lsr:5.5.5.5:0 label:19
        remote binding: lsr:4.4.4.4:0 label:19
30.0.0.2/32
        No local binding
        remote binding: lsr:4.4.4.4:0 label: exp-null
30.0.4.2/32
        No local binding
        remote binding: lsr:5.5.5.5:0 label: exp-null
2.2.2.2/32
        local binding: label: 18
        remote binding: lsr:5.5.5.5:0 label:18
        remote binding: lsr:4.4.4.4:0 label:18
30.0.3.1/32
        local binding: label: exp-null
        No remote binding
30.0.5.1/32
        No local binding
        remote binding: lsr:5.5.5.5:0 label: exp-null
```

```
PE1# show mpls forwarding

MPLS Bindings

Entry Bindings   : 8
Exit Bindings    : 4
Transit Bindings : 6
PHP Mode         : Explicit-Null
QoS Mode         : Uniform
TTL Propagation  : Uniform






Entry Bindings:

Origin Prefix       Ingress  Nexthop    Outgoing  Egress     Egress   Status
                    VRF      Address    Label     Interface  VRF
-------------------------------------------------------------------------------
LDP    2.2.2.2/32   default  30.0.3.2   18        1/1/40     default  operational

LDP    2.2.2.2/32   default  30.0.0.2   18        1/1/39     default  operational

LDP    3.3.3.3/32   default  30.0.3.2   19        1/1/40     default  operational

LDP    3.3.3.3/32   default  30.0.0.2   19        1/1/39     default  operational

BGP    201.0.0.0/24 vrf_1    3.3.3.3    21        1/1/40     default  operational

BGP    202.0.0.0/24 vrf_2    3.3.3.3    22        1/1/40     default  operational


Exit Bindings:

Origin   Prefix              Incoming  Service   Egress            Status
                             Label     Label     VRF
-------------------------------------------------------------------------------
static   n/a                 exp-null  -         default           operational
BGP      n/a                 imp-null  20        vrf_1             operational
BGP      n/a                 imp-null  22        vrf_2             operational
static   n/a                 7         -         default           operational


Transit Bindings:

Origin  Prefix       Incoming  Egress     Egress   Nexthop   Outgoing  Status

                     Label     Interface  VRF      Address   Label

---------------------------------------------------------------------------------
LDP     4.4.4.4/32   16        1/1/39     default  30.0.0.2  exp-null  operational
LDP     5.5.5.5/32   17        1/1/40     default  30.0.3.2  exp-null  operational
LDP     2.2.2.2/32   18        1/1/40     default  30.0.3.2  18        operational
LDP     2.2.2.2/32   18        1/1/39     default  30.0.0.2  18        operational
LDP     3.3.3.3/32   19        1/1/40     default  30.0.3.2  19        operational
LDP     3.3.3.3/32   19        1/1/39     default  30.0.0.2  19        operational
```

```
PE1# show bgp vpnv4 un neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 4.4.4.4 (Internal)
    Description          : Towards RR1{P1}
    Peer-group           :

    Remote Router Id     : 4.4.4.4            Local Router Id    : 1.1.1.1
    Remote AS            : 6488163            Local AS           : 6488163
    Remote Port          : 179                Local Port         : 42093
    State                : Established        Admin Status       : Up
    Conn. Established    : 1                  Conn. Dropped      : 0
    Passive              : No                 Update-Source      : loopback0
    Cfg. Hold Time       : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time       : 180                Neg. Keep Alive    : 60
    Up/Down Time         : 00h:15m:55s        Connect-Retry Time : 120
    Local-AS Prepend     : No                 Alt. Local-AS      : 0
    BFD                  : Disabled
    Password             :
    Last Err Sent        : No Error
    Last SubErr Sent     : No Error
    Last Err Rcvd        : No Error
    Last SubErr Rcvd     : No Error

    Graceful-Restart     : Enabled            Gr. Restart Time   : 120
    Gr. Stalepath Time   : 300                Remove Private-AS  : No
    TTL                  : 255                Local Cluster-ID   :
    Weight               : 0                  Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      3        7
    Keepalives                  18       18
    Route Refresh                0        0
    Total                       22       26

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                   Send Community    : both
    Allow-AS in          : 0                   Advt. Interval    : 30
    Max. Prefix          : 300000              Soft Reconfig In  :
    Nexthop-Self         :                     Default-Originate :
    Cfg. Add-Path        :
    Neg. Add-Path        :
```

```
    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 5.5.5.5 (Internal)
    Description         : Towards RR2{P2}
    Peer-group          :

    Remote Router Id    : 5.5.5.5          Local Router Id    : 1.1.1.1
    Remote AS           : 6488163          Local AS           : 6488163
    Remote Port         : 179              Local Port         : 40029
    State               : Established      Admin Status       : Up
    Conn. Established   : 1                Conn. Dropped      : 0
    Passive             : No               Update-Source      : loopback0
    Cfg. Hold Time      : 180              Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180              Neg. Keep Alive    : 60
    Up/Down Time        : 00h:15m:40s      Connect-Retry Time : 120
    Local-AS Prepend    : No               Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled          Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300              Remove Private-AS  : No
    TTL                 : 255              Local Cluster-ID   :
    Weight              : 0                Fall-over          : No
    Confederation-Peers : No

    Message statistics       Sent     Rcvd
    -------------------     -----    -----
    Open                        1        1
    Notification                0        0
    Updates                     3        5
    Keepalives                 17       16
    Route Refresh               0        0
    Total                      21       22

    Capability                         Advertised      Received
    ----------------------------       -----------     ----------
    Route Refresh                      Yes             Yes
    Graceful Restart                   Yes             Yes
    Add-Path                           No              No
    Four Octet ASN                     Yes             Yes
    Address family IPv4 Unicast        No              No
    Address family IPv6 Unicast        No              No
    Address family VPNv4 Unicast       Yes             Yes
    Address family L2VPN EVPN          No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                Send Community    : both
    Allow-AS in         : 0                 Advt. Interval    : 30
    Max. Prefix         : 300000            Soft Reconfig In  :
    Nexthop-Self        :                   Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :
```

```
    Routemap In          :
    Routemap Out         :
    ORF type             : Prefix-list
    ORF capability       :
```

```
PE1# show bgp vpnv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 1.1.1.1

    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 1:1                  (Label 20)
*>  101.0.0.0/24       0.0.0.0         0          100        0       100 i

Route Distinguisher: 2:1                  (Label 22)
*>  102.0.0.0/24       0.0.0.0         0          100        0       100 i

Route Distinguisher: 1:3                  (Label 21)
*>i 201.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 201.0.0.0/24       3.3.3.3         0          100        0       100 i

Route Distinguisher: 2:3                  (Label 22)
*>i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 6
```

```
PE1# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 1.1.1.1
 Peers                  : 2            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------

Address-family : IPv6 Unicast
-----------------------------

Address-family : VPNv4 Unicast
------------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 4.4.4.4         6488163     43      36      00h:27m:27s  Established   Up
 5.5.5.5         6488163     39      34      00h:27m:12s  Established   Up

Address-family : L2VPN EVPN
-----------------------------

VRF : vrf_1
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 20.3.1.2
 Peers                  : 2            Log Neighbor Changes   : No
```

```
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.1.1.1        100         36      37      00h:28m:34s  Established   Up
 20.3.1.1        100         36      36      00h:28m:34s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

VRF : vrf_2
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 20.3.1.2
 Peers                  : 2            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.1.1.1        100         40      37      00h:28m:33s  Established   Up
 20.3.1.1        100         40      38      00h:28m:42s  Established   Up

Address-family : IPv6 Unicast
-----------------------------
```

```
PE1# show bgp all-vrf all
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 1.1.1.1

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : VPNv4 Unicast
------------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 1:1                   (Label 20)
*>  101.0.0.0/24       0.0.0.0         0          100        0       100 i

Route Distinguisher: 2:1                   (Label 22)
*>  102.0.0.0/24       0.0.0.0         0          100        0       100 i

Route Distinguisher: 1:3                   (Label 21)
*>i 201.0.0.0/24       3.3.3.3         0          100        0       100 i
```

```
* i 201.0.0.0/24       3.3.3.3         0          100        0       100 i

Route Distinguisher: 2:3                  (Label 22)
*>i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 6

Address-family : L2VPN EVPN
-----------------------------
     Network                                                   Nexthop         Metric
   LocPrf    Weight   Path
-----------------------------------------------------------------------------------
--------------------------
Total number of entries 0

VRF : vrf_1
Local Router-ID 20.3.1.2

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*=e 101.0.0.0/24       20.1.1.1        0          100        0       100 i
*>e 101.0.0.0/24       20.3.1.1        0          100        0       100 i
*>  201.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 3

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

VRF : vrf_2
Local Router-ID 20.3.1.2

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*=e 102.0.0.0/24       20.1.1.1        0          100        0       100 i
*>e 102.0.0.0/24       20.3.1.1        0          100        0       100 i
*>  202.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 3

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0
```

```
PE1# show ip route all-vrf

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
```

```
                                                                   Type     Metric
-----------------------------------------------------------------------------------
------------
1.1.1.1/32          -                loopback0     -                 L
[0/0]         -
2.2.2.2/32          30.0.3.2         1/1/40        -                 O
[110/200]     01h:12m:16s
                    30.0.0.2         1/1/39        -
[110/200]     01h:12m:16s
3.3.3.3/32          30.0.3.2         1/1/40        -                 O
[110/150]     01h:11m:52s
                    30.0.0.2         1/1/39        -
[110/150]     01h:11m:52s
4.4.4.4/32          30.0.0.2         1/1/39        -                 O
[110/100]     01h:13m:07s
5.5.5.5/32          30.0.3.2         1/1/40        -                 O
[110/100]     01h:12m:46s
30.0.0.0/24         -                1/1/39        -                 C
[0/0]         -
30.0.0.1/32         -                1/1/39        -                 L
[0/0]         -
30.0.1.0/24         30.0.0.2         1/1/39        -                 O
[110/200]     01h:13m:07s
30.0.2.0/24         30.0.0.2         1/1/39        -                 O
[110/150]     01h:12m:42s
30.0.3.0/24         -                1/1/40        -                 C
[0/0]         -
30.0.3.1/32         -                1/1/40        -                 L
[0/0]         -
30.0.4.0/24         30.0.3.2         1/1/40        -                 O
[110/200]     01h:12m:46s
30.0.5.0/24         30.0.3.2         1/1/40        -                 O
[110/150]     01h:12m:41s

VRF: vrf_1

Prefix               Nexthop            Interface     VRF(egress)   Origin/
Distance/     Age
                                                                   Type     Metric
-----------------------------------------------------------------------------------
------------
20.1.1.0/24         -                vlan2011      -                 C
[0/0]         -
20.1.1.2/32         -                vlan2011      -                 L
[0/0]         -
20.3.1.0/24         -                vlan2031      -                 C
[0/0]         -
20.3.1.2/32         -                vlan2031      -                 L
[0/0]         -
101.0.0.0/24        20.3.1.1         vlan2031      -                 B/E
[20/0]        01h:11m:55s
                    20.1.1.1         vlan2011      -
[20/0]        01h:11m:55s
201.0.0.0/24        3.3.3.3          1/1/40        default           B/V
[200/0]       01h:10m:34s

VRF: vrf_2

Prefix               Nexthop            Interface     VRF(egress)   Origin/
Distance/     Age
                                                                   Type     Metric
-----------------------------------------------------------------------------------
```

```
------------
20.1.1.0/24         -               vlan2012      -                 C
[0/0]         -
20.1.1.2/32         -               vlan2012      -                 L
[0/0]         -
20.3.1.0/24         -               vlan2032      -                 C
[0/0]         -
20.3.1.2/32         -               vlan2032      -                 L
[0/0]         -
102.0.0.0/24        20.3.1.1        vlan2032      -                 B/E
[20/0]        00h:54m:06s
                    20.1.1.1        vlan2012      -
[20/0]        00h:54m:06s
202.0.0.0/24        3.3.3.3         1/1/40        default           B/V
[200/0]       00h:54m:06s

Total Route Count : 25
```

PE2 configuration

```
PE2# show running
Current configuration:
!
!Version ArubaOS-CX LL.10.09.0001BF
!export-password: default
hostname PE2
profile core-spine
logging console severity crit
vrf vrf_1
    rd 1:1
    l3vpn-only
    address-family ipv4 unicast
        route-target export 1:1
        route-target import 1:1
    exit-address-family
vrf vrf_2
    rd 2:1
    l3vpn-only
    address-family ipv4 unicast
        route-target export 2:1
        route-target import 2:1
    exit-address-family
cli-session
    timeout 0
!
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2021-2022,2041-2042
interface mgmt
    no shutdown
    ip dhcp
interface 1/1/43
    no shutdown
    mtu 9000
    description Towards CE1
    no routing
```

```
    vlan trunk native 1
    vlan trunk allowed 2021-2022
interface 1/1/44
    no shutdown
    mtu 9000
    description Towards CE2
    no routing
    vlan trunk native 1
    vlan trunk allowed 2041-2042
interface 1/1/45
    no shutdown
    mtu 9198
    description Towards P1
    ip mtu 9198
    ip address 30.0.1.1/24
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface 1/1/46
    no shutdown
    mtu 9198
    description Towards P2
    ip mtu 9198
    ip address 30.0.4.1/24
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface loopback 0
    ip address 2.2.2.2/32
    ip ospf 1 area 0.0.0.0
interface vlan 2021
    vrf attach vrf_1
    description Towards CE1
    ip mtu 9000
    ip address 20.2.1.2/24
interface vlan 2022
    vrf attach vrf_2
    description Towards CE1
    ip mtu 9000
    ip address 20.2.1.2/24
interface vlan 2041
    vrf attach vrf_1
    description Towards CE2
    ip mtu 9000
    ip address 20.4.1.2/24
interface vlan 2042
    vrf attach vrf_2
    description Towards CE2
    ip mtu 9000
    ip address 20.4.1.2/24
!
ip community-list standard site1_community description For_setting_community_From_
Site1
ip community-list standard site1_community seq 10 permit 99:99
!
!
!
route-map rmap1 permit seq 10
     description For_setting_community_For Site1_Received_EBGP_routes
     set community 99:99
route-map rmap2 deny seq 10
     description For_Denying_routes_with_Site1_community_To_EBGP_Peer
```

```
     match community-list site1_community
route-map rmap2 permit seq 20
     description For_Allowing_all_other_route
!
router ospf 1
    area 0.0.0.0
router bgp 6488163
    neighbor 4.4.4.4 remote-as 6488163
    neighbor 4.4.4.4 description Towards RR1{P1}
    neighbor 4.4.4.4 update-source loopback 0
    neighbor 5.5.5.5 remote-as 6488163
    neighbor 5.5.5.5 description Towards RR2{P2}
    neighbor 5.5.5.5 update-source loopback 0
    address-family vpnv4 unicast
        neighbor 4.4.4.4 activate
        neighbor 4.4.4.4 send-community both
        neighbor 5.5.5.5 activate
        neighbor 5.5.5.5 send-community both
    exit-address-family
!
    vrf vrf_1
        neighbor 20.2.1.1 remote-as 100
        neighbor 20.2.1.1 description Towards CE1
        neighbor 20.4.1.1 remote-as 100
        neighbor 20.4.1.1 description Towards CE2
        address-family ipv4 unicast
            neighbor 20.2.1.1 activate
            neighbor 20.2.1.1 route-map rmap1 in
            neighbor 20.2.1.1 route-map rmap2 out
            neighbor 20.2.1.1 send-community both
            neighbor 20.2.1.1 soft-reconfiguration inbound
            neighbor 20.4.1.1 activate
            neighbor 20.4.1.1 route-map rmap1 in
            neighbor 20.4.1.1 route-map rmap2 out
            neighbor 20.4.1.1 send-community both
            neighbor 20.4.1.1 soft-reconfiguration inbound
        exit-address-family
!
    vrf vrf_2
        neighbor 20.2.1.1 remote-as 100
        neighbor 20.2.1.1 description Towards CE1
        neighbor 20.4.1.1 remote-as 100
        neighbor 20.4.1.1 description Towards CE2
        address-family ipv4 unicast
            neighbor 20.2.1.1 activate
            neighbor 20.2.1.1 route-map rmap1 in
            neighbor 20.2.1.1 route-map rmap2 out
            neighbor 20.2.1.1 send-community both
            neighbor 20.2.1.1 soft-reconfiguration inbound
            neighbor 20.4.1.1 activate
            neighbor 20.4.1.1 route-map rmap1 in
            neighbor 20.4.1.1 route-map rmap2 out
            neighbor 20.4.1.1 send-community both
            neighbor 20.4.1.1 soft-reconfiguration inbound
        exit-address-family
!
https-server vrf mgmt
mpls
    enable
    label-protocol ldp
        enable
        router-id loopback0
```

```
PE2# show ip ospf neighbors
VRF : default                          Process : 1
===================================================

Total Number of Neighbors : 2

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
4.4.4.4          1         FULL/DR           30.0.1.2           1/1/45

5.5.5.5          1         FULL/BDR          30.0.4.2           1/1/46
```

```
PE2#  show mpls ldp neighbor
Local LDP Identifier: 2.2.2.2:0, Peer LDP Identifier: 4.4.4.4:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:13:36
    State: operational
    LDP Discovery Sources: 1/1/45
    Addresses bound to this peer:
         30.0.0.2 30.0.1.2 30.0.2.1 4.4.4.4

Local LDP Identifier: 2.2.2.2:0, Peer LDP Identifier: 5.5.5.5:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:13:14
    State: operational
    LDP Discovery Sources: 1/1/46
    Addresses bound to this peer:
         30.0.3.2 30.0.4.2 30.0.5.1 5.5.5.5
```

```
PE2# show mpls ldp binding
30.0.3.2/32
        No local binding
        remote binding: lsr:5.5.5.5:0 label: exp-null
4.4.4.4/32
        local binding: label: 17
        remote binding: lsr:5.5.5.5:0 label:17
        remote binding: lsr:4.4.4.4:0 label: exp-null
30.0.4.2/32
        No local binding
        remote binding: lsr:5.5.5.5:0 label: exp-null
3.3.3.3/32
        local binding: label: 19
        remote binding: lsr:5.5.5.5:0 label:19
        remote binding: lsr:4.4.4.4:0 label:19
5.5.5.5/32
        local binding: label: 18
        remote binding: lsr:5.5.5.5:0 label: exp-null
        remote binding: lsr:4.4.4.4:0 label:17
30.0.1.1/32
        local binding: label: exp-null
        No remote binding
30.0.1.2/32
        No local binding
        remote binding: lsr:4.4.4.4:0 label: exp-null
30.0.4.1/32
        local binding: label: exp-null
        No remote binding
1.1.1.1/32
```

```
        local binding: label: 16
        remote binding: lsr:5.5.5.5:0 label:16
        remote binding: lsr:4.4.4.4:0 label:16
30.0.0.2/32
        No local binding
        remote binding: lsr:4.4.4.4:0 label: exp-null
2.2.2.2/32
        local binding: label: exp-null
        remote binding: lsr:5.5.5.5:0 label:18
        remote binding: lsr:4.4.4.4:0 label:18
30.0.5.1/32
        No local binding
        remote binding: lsr:5.5.5.5:0 label: exp-null
30.0.2.1/32
        No local binding
        remote binding: lsr:4.4.4.4:0 label: exp-null
```

```
PE2# show mpls forwarding

MPLS Bindings

Entry Bindings    : 8
Exit Bindings     : 4
Transit Bindings : 6
PHP Mode          : Explicit-Null
QoS Mode          : Uniform
TTL Propagation   : Uniform

Entry Bindings:

Origin Prefix        Ingress  Nexthop    Outgoing  Egress     Egress   Status
                     VRF      Address    Label     Interface  VRF
--------------------------------------------------------------------------------
LDP    1.1.1.1/32   default  30.0.1.2   16        1/1/45     default  operational

LDP    1.1.1.1/32   default  30.0.4.2   16        1/1/46     default  operational

LDP    3.3.3.3/32   default  30.0.4.2   19        1/1/46     default  operational
LDP    3.3.3.3/32   default  30.0.1.2   19        1/1/45     default  operational

BGP    201.0.0.0/24 vrf_1    3.3.3.3    21        1/1/46     default  operational

BGP    202.0.0.0/24 vrf_2    3.3.3.3    22        1/1/46     default  operational


Exit Bindings:

Origin   Prefix             Incoming  Service   Egress            Status
                            Label     Label     VRF
--------------------------------------------------------------------------------
static   n/a                exp-null  -         default           operational
BGP      n/a                imp-null  20        vrf_1             operational
BGP      n/a                imp-null  22        vrf_2             operational
static   n/a                7         -         default           operational


Transit Bindings:

Origin  Prefix        Incoming  Egress     Egress   Nexthop   Outgoing  Status

                      Label     Interface  VRF      Address   Label
```

```
-------------------------------------------------------------------------------------
LDP     4.4.4.4/32   16        1/1/39     default  30.0.0.2  exp-null  operational
LDP     1.1.1.1/32   16        1/1/45     default  30.0.1.2  16        operational
LDP     1.1.1.1/32   16        1/1/46     default  30.0.4.2  16        operational
LDP     4.4.4.4/32   17        1/1/45     default  30.0.1.2  exp-null  operational
LDP     5.5.5.5/32   18        1/1/46     default  30.0.4.2  exp-null  operational
LDP     3.3.3.3/32   19        1/1/45     default  30.0.1.2  19        operational
LDP     3.3.3.3/32   19        1/1/46     default  30.0.4.2  19        operational
```

```
PE2# show bgp vpnv4 un neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 4.4.4.4 (Internal)
    Description          : Towards RR1{P1}
    Peer-group           :

    Remote Router Id     : 4.4.4.4           Local Router Id    : 2.2.2.2
    Remote AS            : 6488163           Local AS           : 6488163
    Remote Port          : 179               Local Port         : 44703
    State                : Established       Admin Status       : Up
    Conn. Established    : 1                 Conn. Dropped      : 0
    Passive              : No                Update-Source      : loopback0
    Cfg. Hold Time       : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time       : 180               Neg. Keep Alive    : 60
    Up/Down Time         : 00h:15m:21s       Connect-Retry Time : 120
    Local-AS Prepend     : No                Alt. Local-AS      : 0
    BFD                  : Disabled
    Password             :
    Last Err Sent        : No Error
    Last SubErr Sent     : No Error
    Last Err Rcvd        : No Error
    Last SubErr Rcvd     : No Error

    Graceful-Restart     : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time   : 300               Remove Private-AS  : No
    TTL                  : 255               Local Cluster-ID   :
    Weight               : 0                 Fall-over          : No
    Confederation-Peers  : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      3        5
    Keepalives                  18       17
    Route Refresh                0        0
    Total                       22       23

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No
```

```
    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  :
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 5.5.5.5 (Internal)
    Description         : Towards RR2{P2}
    Peer-group          :

    Remote Router Id    : 5.5.5.5           Local Router Id    : 2.2.2.2
    Remote AS           : 6488163           Local AS           : 6488163
    Remote Port         : 33107             Local Port         : 179
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      : loopback0
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:15m:30s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 255               Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics       Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      3        5
    Keepalives                  17       17
    Route Refresh                0        0
    Total                       21       23

    Capability                       Advertised     Received
    ----------------------------     -----------    ----------
    Route Refresh                    Yes            Yes
    Graceful Restart                 Yes            Yes
    Add-Path                         No             No
    Four Octet ASN                   Yes            Yes
    Address family IPv4 Unicast      No             No
    Address family IPv6 Unicast      No             No
    Address family VPNv4 Unicast     Yes            Yes
    Address family L2VPN EVPN        No             No
```

```
    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                     Send Community    : both
    Allow-AS in         : 0                      Advt. Interval    : 30
    Max. Prefix         : 300000                 Soft Reconfig In  :
    Nexthop-Self        :                        Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :
```

```
PE2# show bgp vpnv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 2.2.2.2

    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 1:1                  (Label 20)
*>  101.0.0.0/24       0.0.0.0         0          100        0       100 i
* i 101.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 101.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 2:1                  (Label 22)
*>  102.0.0.0/24       0.0.0.0         0          100        0       100 i
* i 102.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 102.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 1:3                  (Label 21)
*>i 201.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 201.0.0.0/24       3.3.3.3         0          100        0       100 i

Route Distinguisher: 2:3                  (Label 22)
*>i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 10
```

```
PE2#  show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 2.2.2.2
 Peers                  : 2            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------

Address-family : IPv6 Unicast
-----------------------------
```

```
Address-family : VPNv4 Unicast
------------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 4.4.4.4         6488163     40      37      00h:26m:53s  Established   Up
 5.5.5.5         6488163     41      35      00h:27m:02s  Established   Up

Address-family : L2VPN EVPN
-----------------------------

VRF : vrf_1
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 20.4.1.2
 Peers                  : 2            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.2.1.1        100         35      35      00h:28m:01s  Established   Up
 20.4.1.1        100         35      35      00h:27m:58s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

VRF : vrf_2
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 20.4.1.2
 Peers                  : 2            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.2.1.1        100         38      38      00h:27m:57s  Established   Up
 20.4.1.1        100         38      37      00h:27m:56s  Established   Up

Address-family : IPv6 Unicast
-----------------------------
```

```
PE2# show bgp all-vrf all
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 2.2.2.2

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : IPv6 Unicast
-----------------------------
```

```
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : VPNv4 Unicast
------------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 1:1                   (Label 20)
*>  101.0.0.0/24       0.0.0.0         0          100        0       100 i
* i 101.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 101.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 2:1                   (Label 22)
*>  102.0.0.0/24       0.0.0.0         0          100        0       100 i
* i 102.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 102.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 1:3                   (Label 21)
*>i 201.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 201.0.0.0/24       3.3.3.3         0          100        0       100 i

Route Distinguisher: 2:3                   (Label 22)
*>i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
* i 202.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 10

Address-family : L2VPN EVPN
-----------------------------
     Network                                               Nexthop         Metric
   LocPrf    Weight   Path
----------------------------------------------------------------------------------
--------------------------
Total number of entries 0

VRF : vrf_1
Local Router-ID 20.4.1.2

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*=e 101.0.0.0/24       20.2.1.1        0          100        0       100 i
*>e 101.0.0.0/24       20.4.1.1        0          100        0       100 i
*>  201.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 3

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

VRF : vrf_2
Local Router-ID 20.4.1.2

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*=e 102.0.0.0/24       20.2.1.1        0          100        0       100 i
*>e 102.0.0.0/24       20.4.1.1        0          100        0       100 i
*>  202.0.0.0/24       3.3.3.3         0          100        0       100 i
Total number of entries 3

Address-family : IPv6 Unicast
```

```
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0
```

```
PE2# show ip route all-vrf

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
----------------------------------------------------------------------------------
------------
1.1.1.1/32          30.0.1.2         1/1/45        -                 O
[110/200]    01h:12m:16s
                    30.0.4.2         1/1/46        -
[110/200]    01h:12m:16s
2.2.2.2/32          -                loopback0     -                 L
[0/0]        -
3.3.3.3/32          30.0.1.2         1/1/45        -                 O
[110/150]    01h:11m:52s
                    30.0.4.2         1/1/46        -
[110/150]    01h:11m:52s
4.4.4.4/32          30.0.1.2         1/1/45        -                 O
[110/100]    01h:12m:33s
5.5.5.5/32          30.0.4.2         1/1/46        -                 O
[110/100]    01h:12m:16s
30.0.0.0/24         30.0.1.2         1/1/45        -                 O
[110/200]    01h:12m:33s
30.0.1.0/24         -                1/1/45        -                 C
[0/0]        -
30.0.1.1/32         -                1/1/45        -                 L
[0/0]        -
30.0.2.0/24         30.0.1.2         1/1/45        -                 O
[110/150]    01h:12m:33s
30.0.3.0/24         30.0.4.2         1/1/46        -                 O
[110/200]    01h:12m:16s
30.0.4.0/24         -                1/1/46        -                 C
[0/0]        -
30.0.4.1/32         -                1/1/46        -                 L
[0/0]        -
30.0.5.0/24         30.0.4.2         1/1/46        -                 O
[110/150]    01h:12m:16s

VRF: vrf_1

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
----------------------------------------------------------------------------------
------------
20.2.1.0/24         -                vlan2021      -                 C
[0/0]        -
```

```
20.2.1.2/32         -                vlan2021      -                 L
[0/0]        -
20.4.1.0/24         -                vlan2041      -                 C
[0/0]        -
20.4.1.2/32         -                vlan2041      -                 L
[0/0]        -
101.0.0.0/24        20.4.1.1         vlan2041      -                 B/E
[20/0]       01h:11m:55s
                    20.2.1.1         vlan2021      -
[20/0]       01h:11m:55s
201.0.0.0/24        3.3.3.3          1/1/46        default           B/V
[200/0]      01h:10m:34s

VRF: vrf_2

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
----------------------------------------------------------------------------------
------------
20.2.1.0/24         -                vlan2022      -                 C
[0/0]        -
20.2.1.2/32         -                vlan2022      -                 L
[0/0]        -
20.4.1.0/24         -                vlan2042      -                 C
[0/0]        -
20.4.1.2/32         -                vlan2042      -                 L
[0/0]        -
102.0.0.0/24        20.4.1.1         vlan2042      -                 B/E
[20/0]       00h:54m:06s
                    20.2.1.1         vlan2022      -
[20/0]       00h:54m:06s
202.0.0.0/24        3.3.3.3          1/1/46        default           B/V
[200/0]      00h:54m:06s

Total Route Count : 25
```

PE3 configuration

```
PE3# show running
Current configuration:
!
!Version ArubaOS-CX LL.10.09.0001BF
!export-password: default
hostname PE3
profile core-spine
logging console severity crit
vrf vrf_1
    rd 1:3
    l3vpn-only
    address-family ipv4 unicast
        route-target export 1:1
        route-target import 1:1
    exit-address-family
vrf vrf_2
    rd 2:3
    l3vpn-only
    address-family ipv4 unicast
        route-target export 2:1
        route-target import 2:1
```

```
    exit-address-family
cli-session
    timeout 0
!
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2051-2052
interface mgmt
    no shutdown
    ip dhcp
system interface-group 1 speed 10g
    !interface group 1 contains ports 1/1/1-1/1/4
interface lag 99
    no shutdown
    description Towards P1
    ip mtu 9198
    ip address 30.0.2.2/24
    lacp mode active
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface lag 100
    no shutdown
    description Towards P2
    ip mtu 9198
    ip address 30.0.5.2/24
    lacp mode active
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface 1/1/19
    no shutdown
    mtu 9198
    description Towards P1
    lag 99
interface 1/1/20
    no shutdown
    mtu 9198
    description Towards P1
    lag 99
interface 1/1/21
    no shutdown
    mtu 9000
    description Towards CE3
    no routing
    vlan trunk native 1
    vlan trunk allowed 2051-2052
interface 1/1/22
    no shutdown
    mtu 9198
    description Towards P2
    lag 100
interface 1/1/23
    no shutdown
    mtu 9198
    description Towards P2
    lag 100
interface loopback 0
```

```
    ip address 3.3.3.3/32
    ip ospf 1 area 0.0.0.0
interface vlan 2051
    vrf attach vrf_1
    description Towards CE3
    ip mtu 9000
    ip address 20.5.1.1/24
interface vlan 2052
    vrf attach vrf_2
    description Towards CE3
    ip mtu 9000
    ip address 20.5.1.1/24
!
!
!
!
!
router ospf 1
    area 0.0.0.0
router bgp 6488163
    neighbor 4.4.4.4 remote-as 6488163
    neighbor 4.4.4.4 description Towards RR1{P1}
    neighbor 4.4.4.4 update-source loopback 0
    neighbor 5.5.5.5 remote-as 6488163
    neighbor 5.5.5.5 description Towards RR2{P2}
    neighbor 5.5.5.5 update-source loopback 0
    address-family vpnv4 unicast
        neighbor 4.4.4.4 activate
        neighbor 4.4.4.4 send-community both
        neighbor 5.5.5.5 activate
        neighbor 5.5.5.5 send-community both
    exit-address-family
!
    vrf vrf_1
        neighbor 20.5.1.2 remote-as 100
        neighbor 20.5.1.2 description Towards CE3
        address-family ipv4 unicast
            neighbor 20.5.1.2 activate
            neighbor 20.5.1.2 send-community both
            neighbor 20.5.1.2 soft-reconfiguration inbound
        exit-address-family
!
    vrf vrf_2
        neighbor 20.5.1.2 remote-as 100
        neighbor 20.5.1.2 description Towards CE3
        address-family ipv4 unicast
            neighbor 20.5.1.2 activate
            neighbor 20.5.1.2 send-community both
            neighbor 20.5.1.2 soft-reconfiguration inbound
        exit-address-family
!
https-server vrf mgmt
mpls
    enable
    label-protocol ldp
        enable
        router-id loopback0
```

PE3 verification

```
PE3# show ip ospf neighbors
VRF : default                          Process : 1
===================================================

Total Number of Neighbors : 2

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
5.5.5.5          1         FULL/DR           30.0.5.1           lag100

4.4.4.4          1         FULL/DR           30.0.2.1           lag99
```

```
PE3#  show mpls ldp neighbor
Local LDP Identifier: 3.3.3.3:0, Peer LDP Identifier: 4.4.4.4:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:12:42
    State: operational
    LDP Discovery Sources: lag99
    Addresses bound to this peer:
         30.0.0.2 30.0.1.2 30.0.2.1 4.4.4.4

Local LDP Identifier: 3.3.3.3:0, Peer LDP Identifier: 5.5.5.5:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:12:42
    State: operational
    LDP Discovery Sources: lag100
    Addresses bound to this peer:
         30.0.3.2 30.0.4.2 30.0.5.1 5.5.5.5
```

```
PE3#  show mpls ldp binding
30.0.5.2/32
        local binding: label: exp-null
        No remote binding
2.2.2.2/32
        local binding: label: 17
        remote binding: lsr:5.5.5.5:0 label:18
        remote binding: lsr:4.4.4.4:0 label:18
30.0.1.2/32
        No local binding
        remote binding: lsr:4.4.4.4:0 label: exp-null
4.4.4.4/32
        local binding: label: 18
        remote binding: lsr:5.5.5.5:0 label:17
        remote binding: lsr:4.4.4.4:0 label: exp-null
30.0.4.2/32
        No local binding
        remote binding: lsr:5.5.5.5:0 label: exp-null
30.0.5.1/32
        No local binding
        remote binding: lsr:5.5.5.5:0 label: exp-null
30.0.3.2/32
        No local binding
        remote binding: lsr:5.5.5.5:0 label: exp-null
30.0.2.2/32
        local binding: label: exp-null
        No remote binding
30.0.0.2/32
        No local binding
```

```
        remote binding: lsr:4.4.4.4:0 label: exp-null
1.1.1.1/32
        local binding: label: 16
        remote binding: lsr:5.5.5.5:0 label:16
        remote binding: lsr:4.4.4.4:0 label:16
5.5.5.5/32
        local binding: label: 19
        remote binding: lsr:5.5.5.5:0 label: exp-null
        remote binding: lsr:4.4.4.4:0 label:17
3.3.3.3/32
        local binding: label: exp-null
        remote binding: lsr:5.5.5.5:0 label:19
        remote binding: lsr:4.4.4.4:0 label:19
30.0.2.1/32
        No local binding
        remote binding: lsr:4.4.4.4:0 label: exp-null
```

```
PE3# show mpls forwarding

MPLS Bindings

Entry Bindings   : 8
Exit Bindings    : 4
Transit Bindings : 6
PHP Mode         : Explicit-Null
QoS Mode         : Uniform
TTL Propagation  : Uniform

Entry Bindings:

Origin Prefix       Ingress  Nexthop    Outgoing  Egress     Egress   Status
                    VRF      Address    Label     Interface  VRF
--------------------------------------------------------------------------------
LDP    1.1.1.1/32   default  30.0.5.1   16        lag100     default  operational

LDP    1.1.1.1/32   default  30.0.2.1   16        lag99      default  operational
LDP    2.2.2.2/32   default  30.0.2.1   18        lag99      default  operational

LDP    2.2.2.2/32   default  30.0.5.1   18        lag100     default  operational
BGP    101.0.0.0/24 vrf_1    1.1.1.1    20        lag100     default  operational

BGP    102.0.0.0/24 vrf_2    1.1.1.1    22        lag100     default  operational

Exit Bindings:

Origin   Prefix             Incoming  Service   Egress            Status
                            Label     Label     VRF
--------------------------------------------------------------------------------
static   n/a                exp-null  -         default           operational
BGP      n/a                imp-null  21        vrf_1             operational
BGP      n/a                imp-null  22        vrf_2             operational
static   n/a                7         -         default           operational


Transit Bindings:

Origin  Prefix       Incoming  Egress     Egress   Nexthop   Outgoing  Status

                     Label     Interface  VRF      Address   Label

--------------------------------------------------------------------------------
```

```
LDP     1.1.1.1/32   16         lag100     default  30.0.5.1  16         operational
LDP     1.1.1.1/32   16         lag99      default  30.0.2.1  16         operational
LDP     2.2.2.2/32   17         lag100     default  30.0.5.1  18         operational
LDP     2.2.2.2/32   17         lag99      default  30.0.2.1  18         operational
LDP     4.4.4.4/32   18         lag99      default  30.0.2.1  exp-null   operational
LDP     5.5.5.5/32   19         lag100     default  30.0.5.1  exp-null   operational
```

```
PE3# show bgp vpnv4 un neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 4.4.4.4 (Internal)
    Description          : Towards RR1{P1}
    Peer-group           :

    Remote Router Id     : 4.4.4.4            Local Router Id     : 3.3.3.3
    Remote AS            : 6488163            Local AS            : 6488163
    Remote Port          : 33231              Local Port          : 179
    State                : Established        Admin Status        : Up
    Conn. Established    : 1                  Conn. Dropped       : 0
    Passive              : No                 Update-Source       : loopback0
    Cfg. Hold Time       : 180                Cfg. Keep Alive     : 60
    Neg. Hold Time       : 180                Neg. Keep Alive     : 60
    Up/Down Time         : 00h:14m:47s        Connect-Retry Time : 120
    Local-AS Prepend     : No                 Alt. Local-AS       : 0
    BFD                  : Disabled
    Password             :
    Last Err Sent        : No Error
    Last SubErr Sent     : No Error
    Last Err Rcvd        : No Error
    Last SubErr Rcvd     : No Error

    Graceful-Restart     : Enabled            Gr. Restart Time    : 120
    Gr. Stalepath Time   : 300                Remove Private-AS   : No
    TTL                  : 255                Local Cluster-ID    :
    Weight               : 0                  Fall-over           : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      3        3
    Keepalives                  17       18
    Route Refresh                0        0
    Total                       21       22

    Capability                          Advertised      Received
    ----------------------------        -----------     ----------
    Route Refresh                       Yes             Yes
    Graceful Restart                    Yes             Yes
    Add-Path                            No              No
    Four Octet ASN                      Yes             Yes
    Address family IPv4 Unicast         No              No
    Address family IPv6 Unicast         No              No
    Address family VPNv4 Unicast        Yes             Yes
    Address family L2VPN EVPN           No              No

    Address Family : VPNv4 Unicast
    ------------------------------
```

```
    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  :
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 5.5.5.5 (Internal)
    Description         : Towards RR2{P2}
    Peer-group          :

    Remote Router Id    : 5.5.5.5           Local Router Id    : 3.3.3.3
    Remote AS           : 6488163           Local AS           : 6488163
    Remote Port         : 39781             Local Port         : 179
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      : loopback0
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:14m:45s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 255               Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      3        3
    Keepalives                  17       17
    Route Refresh                0        0
    Total                       21       21

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------
```

```
    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  :
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :
```

```
PE3# show bgp vpnv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 3.3.3.3

    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 1:1                  (Label 20)
*>i 101.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 101.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 2:1                  (Label 22)
*>i 102.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 102.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 1:3                  (Label 21)
*>  201.0.0.0/24       0.0.0.0         0          100        0       100 i

Route Distinguisher: 2:3                  (Label 22)
*>  202.0.0.0/24       0.0.0.0         0          100        0       100 i
Total number of entries 6
```

```
PE3# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 3.3.3.3
 Peers                  : 2            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------

Address-family : IPv6 Unicast
-----------------------------

Address-family : VPNv4 Unicast
------------------------------
 Neighbor         Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 4.4.4.4          6488163     40      35      00h:26m:19s  Established   Up
 5.5.5.5          6488163     39      36      00h:26m:17s  Established   Up
```

```
Address-family : L2VPN EVPN
-----------------------------

VRF : vrf_1
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 20.5.1.1
 Peers                  : 1            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.5.1.2        100         33      34      00h:25m:12s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

VRF : vrf_2
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 20.5.1.1
 Peers                  : 1            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.5.1.2        100         34      34      00h:25m:11s  Established   Up

Address-family : IPv6 Unicast
-----------------------------
```

```
PE3#  show bgp all-vrf all
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 3.3.3.3

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

Address-family : VPNv4 Unicast
------------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 1:1                  (Label 20)
*>i 101.0.0.0/24       1.1.1.1         0          100        0       100 i
```

```
* i 101.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 2:1                   (Label 22)
*>i 102.0.0.0/24       1.1.1.1         0          100        0       100 i
* i 102.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 1:3                   (Label 21)
*>  201.0.0.0/24       0.0.0.0         0          100        0       100 i

Route Distinguisher: 2:3                   (Label 22)
*>  202.0.0.0/24       0.0.0.0         0          100        0       100 i
Total number of entries 6

Address-family : L2VPN EVPN
-----------------------------
     Network                                              Nexthop         Metric
   LocPrf    Weight   Path
-----------------------------------------------------------------------------------
--------------------------
Total number of entries 0

VRF : vrf_1
Local Router-ID 20.5.1.1

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*>  101.0.0.0/24       1.1.1.1         0          100        0       100 i
*>e 201.0.0.0/24       20.5.1.2        0          100        0       100 i
Total number of entries 2

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0

VRF : vrf_2
Local Router-ID 20.5.1.1

Address-family : IPv4 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
*>  102.0.0.0/24       1.1.1.1         0          100        0       100 i
*>e 202.0.0.0/24       20.5.1.2        0          100        0       100 i
Total number of entries 2

Address-family : IPv6 Unicast
-----------------------------
    Network            Nexthop         Metric     LocPrf     Weight Path
Total number of entries 0
```

```
PE3# show ip route all-vrf

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2
```

```
VRF: default

Prefix              Nexthop            Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
---------------------------------------------------------------------------------
------------
1.1.1.1/32         30.0.5.1         lag100        -                O
[110/150]    01h:11m:56s
                   30.0.2.1         lag99         -
[110/150]    01h:11m:56s
2.2.2.2/32         30.0.5.1         lag100        -                O
[110/150]    01h:11m:56s
                   30.0.2.1         lag99         -
[110/150]    01h:11m:56s
3.3.3.3/32         -                loopback0     -                L
[0/0]        -
4.4.4.4/32         30.0.2.1         lag99         -                O
[110/50]     01h:11m:57s
5.5.5.5/32         30.0.5.1         lag100        -                O
[110/50]     01h:11m:56s
30.0.0.0/24        30.0.2.1         lag99         -                O
[110/150]    01h:11m:57s
30.0.1.0/24        30.0.2.1         lag99         -                O
[110/150]    01h:11m:57s
30.0.2.0/24        -                lag99         -                C
[0/0]        -
30.0.2.2/32        -                lag99         -                L
[0/0]        -
30.0.3.0/24        30.0.5.1         lag100        -                O
[110/150]    01h:11m:56s
30.0.4.0/24        30.0.5.1         lag100        -                O
[110/150]    01h:11m:56s
30.0.5.0/24        -                lag100        -                C
[0/0]        -
30.0.5.2/32        -                lag100        -                L
[0/0]        -

VRF: vrf_1

Prefix              Nexthop            Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
---------------------------------------------------------------------------------
------------
20.5.1.0/24        -                vlan2051      -                C
[0/0]        -
20.5.1.1/32        -                vlan2051      -                L
[0/0]        -
101.0.0.0/24       1.1.1.1          lag100        default          B/V
[200/0]      01h:11m:49s
201.0.0.0/24       20.5.1.2         vlan2051      -                B/E
[20/0]       01h:10m:34s

VRF: vrf_2

Prefix              Nexthop            Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
---------------------------------------------------------------------------------
------------
20.5.1.0/24        -                vlan2052      -                C
```

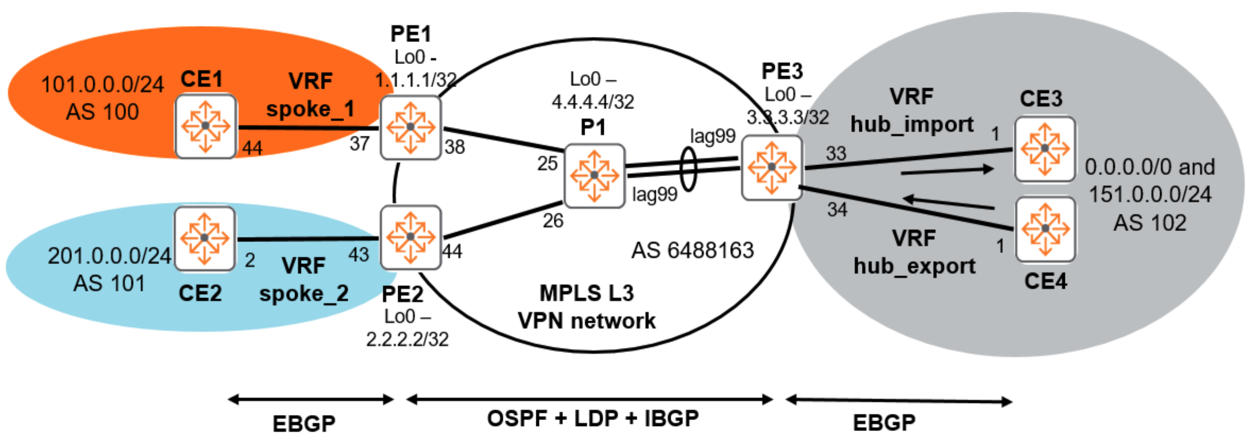
```
[0/0]          -
20.5.1.1/32          -               vlan2052      -                L
[0/0]          -
102.0.0.0/24         1.1.1.1         lag100        default          B/V
[200/0]        00h:54m:06s
202.0.0.0/24         20.5.1.2        vlan2052      -                B/E
[20/0]         00h:54m:06s

Total Route Count : 21
```

Use case 2: MPLS L3 VPN with hub and spoke VRFs

This use case covers:

- An MPLS L3 VPN network with a shared hub VRF that has connectivity to spoke VRFs, while spoke VRFs that are isolated between each other
- CEs in hub VRF are expected to provide hub connectivity between CE spoke subnets (101.0.0.0/24 and 201.0.0.0/24) and hub subnets (151.0.0.0/24 and 0.0.0.0/0 default route)
- PE3 should not allow direct spoke to spoke connectivity, this is achieved through:
  - Using hub_import VRF to advertise spoke CE subnets from PE1/PE2 to CE3
  - Using hub_export VRF to advertise hub and spoke CE subnets from CE4 to PE1/PE2
- Sample configurations
- Relevant verification commands

MPLS L3 VPN with hub and spoke VRFs:



CE1 Configuration

```
CE1# show run
Current configuration:
!
!Version ArubaOS-CX TL.10.09.0001AX
!export-password: default
hostname CE1
profile l3-core
logging console severity crit
cli-session
    timeout 0
```

```
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2001,2071
interface mgmt
    no shutdown
    ip dhcp
interface 1/1/43
    no shutdown
    description Towards 101 subnet
    no routing
    vlan trunk native 1
    vlan trunk allowed 2071
interface 1/1/44
    no shutdown
    description Towards PE1
    no routing
    vlan trunk native 1
    vlan trunk allowed 2001
interface vlan 2001
    description Towards PE1
    ip address 20.0.1.1/24
interface vlan 2071
    description Towards 101 subnet
    ip address 20.7.1.2/24
    ip ospf 1 area 0.0.0.0
!
!
!
!
!
router ospf 1
    area 0.0.0.0
router bgp 100
    neighbor 20.0.1.2 remote-as 6488163
    neighbor 20.0.1.2 description Towards PE1
    neighbor 20.7.1.1 remote-as 100
    neighbor 20.7.1.1 description Towards 101 subnet
    address-family ipv4 unicast
        neighbor 20.0.1.2 activate
        neighbor 20.0.1.2 send-community both
        neighbor 20.0.1.2 soft-reconfiguration inbound
        neighbor 20.7.1.1 activate
        neighbor 20.7.1.1 next-hop-self
    exit-address-family
!
https-server vrf mgmt
```

CE1 Verification

```
CE1# show bgp ipv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 20.0.1.2 (External)
    Description         : Towards PE1
    Peer-group          :
```

```
    Remote Router Id    : 20.0.1.2          Local Router Id    : 20.7.1.2
    Remote AS           : 6488163           Local AS           : 100
    Remote Port         : 57770             Local Port         : 179
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      :
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:23m:45s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : Cease
    Last SubErr Sent    : Administrative Shutdown
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 1                 Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      2        9
    Keepalives                  27       23
    Route Refresh                0        0
    Total                       30       33

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          Yes             Yes
    Address family IPv6 Unicast          No              No

    Address Family : IPv4 Unicast
    -----------------------------

    Rt. Reflect. Client : No                 Send Community     : both
    Allow-AS in         : 0                  Advt. Interval     : 30
    Max. Prefix         : 128000             Soft Reconfig In   : Yes
    Nexthop-Self        :                    Default-Originate  :
    Cfg. Add-Path       :
    Neg. Add-Path       : Disable

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 20.7.1.1 (Internal)
    Description         : Towards Ixia1-SPOKE1
    Peer-group          :

    Remote Router Id    : 196.0.0.1         Local Router Id    : 20.7.1.2
```

```
    Remote AS           : 100                 Local AS           : 100
    Remote Port         : 32802               Local Port         : 179
    State               : Established         Admin Status       : Up
    Conn. Established   : 1                   Conn. Dropped      : 0
    Passive             : No                  Update-Source      :
    Cfg. Hold Time      : 180                 Cfg. Keep Alive    : 60
    Neg. Hold Time      : 90                  Neg. Keep Alive    : 30
    Up/Down Time        : 00h:20m:28s         Connect-Retry Time : 120
    Local-AS Prepend    : No                  Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled             Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300                 Remove Private-AS  : No
    TTL                 : 255                 Local Cluster-ID   :
    Weight              : 0                   Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      8        1
    Keepalives                  45       42
    Route Refresh                0        0
    Total                       54       44

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             No
    Add-Path                             No              No
    Four Octet ASN                       Yes             No
    Address family IPv4 Unicast          Yes             Yes
    Address family IPv6 Unicast          No              Yes

    Address Family : IPv4 Unicast
    -----------------------------

    Rt. Reflect. Client : No                   Send Community     :
    Allow-AS in         : 0                    Advt. Interval     : 30
    Max. Prefix         : 128000               Soft Reconfig In   :
    Nexthop-Self        : Yes                  Default-Originate  :
    Cfg. Add-Path       :
    Neg. Add-Path       : Disable

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

CE1# show bgp ipv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 20.7.1.2
```

```
    Network             Nexthop          Metric     LocPrf     Weight Path
*>e 0.0.0.0/0           20.0.1.2         0          100        0       6488163 102 ?
*>e 20.4.1.0/24         20.0.1.2         0          100        0       6488163 102 ?
*>e 20.5.1.0/24         20.0.1.2         0          100        0       6488163 102 ?
*>e 20.6.1.0/24         20.0.1.2         0          100        0       6488163 102 ?
*>i 101.0.0.0/24        20.7.1.1         0          100        0       i
*>e 151.0.0.0/24        20.0.1.2         0          100        0       6488163 102 ?
*>e 201.0.0.0/24        20.0.1.2         0          100        0       6488163 102 ?
Total number of entries 7

CE1#

CE1# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS                : 100          BGP Router Identifier  : 20.7.1.2
 Peers                   : 2            Log Neighbor Changes   : No
 Cfg. Hold Time          : 180          Cfg. Keep Alive        : 60
 Confederation Id        : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.0.1.2        6488163     40      37      00h:29m:45s  Established   Up
 20.7.1.1        100         56      68      00h:26m:28s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

CE1#

CE1# show ip route all-vrfs

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
--------------------------------------------------------------------------------
------------
0.0.0.0/0           20.0.1.2         vlan2001      -                 B/E
[20/0]       00h:23m:22s
20.0.1.0/24         -                vlan2001      -                 C
[0/0]        -
20.0.1.1/32         -                vlan2001      -                 L
[0/0]        -
20.4.1.0/24         20.0.1.2         vlan2001      -                 B/E
[20/0]       00h:29m:54s
20.5.1.0/24         20.0.1.2         vlan2001      -                 B/E
[20/0]       00h:30m:36s
20.6.1.0/24         20.0.1.2         vlan2001      -                 B/E
```

```
[20/0]       00h:24m:12s
20.7.1.0/24         -                vlan2071      -                 C
[0/0]        -
20.7.1.2/32         -                vlan2071      -                 L
[0/0]        -
101.0.0.0/24        20.7.1.1         vlan2071      -                 B/I
[200/0]      00h:28m:22s
151.0.0.0/24        20.0.1.2         vlan2001      -                 B/E
[20/0]       00h:23m:22s
201.0.0.0/24        20.0.1.2         vlan2001      -                 B/E
[20/0]       00h:28m:22s

Total Route Count : 11
```

CE2 Configuration

```
CE2# show run
Current configuration:
!
!Version ArubaOS-CX TL.10.09.0001AX
!export-password: default
hostname CE2
profile l3-core
logging console severity crit
cli-session
    timeout 0
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2011,2081
interface mgmt
    no shutdown
    ip dhcp
interface 1/1/1
    no shutdown
    description Towards 201 subnet
    no routing
    vlan trunk native 1
    vlan trunk allowed 2081
interface 1/1/2
    no shutdown
    description Towards PE2
    no routing
    vlan trunk native 1
    vlan trunk allowed 2011
interface vlan 2011
    description Towards PE2
    ip address 20.1.1.1/24
interface vlan 2081
    description Towards 201 subnet
    ip address 20.8.1.2/24
    ip ospf 1 area 0.0.0.0
!
!
!
!
!
router ospf 1
```

```
    area 0.0.0.0
router bgp 101
    neighbor 20.1.1.2 remote-as 6488163
    neighbor 20.1.1.2 description Towards PE2
    neighbor 20.8.1.1 remote-as 101
    neighbor 20.8.1.1 description Towards 201 subnet
    address-family ipv4 unicast
        neighbor 20.1.1.2 activate
        neighbor 20.1.1.2 send-community both
        neighbor 20.1.1.2 soft-reconfiguration inbound
        neighbor 20.8.1.1 activate
        neighbor 20.8.1.1 next-hop-self
    exit-address-family
!
https-server vrf mgmt
```

CE2 Verification

```
CE2# show bgp ipv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 20.1.1.2 (External)
    Description          : Towards PE2
    Peer-group           :

    Remote Router Id     : 20.1.1.2           Local Router Id    : 20.8.1.2
    Remote AS            : 6488163            Local AS           : 101
    Remote Port          : 60400              Local Port         : 179
    State                : Established        Admin Status       : Up
    Conn. Established    : 1                  Conn. Dropped      : 0
    Passive              : No                 Update-Source      :
    Cfg. Hold Time       : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time       : 180                Neg. Keep Alive    : 60
    Up/Down Time         : 00h:23m:27s        Connect-Retry Time : 120
    Local-AS Prepend     : No                 Alt. Local-AS      : 0
    BFD                  : Disabled
    Password             :
    Last Err Sent        : Cease
    Last SubErr Sent     : Administrative Shutdown
    Last Err Rcvd        : No Error
    Last SubErr Rcvd     : No Error

    Graceful-Restart     : Enabled            Gr. Restart Time   : 120
    Gr. Stalepath Time   : 300                Remove Private-AS  : No
    TTL                  : 1                  Local Cluster-ID   :
    Weight               : 0                  Fall-over          : No
    Confederation-Peers  : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      2        9
    Keepalives                  26       22
    Route Refresh                0        0
    Total                       29       32

    Capability                          Advertised      Received
    ----------------------------        -----------     ----------
```

```
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          Yes             Yes
    Address family IPv6 Unicast          No              No

    Address Family : IPv4 Unicast
    -----------------------------

    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 128000              Soft Reconfig In  : Yes
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       : Disable

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 20.8.1.1 (Internal)
    Description         : Towards Ixia2-SPOKE2
    Peer-group          :

    Remote Router Id    : 195.0.0.1          Local Router Id    : 20.8.1.2
    Remote AS           : 101                Local AS           : 101
    Remote Port         : 32798              Local Port         : 179
    State               : Established        Admin Status       : Up
    Conn. Established   : 1                  Conn. Dropped      : 0
    Passive             : No                 Update-Source      :
    Cfg. Hold Time      : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time      : 90                 Neg. Keep Alive    : 30
    Up/Down Time        : 00h:20m:27s        Connect-Retry Time : 120
    Local-AS Prepend    : No                 Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled            Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300                Remove Private-AS  : No
    TTL                 : 255                Local Cluster-ID   :
    Weight              : 0                  Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      8        1
    Keepalives                  44       42
    Route Refresh                0        0
    Total                       53       44

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             No
```

```
    Add-Path                              No              No
    Four Octet ASN                        Yes             No
    Address family IPv4 Unicast           Yes             Yes
    Address family IPv6 Unicast           No              Yes

    Address Family : IPv4 Unicast
    -----------------------------

    Rt. Reflect. Client : No                   Send Community    :
    Allow-AS in         : 0                    Advt. Interval    : 30
    Max. Prefix         : 128000               Soft Reconfig In  :
    Nexthop-Self        : Yes                  Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       : Disable

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

CE2# show bgp ipv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 20.8.1.2

    Network            Nexthop         Metric     LocPrf     Weight Path
*>e 0.0.0.0/0          20.1.1.2        0          100        0       6488163 102 ?
*>e 20.4.1.0/24        20.1.1.2        0          100        0       6488163 102 ?
*>e 20.5.1.0/24        20.1.1.2        0          100        0       6488163 102 ?
*>e 20.6.1.0/24        20.1.1.2        0          100        0       6488163 102 ?
*>e 101.0.0.0/24       20.1.1.2        0          100        0       6488163 102 ?
*>e 151.0.0.0/24       20.1.1.2        0          100        0       6488163 102 ?
*>i 201.0.0.0/24       20.8.1.1        0          100        0       i
Total number of entries 7

CE2#
CE2# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS                : 101          BGP Router Identifier  : 20.8.1.2
 Peers                   : 2            Log Neighbor Changes   : No
 Cfg. Hold Time          : 180          Cfg. Keep Alive        : 60
 Confederation Id        : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.1.1.2        6488163     39      36      00h:29m:27s  Established   Up
 20.8.1.1        101         56      67      00h:26m:28s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

CE2#

CE2# show ip route all-vrfs
```

```
Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
----------------------------------------------------------------------------------
------------
0.0.0.0/0           20.1.1.2         vlan2011      -                 B/E
[20/0]       00h:23m:22s
20.1.1.0/24         -                vlan2011      -                 C
[0/0]        -
20.1.1.1/32         -                vlan2011      -                 L
[0/0]        -
20.4.1.0/24         20.1.1.2         vlan2011      -                 B/E
[20/0]       00h:29m:54s
20.5.1.0/24         20.1.1.2         vlan2011      -                 B/E
[20/0]       00h:30m:36s
20.6.1.0/24         20.1.1.2         vlan2011      -                 B/E
[20/0]       00h:24m:12s
20.8.1.0/24         -                vlan2081      -                 C
[0/0]        -
20.8.1.2/32         -                vlan2081      -                 L
[0/0]        -
101.0.0.0/24        20.1.1.2         vlan2011      -                 B/E
[20/0]       00h:28m:22s
151.0.0.0/24        20.1.1.2         vlan2011      -                 B/E
[20/0]       00h:23m:22s
201.0.0.0/24        20.8.1.1         vlan2081      -                 B/I
[200/0]      00h:28m:22s

Total Route Count : 11
```

CE3 Configuration

```
CE3# show run
Current configuration:
!
!Version ArubaOS-CX TL.10.09.0001AX
!export-password: default
hostname CE3
profile l3-core
logging console severity crit
cli-session
    timeout 0
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2021,2041
interface mgmt
    no shutdown
```

```
    ip dhcp
interface 1/1/1
    no shutdown
    description Towards PE3
    no routing
    vlan trunk native 1
    vlan trunk allowed 2021
interface 1/1/2
    no shutdown
    description Towards 151 and default subnet
    no routing
    vlan trunk native 1
    vlan trunk allowed 2041
interface vlan 2021
    description Towards PE3
    ip address 20.2.1.2/24
interface vlan 2041
    description Towards 151 and default subnet
    ip address 20.4.1.1/24
    ip ospf 1 area 0.0.0.0
!
!
!
!
!
router ospf 1
    redistribute bgp
    area 0.0.0.0
router bgp 102
    neighbor 20.2.1.1 remote-as 6488163
    neighbor 20.2.1.1 description Towards PE3
    address-family ipv4 unicast
        neighbor 20.2.1.1 activate
        neighbor 20.2.1.1 send-community both
        neighbor 20.2.1.1 soft-reconfiguration inbound
        redistribute ospf
    exit-address-family
!
https-server vrf mgmt
CE3#

CE3# show ip ospf neighbors
VRF : default                          Process : 1
===================================================

Total Number of Neighbors : 1

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
20.4.1.2         1         FULL/BDR          20.4.1.2           vlan2041
```

CE3 Verification

```
CE3# show bgp ipv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 20.2.1.1 (External)
    Description         : Towards PE3
    Peer-group          :
```

```
    Remote Router Id    : 20.2.1.1          Local Router Id    : 20.4.1.1
    Remote AS           : 6488163           Local AS           : 102
    Remote Port         : 179               Local Port         : 48554
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      :
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:23m:06s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : Cease
    Last SubErr Sent    : Administrative Shutdown
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 1                 Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      8        3
    Keepalives                  23       27
    Route Refresh                0        0
    Total                       32       31

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          Yes             Yes
    Address family IPv6 Unicast          No              No

    Address Family : IPv4 Unicast
    -----------------------------

    Rt. Reflect. Client : No                 Send Community    : both
    Allow-AS in         : 5                  Advt. Interval    : 30
    Max. Prefix         : 128000             Soft Reconfig In  : Yes
    Nexthop-Self        :                    Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       : Disable

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :


CE3# show bgp ipv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete
```

```
VRF : default
Local Router-ID 20.4.1.1

    Network            Nexthop         Metric     LocPrf     Weight Path
*>  0.0.0.0/0          20.4.1.2        0          100        0       ?
*>  20.4.1.0/24        0.0.0.0         0          100        0       ?
*>  20.5.1.0/24        20.4.1.2        0          100        0       ?
*>  20.6.1.0/24        20.4.1.2        0          100        0       ?
*>e 101.0.0.0/24       20.2.1.1        0          100        0       6488163 100 i
*>  151.0.0.0/24       20.4.1.2        0          100        0       ?
*>e 201.0.0.0/24       20.2.1.1        0          100        0       6488163 101 i
Total number of entries 7

CE3# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS                : 102          BGP Router Identifier  : 20.4.1.1
 Peers                   : 1            Log Neighbor Changes   : No
 Cfg. Hold Time          : 180          Cfg. Keep Alive        : 60
 Confederation Id        : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.2.1.1        6488163     38      39      00h:29m:07s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

CE3# show ip route all-vrfs

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix                Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                    Type     Metric
----------------------------------------------------------------------------------
------------
0.0.0.0/0            20.4.1.2         vlan2041      -               O/IA
[110/200]    00h:23m:22s
20.2.1.0/24          -                vlan2021      -               C
[0/0]        -
20.2.1.2/32          -                vlan2021      -               L
[0/0]        -
20.4.1.0/24          -                vlan2041      -               C
[0/0]        -
20.4.1.1/32          -                vlan2041      -               L
[0/0]        -
20.5.1.0/24          20.4.1.2         vlan2041      -               O
[110/200]    00h:29m:54s
20.6.1.0/24          20.4.1.2         vlan2041      -               O
```

```
[110/200]    00h:24m:12s
101.0.0.0/24        20.2.1.1         vlan2021      -                 B/E
[20/0]       00h:28m:23s
151.0.0.0/24        20.4.1.2         vlan2041      -                 O/IA
[110/200]    00h:23m:22s
201.0.0.0/24        20.2.1.1         vlan2021      -                 B/E
[20/0]       00h:28m:22s

Total Route Count : 10
```

CE4 Configuration

```
CE4# show run
Current configuration:
!
!Version ArubaOS-CX TL.10.09.0001AX
!export-password: default
hostname CE4
profile l3-core
logging console severity crit
cli-session
    timeout 0
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2031,2051
interface mgmt
    no shutdown
    ip dhcp
interface 1/1/1
    no shutdown
    description Towards PE3
    no routing
    vlan trunk native 1
    vlan trunk allowed 2031
interface 1/1/2
    no shutdown
    description Towards 151 and default subnet
    no routing
    vlan trunk native 1
    vlan trunk allowed 2051
interface vlan 2031
    description Towards PE3
    ip address 20.3.1.2/24
interface vlan 2051
    description Towards 151 and default subnet
    ip address 20.5.1.1/24
    ip ospf 1 area 0.0.0.0
!
!
!
!
!
router ospf 1
    redistribute bgp
    area 0.0.0.0
router bgp 102
    neighbor 20.3.1.1 remote-as 6488163
```

```
    neighbor 20.3.1.1 description Towards PE3
    address-family ipv4 unicast
        neighbor 20.3.1.1 activate
        neighbor 20.3.1.1 send-community both
        neighbor 20.3.1.1 soft-reconfiguration inbound
        redistribute ospf
    exit-address-family
!
https-server vrf mgmt
CE4#

CE4# show ip ospf neighbors
VRF : default                          Process : 1
===================================================

Total Number of Neighbors : 1

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
20.4.1.2         1         FULL/DR           20.5.1.2           vlan2051
```

CE4 Verification

```
CE4# show bgp ipv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 20.3.1.1 (External)
    Description          : Towards PE3
    Peer-group           :

    Remote Router Id     : 20.3.1.1           Local Router Id    : 20.5.1.1
    Remote AS            : 6488163            Local AS           : 102
    Remote Port          : 52062              Local Port         : 179
    State                : Established        Admin Status       : Up
    Conn. Established    : 1                  Conn. Dropped      : 0
    Passive              : No                 Update-Source      :
    Cfg. Hold Time       : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time       : 180                Neg. Keep Alive    : 60
    Up/Down Time         : 00h:22m:47s        Connect-Retry Time : 120
    Local-AS Prepend     : No                 Alt. Local-AS      : 0
    BFD                  : Disabled
    Password             :
    Last Err Sent        : Cease
    Last SubErr Sent     : Administrative Shutdown
    Last Err Rcvd        : No Error
    Last SubErr Rcvd     : No Error

    Graceful-Restart     : Enabled            Gr. Restart Time   : 120
    Gr. Stalepath Time   : 300                Remove Private-AS  : No
    TTL                  : 1                  Local Cluster-ID   :
    Weight               : 0                  Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent      Rcvd
    -------------------      -----     -----
    Open                         1         1
    Notification                 0         0
    Updates                      9         1
    Keepalives                  22        26
```

```
    Route Refresh                0        0
    Total                       32       28

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          Yes             Yes
    Address family IPv6 Unicast          No              No

    Address Family : IPv4 Unicast
    -----------------------------

    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 5                   Advt. Interval    : 30
    Max. Prefix         : 128000              Soft Reconfig In  : Yes
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       : Disable

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :
CE4#

CE4# show bgp ipv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 20.5.1.1

    Network            Nexthop         Metric     LocPrf     Weight Path
*>  0.0.0.0/0          20.5.1.2        0          100        0       ?
*>  20.4.1.0/24        20.5.1.2        0          100        0       ?
*>  20.5.1.0/24        0.0.0.0         0          100        0       ?
*>  20.6.1.0/24        20.5.1.2        0          100        0       ?
*>  101.0.0.0/24       20.5.1.2        0          100        0       ?
*>  151.0.0.0/24       20.5.1.2        0          100        0       ?
*>  201.0.0.0/24       20.5.1.2        0          100        0       ?
Total number of entries 7

CE4#

CE4# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS               : 102          BGP Router Identifier  : 20.5.1.1
 Peers                  : 1            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.3.1.1        6488163     35      39      00h:28m:48s  Established   Up
```

```
Address-family : IPv6 Unicast
-----------------------------

CE4# show ip route all-vrfs

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix              Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                  Type     Metric
--------------------------------------------------------------------------------
------------
0.0.0.0/0          20.5.1.2         vlan2051      -                 O/IA
[110/200]    00h:23m:22s
20.3.1.0/24        -                vlan2031      -                 C
[0/0]        -
20.3.1.2/32        -                vlan2031      -                 L
[0/0]        -
20.4.1.0/24        20.5.1.2         vlan2051      -                 O
[110/200]    00h:29m:55s
20.5.1.0/24        -                vlan2051      -                 C
[0/0]        -
20.5.1.1/32        -                vlan2051      -                 L
[0/0]        -
20.6.1.0/24        20.5.1.2         vlan2051      -                 O
[110/200]    00h:24m:12s
101.0.0.0/24       20.5.1.2         vlan2051      -                 O/E2
[110/25]     00h:28m:22s
151.0.0.0/24       20.5.1.2         vlan2051      -                 O/IA
[110/200]    00h:23m:22s
201.0.0.0/24       20.5.1.2         vlan2051      -                 O/E2
[110/25]     00h:28m:22s

Total Route Count : 10
```

PE1 Configuration

```
PE1# show run
Current configuration:
!
!Version ArubaOS-CX LL.10.09.0001AX
!export-password: default
hostname PE1
profile core-spine
logging console severity crit
vrf spoke_1
    rd 1:1
    l3vpn-only
    address-family ipv4 unicast
        route-target export 1:1
        route-target import 1:0
    exit-address-family
```

```
cli-session
    timeout 0
!
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2001
interface mgmt
    no shutdown
    ip dhcp
interface 1/1/37
    no shutdown
    description Towards CE1
    no routing
    vlan trunk native 1
    vlan trunk allowed 2001
interface 1/1/38
    no shutdown
    mtu 9198
    description Towards P1
    ip address 30.0.1.1/24
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface loopback 0
    ip address 1.1.1.1/32
    ip ospf 1 area 0.0.0.0
interface vlan 2001
    vrf attach spoke_1
    description Towards CE1
    ip address 20.0.1.2/24
!
!
!
!
!
router ospf 1
    area 0.0.0.0
router bgp 6488163
    neighbor 2.2.2.2 remote-as 6488163
    neighbor 2.2.2.2 description Towards PE2-SPOKE2
    neighbor 2.2.2.2 update-source loopback 0
    neighbor 3.3.3.3 remote-as 6488163
    neighbor 3.3.3.3 description Towards PE3-HUB
    neighbor 3.3.3.3 update-source loopback 0
    address-family vpnv4 unicast
        neighbor 2.2.2.2 activate
        neighbor 2.2.2.2 send-community both
        neighbor 3.3.3.3 activate
        neighbor 3.3.3.3 send-community both
    exit-address-family
!
    vrf spoke_1
        neighbor 20.0.1.1 remote-as 100
        neighbor 20.0.1.1 description Towards CE1
        address-family ipv4 unicast
            neighbor 20.0.1.1 activate
            neighbor 20.0.1.1 send-community both
            neighbor 20.0.1.1 soft-reconfiguration inbound
```

```
        exit-address-family
!
https-server vrf mgmt
mpls
    enable
    label-protocol ldp
        enable
        router-id loopback0
```

PE1 Verification

```
PE1# show ip ospf neighbors
VRF : default                          Process : 1
===================================================

Total Number of Neighbors : 1

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
4.4.4.4          1         FULL/BDR          30.0.1.2           1/1/38

PE1# show mpls ldp neighbor
Local LDP Identifier: 1.1.1.1:0, Peer LDP Identifier: 4.4.4.4:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:17:58
    State: operational
    LDP Discovery Sources: 1/1/38
    Addresses bound to this peer:
         30.0.1.2 30.0.2.2 30.0.3.1 4.4.4.4

PE1# show mpls forwarding

MPLS Bindings

Entry Bindings   : 9
Exit Bindings    : 3
Transit Bindings : 3
PHP Mode         : Explicit-Null
QoS Mode         : Uniform
TTL Propagation  : Uniform

Entry Bindings:

Origin Prefix       Ingress  Nexthop    Outgoing  Egress     Egress   Status
                    VRF      Address    Label     Interface  VRF
-------------------------------------------------------------------------------
LDP    2.2.2.2/32   default  30.0.1.2   17        1/1/38     default  operational

LDP    3.3.3.3/32   default  30.0.1.2   18        1/1/38     default  operational

BGP    0.0.0.0/0    spoke_1  3.3.3.3    19        1/1/38     default  operational

BGP    20.4.1.0/24  spoke_1  3.3.3.3    19        1/1/38     default  operational

BGP    20.5.1.0/24  spoke_1  3.3.3.3    19        1/1/38     default  operational

BGP    20.6.1.0/24  spoke_1  3.3.3.3    19        1/1/38     default  operational

BGP    151.0.0.0/24 spoke_1  3.3.3.3    19        1/1/38     default  operational
```

```
BGP    201.0.0.0/24 spoke_1  3.3.3.3    19        1/1/38     default  operational

Exit Bindings:

Origin   Prefix                 Incoming  Service   Egress            Status
                                Label     Label     VRF
-------------------------------------------------------------------------------
static   n/a                    exp-null  -         default           operational
BGP      n/a                    imp-null  19        spoke_1           operational
static   n/a                    7         -         default           operational



Transit Bindings:

Origin  Prefix        Incoming  Egress     Egress   Nexthop   Outgoing  Status

                      Label     Interface  VRF      Address   Label

-------------------------------------------------------------------------------
LDP     4.4.4.4/32   16        1/1/38     default  30.0.1.2  exp-null  operational
LDP     2.2.2.2/32   17        1/1/38     default  30.0.1.2  17        operational
LDP     3.3.3.3/32   18        1/1/38     default  30.0.1.2  18        operational

PE1# show bgp vpnv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 2.2.2.2 (Internal)
    Description         : Towards PE2-SPOKE2
    Peer-group          :

    Remote Router Id    : 2.2.2.2           Local Router Id    : 1.1.1.1
    Remote AS           : 6488163           Local AS           : 6488163
    Remote Port         : 179               Local Port         : 39729
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      : loopback0
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:19m:46s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 255               Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      2        2
    Keepalives                  22       22
```

```
    Route Refresh                0        0
    Total                       25       25

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  :
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 3.3.3.3 (Internal)
    Description         : Towards PE3-HUB
    Peer-group          :

    Remote Router Id    : 3.3.3.3            Local Router Id    : 1.1.1.1
    Remote AS           : 6488163            Local AS           : 6488163
    Remote Port         : 43547              Local Port         : 179
    State               : Established        Admin Status       : Up
    Conn. Established   : 1                  Conn. Dropped      : 0
    Passive             : No                 Update-Source      : loopback0
    Cfg. Hold Time      : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180                Neg. Keep Alive    : 60
    Up/Down Time        : 00h:19m:18s        Connect-Retry Time : 120
    Local-AS Prepend    : No                 Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled            Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300                Remove Private-AS  : No
    TTL                 : 255                Local Cluster-ID   :
    Weight              : 0                  Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      2        9
    Keepalives                  22       18
```

```
    Route Refresh                0        0
    Total                       25       28

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  :
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

PE1# show bgp vrf spoke_1 ipv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : spoke_1

BGP Neighbor 20.0.1.1 (External)
    Description         : Towards CE1
    Peer-group          :

    Remote Router Id    : 20.7.1.2          Local Router Id    : 20.0.1.2
    Remote AS           : 100               Local AS           : 6488163
    Remote Port         : 179               Local Port         : 57770
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      :
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:21m:38s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : Cease
    Last SubErr Sent    : Administrative Shutdown
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 1                 Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
```

```
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      9        2
    Keepalives                  21       25
    Route Refresh                0        0
    Total                       31       28

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          Yes             Yes
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         No              No
    Address family L2VPN EVPN            No              No

    Address Family : IPv4 Unicast
    -----------------------------

    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  : Yes
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       : Disable

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :


PE1# show bgp vrf spoke_1 ipv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : spoke_1
Local Router-ID 20.0.1.2

    Network            Nexthop         Metric     LocPrf     Weight Path
    Route Distinguisher: 1:1
*>  0.0.0.0/0          3.3.3.3         0          100        0       102 ?
*>  20.4.1.0/24        3.3.3.3         0          100        0       102 ?
*>  20.5.1.0/24        3.3.3.3         0          100        0       102 ?
*>  20.6.1.0/24        3.3.3.3         0          100        0       102 ?
*>e 101.0.0.0/24       20.0.1.1        0          100        0       100 i
*>  151.0.0.0/24       3.3.3.3         0          100        0       102 ?
*>  201.0.0.0/24       3.3.3.3         0          100        0       102 ?
Total number of entries 7

PE1# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 1.1.1.1
 Peers                  : 2            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0
```

```
Address-family : IPv4 Unicast
-----------------------------

Address-family : IPv6 Unicast
-----------------------------

Address-family : VPNv4 Unicast
------------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 2.2.2.2         6488163     37      36      00h:29m:42s  Established   Up
 3.3.3.3         6488163     40      36      00h:29m:14s  Established   Up

Address-family : L2VPN EVPN
-----------------------------

VRF : spoke_1
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 20.0.1.2
 Peers                  : 1            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.0.1.1        100         37      40      00h:29m:45s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

PE1# show ip route all-vrfs

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
---------------------------------------------------------------------------------
------------
1.1.1.1/32          -                loopback0     -                 L
[0/0]        -
2.2.2.2/32          30.0.1.2         1/1/38        -                 O
[110/200]    00h:31m:42s
3.3.3.3/32          30.0.1.2         1/1/38        -                 O
[110/150]    00h:31m:21s
4.4.4.4/32          30.0.1.2         1/1/38        -                 O
[110/100]    00h:32m:18s
30.0.1.0/24         -                1/1/38        -                 C
[0/0]        -
30.0.1.1/32         -                1/1/38        -                 L
```

```
[0/0]         -
30.0.2.0/24         30.0.1.2         1/1/38        -                 O
[110/200]     00h:32m:18s
30.0.3.0/24         30.0.1.2         1/1/38        -                 O
[110/150]     00h:32m:06s

VRF: spoke_1

Prefix               Nexthop           Interface     VRF(egress)    Origin/
Distance/     Age
                                                                    Type     Metric
--------------------------------------------------------------------------------
------------
0.0.0.0/0           3.3.3.3          1/1/38        default           B/V
[200/0]       00h:23m:22s
20.0.1.0/24         -                vlan2001      -                 C
[0/0]         -
20.0.1.2/32         -                vlan2001      -                 L
[0/0]         -
20.4.1.0/24         3.3.3.3          1/1/38        default           B/V
[200/0]       00h:29m:55s
20.5.1.0/24         3.3.3.3          1/1/38        default           B/V
[200/0]       00h:30m:37s
20.6.1.0/24         3.3.3.3          1/1/38        default           B/V
[200/0]       00h:24m:12s
101.0.0.0/24        20.0.1.1         vlan2001      -                 B/E
[20/0]        00h:28m:23s
151.0.0.0/24        3.3.3.3          1/1/38        default           B/V
[200/0]       00h:23m:22s
201.0.0.0/24        3.3.3.3          1/1/38        default           B/V
[200/0]       00h:28m:22s

Total Route Count : 17
```

PE2 Configuration

```
PE2# show running-config
Current configuration:
!
!Version ArubaOS-CX LL.10.09.0001AX
!export-password: default
hostname PE2
profile core-spine
logging console severity crit
vrf spoke_2
    rd 1:1
    l3vpn-only
    address-family ipv4 unicast
        route-target export 1:1
        route-target import 1:0
    exit-address-family
cli-session
    timeout 0
!
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2011
```

```
interface mgmt
    no shutdown
    ip dhcp
interface 1/1/43
    no shutdown
    description Towards CE1
    no routing
    vlan trunk native 1
    vlan trunk allowed 2011
interface 1/1/44
    no shutdown
    mtu 9198
    description Towards P1
    ip address 30.0.2.1/24
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface loopback 0
    ip address 2.2.2.2/32
    ip ospf 1 area 0.0.0.0
interface vlan 2011
    vrf attach spoke_2
    description Towards CE1
    ip address 20.1.1.2/24
!
!
!
!
!
router ospf 1
    area 0.0.0.0
router bgp 6488163
    neighbor 1.1.1.1 remote-as 6488163
    neighbor 1.1.1.1 description Towards PE1-SPOKE1
    neighbor 1.1.1.1 update-source loopback 0
    neighbor 3.3.3.3 remote-as 6488163
    neighbor 3.3.3.3 description Towards PE3-HUB
    neighbor 3.3.3.3 update-source loopback 0
    address-family vpnv4 unicast
        neighbor 1.1.1.1 activate
        neighbor 1.1.1.1 send-community both
        neighbor 3.3.3.3 activate
        neighbor 3.3.3.3 send-community both
    exit-address-family
!
    vrf spoke_2
        neighbor 20.1.1.1 remote-as 101
        neighbor 20.1.1.1 description Towards CE2
        address-family ipv4 unicast
            neighbor 20.1.1.1 activate
            neighbor 20.1.1.1 send-community both
            neighbor 20.1.1.1 soft-reconfiguration inbound
        exit-address-family
!
https-server vrf mgmt
mpls
    enable
    label-protocol ldp
        enable
        router-id loopback0

PE2# show ip ospf neighbors
```

```
VRF : default                          Process : 1
===================================================

Total Number of Neighbors : 1

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
4.4.4.4          1         FULL/DR           30.0.2.2           1/1/44
```

PE2 Verification

```
PE2# show mpls ldp neighbor
Local LDP Identifier: 2.2.2.2:0, Peer LDP Identifier: 4.4.4.4:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:17:57
    State: operational
    LDP Discovery Sources: 1/1/44
    Addresses bound to this peer:
         30.0.1.2 30.0.2.2 30.0.3.1 4.4.4.4

PE2# show mpls forwarding

MPLS Bindings

Entry Bindings   : 9
Exit Bindings    : 3
Transit Bindings : 3
PHP Mode         : Explicit-Null
QoS Mode         : Uniform
TTL Propagation  : Uniform


Entry Bindings:

Origin Prefix        Ingress  Nexthop    Outgoing  Egress     Egress   Status
                     VRF      Address    Label     Interface  VRF
--------------------------------------------------------------------------------
LDP    1.1.1.1/32    default  30.0.2.2   16        1/1/44     default  operational

LDP    3.3.3.3/32    default  30.0.2.2   18        1/1/44     default  operational

BGP    0.0.0.0/0     spoke_2  3.3.3.3    19        1/1/44     default  operational

BGP    20.4.1.0/24   spoke_2  3.3.3.3    19        1/1/44     default  operational

BGP    20.5.1.0/24   spoke_2  3.3.3.3    19        1/1/44     default  operational

BGP    20.6.1.0/24   spoke_2  3.3.3.3    19        1/1/44     default  operational

BGP    101.0.0.0/24 spoke_2  3.3.3.3    19        1/1/44     default  operational

BGP    151.0.0.0/24 spoke_2  3.3.3.3    19        1/1/44     default  operational

Exit Bindings:

Origin   Prefix             Incoming  Service   Egress            Status
                            Label     Label     VRF
--------------------------------------------------------------------------------
static   n/a                exp-null  -         default           operational
BGP      n/a                imp-null  19        spoke_2           operational
```

```
static   n/a                 7         -         default          operational

Transit Bindings:

Origin  Prefix       Incoming  Egress     Egress   Nexthop   Outgoing  Status

                     Label     Interface  VRF      Address   Label

-------------------------------------------------------------------------------
LDP     1.1.1.1/32   16        1/1/44     default  30.0.2.2  16        operational
LDP     4.4.4.4/32   17        1/1/44     default  30.0.2.2  exp-null  operational
LDP     3.3.3.3/32   18        1/1/44     default  30.0.2.2  18        operational


PE2# show bgp vpnv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 1.1.1.1 (Internal)
    Description          : Towards PE1-SPOKE1
    Peer-group           :

    Remote Router Id     : 1.1.1.1            Local Router Id    : 2.2.2.2
    Remote AS            : 6488163            Local AS           : 6488163
    Remote Port          : 39729              Local Port         : 179
    State                : Established        Admin Status       : Up
    Conn. Established    : 1                  Conn. Dropped      : 0
    Passive              : No                 Update-Source      : loopback0
    Cfg. Hold Time       : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time       : 180                Neg. Keep Alive    : 60
    Up/Down Time         : 00h:19m:46s        Connect-Retry Time : 120
    Local-AS Prepend     : No                 Alt. Local-AS      : 0
    BFD                  : Disabled
    Password             :
    Last Err Sent        : No Error
    Last SubErr Sent     : No Error
    Last Err Rcvd        : No Error
    Last SubErr Rcvd     : No Error

    Graceful-Restart     : Enabled            Gr. Restart Time   : 120
    Gr. Stalepath Time   : 300                Remove Private-AS  : No
    TTL                  : 255                Local Cluster-ID   :
    Weight               : 0                  Fall-over          : No
    Confederation-Peers  : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      2        2
    Keepalives                  22       22
    Route Refresh                0        0
    Total                       25       25

    Capability                          Advertised      Received
    ----------------------------        -----------     ----------
    Route Refresh                       Yes             Yes
    Graceful Restart                    Yes             Yes
    Add-Path                            No              No
    Four Octet ASN                      Yes             Yes
    Address family IPv4 Unicast         No              No
```

```
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  :
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 3.3.3.3 (Internal)
    Description         : Towards PE3-HUB
    Peer-group          :

    Remote Router Id    : 3.3.3.3            Local Router Id    : 2.2.2.2
    Remote AS           : 6488163            Local AS           : 6488163
    Remote Port         : 179                Local Port         : 38293
    State               : Established        Admin Status       : Up
    Conn. Established   : 1                  Conn. Dropped      : 0
    Passive             : No                 Update-Source      : loopback0
    Cfg. Hold Time      : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180                Neg. Keep Alive    : 60
    Up/Down Time        : 00h:19m:28s        Connect-Retry Time : 120
    Local-AS Prepend    : No                 Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled            Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300                Remove Private-AS  : No
    TTL                 : 255                Local Cluster-ID   :
    Weight              : 0                  Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      2        9
    Keepalives                  22       17
    Route Refresh                0        0
    Total                       25       27

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
```

```
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                    Send Community    : both
    Allow-AS in         : 0                     Advt. Interval    : 30
    Max. Prefix         : 300000                Soft Reconfig In  :
    Nexthop-Self        :                       Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

PE2# show bgp vrf spoke_2 ipv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : spoke_2

BGP Neighbor 20.1.1.1 (External)
    Description         : Towards CE2
    Peer-group          :

    Remote Router Id    : 20.8.1.2             Local Router Id    : 20.1.1.2
    Remote AS           : 101                  Local AS           : 6488163
    Remote Port         : 179                  Local Port         : 60400
    State               : Established          Admin Status       : Up
    Conn. Established   : 1                    Conn. Dropped      : 0
    Passive             : No                   Update-Source      :
    Cfg. Hold Time      : 180                  Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180                  Neg. Keep Alive    : 60
    Up/Down Time        : 00h:21m:33s          Connect-Retry Time : 120
    Local-AS Prepend    : No                   Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : Cease
    Last SubErr Sent    : Administrative Shutdown
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled              Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300                  Remove Private-AS  : No
    TTL                 : 1                    Local Cluster-ID   :
    Weight              : 0                    Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      9        2
    Keepalives                  20       24
    Route Refresh                0        0
    Total                       30       27

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
```

```
    Route Refresh                      Yes            Yes
    Graceful Restart                   Yes            Yes
    Add-Path                           No             No
    Four Octet ASN                     Yes            Yes
    Address family IPv4 Unicast        Yes            Yes
    Address family IPv6 Unicast        No             No
    Address family VPNv4 Unicast       No             No
    Address family L2VPN EVPN          No             No

    Address Family : IPv4 Unicast
    -----------------------------

    Rt. Reflect. Client : No                   Send Community    : both
    Allow-AS in         : 0                    Advt. Interval    : 30
    Max. Prefix         : 300000               Soft Reconfig In  : Yes
    Nexthop-Self        :                      Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       : Disable

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

PE2# show bgp vrf spoke_2 ipv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : spoke_2
Local Router-ID 20.1.1.2

    Network            Nexthop         Metric     LocPrf     Weight Path
    Route Distinguisher: 1:1
*>  0.0.0.0/0          3.3.3.3         0          100        0       102 ?
*>  20.4.1.0/24        3.3.3.3         0          100        0       102 ?
*>  20.5.1.0/24        3.3.3.3         0          100        0       102 ?
*>  20.6.1.0/24        3.3.3.3         0          100        0       102 ?
*>  101.0.0.0/24       3.3.3.3         0          100        0       102 ?
*>  151.0.0.0/24       3.3.3.3         0          100        0       102 ?
*>e 201.0.0.0/24       20.1.1.1        0          100        0       101 i
Total number of entries 7

PE2# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 2.2.2.2
 Peers                  : 2            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------

Address-family : IPv6 Unicast
-----------------------------

Address-family : VPNv4 Unicast
------------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
```

```
 1.1.1.1         6488163     36      37      00h:29m:42s  Established   Up
 3.3.3.3         6488163     39      37      00h:29m:24s  Established   Up

Address-family : L2VPN EVPN
-----------------------------

VRF : spoke_2
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 20.1.1.2
 Peers                  : 1            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.1.1.1        101         36      39      00h:29m:27s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

PE2# show ip route all-vrfs

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix                Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                    Type     Metric
-----------------------------------------------------------------------------------
------------
1.1.1.1/32           30.0.2.2         1/1/44        -               O
[110/200]    00h:31m:42s
2.2.2.2/32           -                loopback0     -               L
[0/0]        -
3.3.3.3/32           30.0.2.2         1/1/44        -               O
[110/150]    00h:31m:21s
4.4.4.4/32           30.0.2.2         1/1/44        -               O
[110/100]    00h:31m:42s
30.0.1.0/24          30.0.2.2         1/1/44        -               O
[110/200]    00h:31m:42s
30.0.2.0/24          -                1/1/44        -               C
[0/0]        -
30.0.2.1/32          -                1/1/44        -               L
[0/0]        -
30.0.3.0/24          30.0.2.2         1/1/44        -               O
[110/150]    00h:31m:42s

VRF: spoke_2

Prefix                Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                    Type     Metric
```

```
--------------------------------------------------------------------------------
------------
0.0.0.0/0           3.3.3.3          1/1/44        default            B/V
[200/0]       00h:23m:22s
20.1.1.0/24         -                vlan2011      -                  C
[0/0]         -
20.1.1.2/32         -                vlan2011      -                  L
[0/0]         -
20.4.1.0/24         3.3.3.3          1/1/44        default            B/V
[200/0]       00h:29m:55s
20.5.1.0/24         3.3.3.3          1/1/44        default            B/V
[200/0]       00h:30m:37s
20.6.1.0/24         3.3.3.3          1/1/44        default            B/V
[200/0]       00h:24m:12s
101.0.0.0/24        3.3.3.3          1/1/44        default            B/V
[200/0]       00h:28m:22s
151.0.0.0/24        3.3.3.3          1/1/44        default            B/V
[200/0]       00h:23m:22s
201.0.0.0/24        20.1.1.1         vlan2011      -                  B/E
[20/0]        00h:28m:22s

Total Route Count : 17
```

PE3 Configuration

```
PE3# show running-config
Current configuration:
!
!Version ArubaOS-CX LL.10.09.0001AX
!export-password: default
hostname PE3
profile core-spine
logging console severity crit
vrf hub_export
    rd 1:0
    l3vpn-only
    address-family ipv4 unicast
        route-target export 1:0
    exit-address-family
vrf hub_import
    rd 0:1
    l3vpn-only
    address-family ipv4 unicast
        route-target import 1:1
    exit-address-family
cli-session
    timeout 0
!
!
!
!
!
!
ssh server vrf mgmt
vlan 1,2021,2031
interface mgmt
    no shutdown
    ip dhcp
system interface-group 1 speed 10g
    !interface group 1 contains ports 1/1/1-1/1/4
interface lag 99
```

```
    no shutdown
    description Towards P1
    ip address 30.0.3.2/24
    lacp mode active
    ip ospf 1 area 0.0.0.0
    mpls enable
    mpls ldp enable
interface 1/1/31
    no shutdown
    mtu 9198
    description Towards P1 LAG 99
    lag 99
interface 1/1/32
    no shutdown
    mtu 9198
    description Towards P1 LAG 99
    lag 99
interface 1/1/33
    no shutdown
    description Towards CE3
    no routing
    vlan trunk native 1
    vlan trunk allowed 2021
interface 1/1/34
    no shutdown
    description Towards CE4
    no routing
    vlan trunk native 1
    vlan trunk allowed 2031
interface loopback 0
    ip address 3.3.3.3/32
    ip ospf 1 area 0.0.0.0
interface vlan 2021
    vrf attach hub_import
    description Towards CE3
    ip address 20.2.1.1/24
interface vlan 2031
    vrf attach hub_export
    description Towards CE4
    ip address 20.3.1.1/24
!
!
!
!
!
router ospf 1
    area 0.0.0.0
router bgp 6488163
    neighbor 1.1.1.1 remote-as 6488163
    neighbor 1.1.1.1 description Towards PE1-SPOKE1
    neighbor 1.1.1.1 update-source loopback 0
    neighbor 2.2.2.2 remote-as 6488163
    neighbor 2.2.2.2 description Towards PE2-SPOKE2
    neighbor 2.2.2.2 update-source loopback 0
    address-family vpnv4 unicast
        neighbor 1.1.1.1 activate
        neighbor 1.1.1.1 send-community both
        neighbor 2.2.2.2 activate
        neighbor 2.2.2.2 send-community both
    exit-address-family
!
    vrf hub_export
```

```
        neighbor 20.3.1.2 remote-as 102
        neighbor 20.3.1.2 description Towards CE3
        address-family ipv4 unicast
            neighbor 20.3.1.2 activate
            neighbor 20.3.1.2 send-community both
            neighbor 20.3.1.2 soft-reconfiguration inbound
        exit-address-family
!
    vrf hub_import
        neighbor 20.2.1.2 remote-as 102
        neighbor 20.2.1.2 description Towards CE4
        address-family ipv4 unicast
            neighbor 20.2.1.2 activate
            neighbor 20.2.1.2 send-community both
            neighbor 20.2.1.2 soft-reconfiguration inbound
        exit-address-family
!
https-server vrf mgmt
mpls
    enable
    label-protocol ldp
        enable
        router-id loopback0

PE3# show ip ospf neighbors
VRF : default                          Process : 1
===================================================

Total Number of Neighbors : 1

Neighbor ID      Priority  State             Nbr Address       Interface
-------------------------------------------------------------------------
4.4.4.4          1         FULL/DR           30.0.3.1           lag99
```

PE3 Verification

```
PE3# show mpls ldp neighbor
Local LDP Identifier: 3.3.3.3:0, Peer LDP Identifier: 4.4.4.4:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:17:21
    State: operational
    LDP Discovery Sources: lag99
    Addresses bound to this peer:
         30.0.1.2 30.0.2.2 30.0.3.1 4.4.4.4


PE3# show mpls forwarding

MPLS Bindings

Entry Bindings   : 5
Exit Bindings    : 3
Transit Bindings : 3
PHP Mode         : Explicit-Null
QoS Mode         : Uniform
TTL Propagation  : Uniform

Entry Bindings:

Origin Prefix       Ingress   Nexthop    Outgoing  Egress     Egress   Status
```

```
                    VRF       Address    Label     Interface  VRF
-------------------------------------------------------------------------------
LDP    1.1.1.1/32   default   30.0.3.1   16        lag99      default  operational

LDP    2.2.2.2/32   default   30.0.3.1   17        lag99      default  operational

LDP    4.4.4.4/32   default   30.0.3.1   exp-null  lag99      default  operational

BGP    101.0.0.0/24 hub_import 1.1.1.1   19        lag99      default  operational
BGP    201.0.0.0/24 hub_import 2.2.2.2   19        lag99      default  operational


Exit Bindings:

Origin   Prefix              Incoming  Service   Egress            Status
                             Label     Label     VRF
-------------------------------------------------------------------------------
static   n/a                 exp-null  -         default           operational
BGP      n/a                 imp-null  19        hub_export        operational
static   n/a                 7         -         default           operational

Transit Bindings:

Origin  Prefix       Incoming  Egress     Egress   Nexthop   Outgoing  Status

                     Label     Interface  VRF      Address   Label

-------------------------------------------------------------------------------
LDP     1.1.1.1/32   16        lag99      default  30.0.3.1  16        operational
LDP     2.2.2.2/32   17        lag99      default  30.0.3.1  17        operational
LDP     4.4.4.4/32   18        lag99      default  30.0.3.1  exp-null  operational

PE3# show bgp vpnv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 1.1.1.1 (Internal)
    Description          : Towards PE1-SPOKE1
    Peer-group           :

    Remote Router Id     : 1.1.1.1            Local Router Id    : 3.3.3.3
    Remote AS            : 6488163            Local AS           : 6488163
    Remote Port          : 179                Local Port         : 43547
    State                : Established        Admin Status       : Up
    Conn. Established    : 1                  Conn. Dropped      : 0
    Passive              : No                 Update-Source      : loopback0
    Cfg. Hold Time       : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time       : 180                Neg. Keep Alive    : 60
    Up/Down Time         : 00h:19m:18s        Connect-Retry Time : 120
    Local-AS Prepend     : No                 Alt. Local-AS      : 0
    BFD                  : Disabled
    Password             :
    Last Err Sent        : No Error
    Last SubErr Sent     : No Error
    Last Err Rcvd        : No Error
    Last SubErr Rcvd     : No Error

    Graceful-Restart     : Enabled            Gr. Restart Time   : 120
    Gr. Stalepath Time   : 300                Remove Private-AS  : No
    TTL                  : 255                Local Cluster-ID   :
    Weight               : 0                  Fall-over          : No
```

```
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      9        2
    Keepalives                  18       22
    Route Refresh                0        0
    Total                       28       25

    Capability                            Advertised      Received
    ----------------------------          -----------     ----------
    Route Refresh                         Yes             Yes
    Graceful Restart                      Yes             Yes
    Add-Path                              No              No
    Four Octet ASN                        Yes             Yes
    Address family IPv4 Unicast           No              No
    Address family IPv6 Unicast           No              No
    Address family VPNv4 Unicast          Yes             Yes
    Address family L2VPN EVPN             No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                   Send Community    : both
    Allow-AS in         : 0                    Advt. Interval    : 30
    Max. Prefix         : 300000               Soft Reconfig In  :
    Nexthop-Self        :                      Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

BGP Neighbor 2.2.2.2 (Internal)
    Description         : Towards PE2-SPOKE2
    Peer-group          :

    Remote Router Id    : 2.2.2.2            Local Router Id    : 3.3.3.3
    Remote AS           : 6488163            Local AS           : 6488163
    Remote Port         : 38293              Local Port         : 179
    State               : Established        Admin Status       : Up
    Conn. Established   : 1                  Conn. Dropped      : 0
    Passive             : No                 Update-Source      : loopback0
    Cfg. Hold Time      : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180                Neg. Keep Alive    : 60
    Up/Down Time        : 00h:19m:28s        Connect-Retry Time : 120
    Local-AS Prepend    : No                 Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled            Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300                Remove Private-AS  : No
    TTL                 : 255                Local Cluster-ID   :
    Weight              : 0                  Fall-over          : No
```

```
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      9        2
    Keepalives                  17       22
    Route Refresh                0        0
    Total                       27       25

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  :
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

PE3# show bgp vrf hub_import ipv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : hub_import

BGP Neighbor 20.2.1.2 (External)
    Description         : Towards CE4
    Peer-group          :

    Remote Router Id    : 20.4.1.1          Local Router Id    : 20.2.1.1
    Remote AS           : 102               Local AS           : 6488163
    Remote Port         : 48554             Local Port         : 179
    State               : Established       Admin Status       : Up
    Conn. Established   : 1                 Conn. Dropped      : 0
    Passive             : No                Update-Source      :
    Cfg. Hold Time      : 180               Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180               Neg. Keep Alive    : 60
    Up/Down Time        : 00h:22m:06s       Connect-Retry Time : 120
    Local-AS Prepend    : No                Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : Cease
    Last SubErr Sent    : Administrative Shutdown
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error
```

```
    Graceful-Restart    : Enabled            Gr. Restart Time    : 120
    Gr. Stalepath Time  : 300                Remove Private-AS   : No
    TTL                 : 1                  Local Cluster-ID    :
    Weight              : 0                  Fall-over           : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      3        8
    Keepalives                  26       22
    Route Refresh                0        0
    Total                       30       31

    Capability                           Advertised      Received
    ----------------------------         -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          Yes             Yes
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         No              No
    Address family L2VPN EVPN            No              No

    Address Family : IPv4 Unicast
    -----------------------------

    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 300000              Soft Reconfig In  : Yes
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       : Disable

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :

PE3# show bgp vrf hub_export ipv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : hub_export

BGP Neighbor 20.3.1.2 (External)
    Description         : Towards CE3
    Peer-group          :

    Remote Router Id    : 20.5.1.1           Local Router Id    : 20.3.1.1
    Remote AS           : 102                Local AS           : 6488163
    Remote Port         : 179                Local Port         : 52062
    State               : Established        Admin Status       : Up
    Conn. Established   : 1                  Conn. Dropped      : 0
    Passive             : No                 Update-Source      :
    Cfg. Hold Time      : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180                Neg. Keep Alive    : 60
    Up/Down Time        : 00h:21m:56s        Connect-Retry Time : 120
    Local-AS Prepend    : No                 Alt. Local-AS      : 0
    BFD                 : Disabled
```

```
    Password            :
    Last Err Sent       : Cease
    Last SubErr Sent    : Administrative Shutdown
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled           Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300               Remove Private-AS  : No
    TTL                 : 1                 Local Cluster-ID   :
    Weight              : 0                 Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      1        9
    Keepalives                  25       21
    Route Refresh                0        0
    Total                       27       31

    Capability                          Advertised      Received
    ----------------------------        -----------     ----------
    Route Refresh                       Yes             Yes
    Graceful Restart                    Yes             Yes
    Add-Path                            No              No
    Four Octet ASN                      Yes             Yes
    Address family IPv4 Unicast         Yes             Yes
    Address family IPv6 Unicast         No              No
    Address family VPNv4 Unicast        No              No
    Address family L2VPN EVPN           No              No

    Address Family : IPv4 Unicast
    -----------------------------

    Rt. Reflect. Client : No                 Send Community    : both
    Allow-AS in         : 0                  Advt. Interval    : 30
    Max. Prefix         : 300000             Soft Reconfig In  : Yes
    Nexthop-Self        :                    Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       : Disable

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :


PE3# show bgp vrf hub_import ipv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : hub_import
Local Router-ID 20.2.1.1

    Network            Nexthop         Metric     LocPrf     Weight Path
    Route Distinguisher: 0:1
*>e 0.0.0.0/0          20.2.1.2        0          100        0       102 ?
*>e 20.4.1.0/24        20.2.1.2        0          100        0       102 ?
*>e 20.5.1.0/24        20.2.1.2        0          100        0       102 ?
*>e 20.6.1.0/24        20.2.1.2        0          100        0       102 ?
```

```
*>  101.0.0.0/24       1.1.1.1         0          100        0       100 i
*>e 151.0.0.0/24       20.2.1.2        0          100        0       102 ?
*>  201.0.0.0/24       2.2.2.2         0          100        0       101 i
Total number of entries 7

PE3# show bgp vrf hub_export ipv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : hub_export
Local Router-ID 20.3.1.1

    Network            Nexthop         Metric     LocPrf     Weight Path
    Route Distinguisher: 1:0
*>e 0.0.0.0/0          20.3.1.2        0          100        0       102 ?
*>e 20.4.1.0/24        20.3.1.2        0          100        0       102 ?
*>e 20.5.1.0/24        20.3.1.2        0          100        0       102 ?
*>e 20.6.1.0/24        20.3.1.2        0          100        0       102 ?
*>e 101.0.0.0/24       20.3.1.2        0          100        0       102 ?
*>e 151.0.0.0/24       20.3.1.2        0          100        0       102 ?
*>e 201.0.0.0/24       20.3.1.2        0          100        0       102 ?
Total number of entries 7


PE3# show bgp all-vrf all summary
VRF : default
BGP Summary
-----------
 Local AS              : 6488163      BGP Router Identifier  : 3.3.3.3
 Peers                 : 2            Log Neighbor Changes   : No
 Cfg. Hold Time        : 180          Cfg. Keep Alive        : 60
 Confederation Id      : 0

Address-family : IPv4 Unicast
-----------------------------

Address-family : IPv6 Unicast
-----------------------------

Address-family : VPNv4 Unicast
------------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 1.1.1.1         6488163     36      40      00h:29m:14s  Established   Up
 2.2.2.2         6488163     37      39      00h:29m:24s  Established   Up

Address-family : L2VPN EVPN
-----------------------------

VRF : hub_export
BGP Summary
-----------
 Local AS              : 6488163      BGP Router Identifier  : 20.3.1.1
 Peers                 : 1            Log Neighbor Changes   : No
 Cfg. Hold Time        : 180          Cfg. Keep Alive        : 60
 Confederation Id      : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
```

```
 20.3.1.2        102         39      35      00h:28m:48s  Established   Up

Address-family : IPv6 Unicast
-----------------------------

VRF : hub_import
BGP Summary
-----------
 Local AS               : 6488163      BGP Router Identifier  : 20.2.1.1
 Peers                  : 1            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

Address-family : IPv4 Unicast
-----------------------------
 Neighbor        Remote-AS   MsgRcvd MsgSent Up/Down Time State
AdminStatus
 20.2.1.2        102         39      38      00h:29m:07s  Established   Up

Address-family : IPv6 Unicast
-----------------------------


PE3# show ip route all-vrfs

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
----------------------------------------------------------------------------------
------------
1.1.1.1/32          30.0.3.1         lag99        -               O
[110/150]    00h:31m:22s
2.2.2.2/32          30.0.3.1         lag99        -               O
[110/150]    00h:31m:22s
3.3.3.3/32          -                loopback0    -               L
[0/0]        -
4.4.4.4/32          30.0.3.1         lag99        -               O
[110/50]     00h:31m:22s
30.0.1.0/24         30.0.3.1         lag99        -               O
[110/150]    00h:31m:22s
30.0.2.0/24         30.0.3.1         lag99        -               O
[110/150]    00h:31m:22s
30.0.3.0/24         -                lag99        -               C
[0/0]        -
30.0.3.2/32         -                lag99        -               L
[0/0]        -

VRF: hub_export

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
```

```
--------------------------------------------------------------------------------
------------
0.0.0.0/0           20.3.1.2         vlan2031      -                 B/E
[20/0]        00h:23m:22s
20.3.1.0/24         -                vlan2031      -                 C
[0/0]         -
20.3.1.1/32         -                vlan2031      -                 L
[0/0]         -
20.4.1.0/24         20.3.1.2         vlan2031      -                 B/E
[20/0]        00h:29m:55s
20.5.1.0/24         20.3.1.2         vlan2031      -                 B/E
[20/0]        00h:30m:37s
20.6.1.0/24         20.3.1.2         vlan2031      -                 B/E
[20/0]        00h:24m:12s
101.0.0.0/24        20.3.1.2         vlan2031      -                 B/E
[20/0]        00h:28m:22s
151.0.0.0/24        20.3.1.2         vlan2031      -                 B/E
[20/0]        00h:23m:22s
201.0.0.0/24        20.3.1.2         vlan2031      -                 B/E
[20/0]        00h:28m:22s

VRF: hub_import

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/     Age
                                                                   Type      Metric
--------------------------------------------------------------------------------
------------
0.0.0.0/0           20.2.1.2         vlan2021      -                 B/E
[20/0]        00h:23m:22s
20.2.1.0/24         -                vlan2021      -                 C
[0/0]         -
20.2.1.1/32         -                vlan2021      -                 L
[0/0]         -
20.4.1.0/24         20.2.1.2         vlan2021      -                 B/E
[20/0]        00h:30m:57s
20.5.1.0/24         20.2.1.2         vlan2021      -                 B/E
[20/0]        00h:29m:55s
20.6.1.0/24         20.2.1.2         vlan2021      -                 B/E
[20/0]        00h:24m:12s
101.0.0.0/24        1.1.1.1          lag99         default           B/V
[200/0]       00h:28m:23s
151.0.0.0/24        20.2.1.2         vlan2021      -                 B/E
[20/0]        00h:23m:22s
201.0.0.0/24        2.2.2.2          lag99         default           B/V
[200/0]       00h:28m:23s

Total Route Count : 26
```

bind ipv4 (lsp label imposition)

```
bind ipv4 {<IP-ADDR>/<MASK> | <IP-ADDR> <MASK>} output <IFNAME> <IP-ADDR> <OUT-LABEL>
no bind ipv4 {<IP-ADDR>/<MASK> | <IP-ADDR><MASK>} output <IFNAME><IP-ADDR><OUT-LABEL>
```

Description

Performs LSP label imposition by adding label to an ingress packet (push operation).

The `no` form of this command removes the ingress packet label.

Parameter	Description
`ipv4 <IP-ADDR>/<MASK>`	Specifies the IPv4 destination in x.x.x.x format, where x is a decimal value from 0 to 255 and the number of bits in an IPv4 address mask in CIDR format (x), where x is a decimal number from 0 to 32.
`ipv4 <IP-ADDR> <MASK>`	Specifies the IPv4 destination in x.x.x.x format, where x is a decimal value from 0 to 255 and the destination IP subnet mask in x.x.x.x format, where x is a decimal value from 0 to 255.
`<IFNAME>`	Specifies the egress interface of the binding.
`<IP-ADDR>`	Specifies he next hop IP address of the binding.
`<OUT-LABEL>`	pecifies the MPLS label to apply. Range: 16-1048575.

Usage

- The `no` form of both the `mpls` and `static-lsp` commands deletes all static LSP bindings.
- The static LSP label range must be allocated before configuring static LSP bindings.
- Specifying an outgoing label outside the range of 16-1048575 is not allowed. An outgoing label is not bound by allocated static LSP label range.
- Types of valid egress interfaces are: System, LAG, VLAN, and Tunnel.
  - Routing must be enabled for egress interfaces.
  - Interfaces must be configured before performing the bind command.
  - LAG member interfaces are not allowed as egress interfaces.

Examples

Configuring binding:

```
switch(config-mpls-static-lsp)# bind ipv4 2.2.2.0/24 output 1/1/1 20.0.0.2 20
```

Unconfiguring binding:

```
switch(config-mpls-static-lsp)# no bind ipv4 2.2.2.0/24 output 1/1/1 20.0.0.2 20
```

Configuring binding with an invalid egress interface:

```
sswitch(config-mpls-static-lsp)# bind ipv4 2.2.2.0/24 output 1/1/1 20.0.0.2 20
 The output must be a layer 3 interface with routing enabled.
```

Configuring binding with an interface that does not have an IP address assigned:

```
switch(config-mpls-static-lsp)# bind ipv4 2.2.2.0/24 output 1/1/1 20.0.0.2 20
 The egress interface must have an IP address assigned.
```

Configuring binding with a next hop IP that is not in the same subnet as egress interface:

```
switch(config-if)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# ip address 10.0.0.1/24
switch(config-if)# mpls enable
switch(config-if)# mpls
switch(config-mpls)# static-lsp
switch(config-mpls-static-lsp)# bind ipv4 2.2.2.0/24 output 1/1/1 60.0.0.20 40
 The next hop IP address must be in the same subnet as interface 1/1/1.
```

Configuring binding with a next hop IP that is the same as the egress interface IP:

```
switch(config-if)# int 1/1/1
switch(config-if)# no shutdown
switch(config-if)# ip address 10.0.0.1/24
switch(config-if)# mpls enable
switch(config)# mpls
switch(config-mpls)# static-lsp
switch(config-mpls-static-lsp)# bind ipv4 2.2.2.0/24 output 1/1/1 10.0.0.1  40
 The next hop IP address cannot be the same as any interface 1/1/1 primary or
secondary addresses.
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-mpls-static-lsp`	Administrators or local user group members with execution rights for this command.

bind ipv4 input (static lsp binding)

```
bind ipv4 input <in-label>
no bind ipv4 input <in-label>
```

Description

Performs label disposition by removing label from an egress packet (pop operation).

The `no` form of this command removes the static LSP binding configuration.

Parameter	Description
<in-label>	Specifies the MPLS label to bind. Range: 16-1048575.

Usage

- The `no` form of both the `mpls` and `static-lsp` commands deletes all MPLS binding configurations.
- The static LSP label range must be allocated before configuring static LSP bindings.
- Specifying an incoming label outside the range of 16-1048575 is not allowed. An incoming label is bound by the allocated static LSP label range.

Examples

Configuring static LSP binding for label disposition:

```
switch(config)# mpls
switch(config-mpls)# static-lsp
switch(config-mpls-static-lsp)# bind ipv4 input 20
```

Removing the configuration for static LSP binding:

```
switch(config)# mpls
switch(config-mpls)# static-lsp
switch(config-mpls-static-lsp)# no bind ipv4 input 20
```

Configuring static LSP binding outside the label range:

```
switch(config-mpls-static-lsp)# bind ipv4 input 200
 The input label must be within the range specified by label-range.
```

Configuring static LSP binding without first allocating a label range:

```
switch(config-mpls-static-lsp)# bind ipv4 input 20
 A label range must be allocated before configuring bindings.
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-hpe_anw-central`	Administrators or local user group members with execution rights for this command.

clear mpls statistics

```
clear mpls statistics {ingress | egress} <LABEL>
no syntax
```

Description

Clears MPLS statistics per label for all sessions.

Parameter	Description
`ingress`	Selects ingress statistics.
egress	Selects egress statistics.
<LABEL>	Specifies the label for which statistics will be cleared.

Examples

Clearing ingress MPLS statistics for a specific label:

```
switch# clear mpls statistics ingress 20
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

crossconnect input (static lsp binding label swap)

```
crossconnect input <in-label> output <IFNAME> <ID-ADDR> {<out-label> | explicit-null}
no crossconnect input <in-label> output <IFNAME> <ID-ADDR> {<out-label> | explicit-null}
```

Description

Configures a static LSP binding to swap labels and route to the given next hop.

The `no` form of this command removes the static LSP binding label swap configuration.

Parameter	Description
<in-label>	Specifies the MPLS label to bind. Range: 16-1048575.
<IFNAME>	Specifies the egress interface of the binding.
<IP-ADDR>	Specifies the next hop IP address of the binding.
<out-label>	Specifies the MPLS label to apply. Range: 16-1048575.
explicit-null	Specifies an IETF MPLS IPv4 explicit null label (0).

Usage

- A static LSP label range must be allocated before configuring static LSP bindings.
- An incoming label must be within the allocated static LSP label range. Outgoing labels are not bound by the allocated static LSP label range, but must still be within the range of 16-1048575.
- The types of valid outgoing interfaces are: System, LAG, VLAN, and Tunnel.
  - Routing must be enabled for egress interfaces.
  - LAG member interfaces cannot be used with this command.
- Next hop and outgoing label pairs must be unique for each crossconnect binding.

Examples

Configuring crossconnect with an incoming and outgoing label:

```
switch(config)# mpls
switch(config-mpls)# static-lsp
switch(config-mpls-static-lsp)# crossconnect input 20 output 1/1/2 11.0.3.2 21
```

Configuring explicit-null PHP:

```
switch(config-mpls-static-lsp)# crossconnect input 20 output 1/1/2 11.0.3.2
explicit-null
```

Removing crossconnect binding:

```
switch(config-mpls-static-lsp)# no crossconnect input 20 output 1/1/2 11.0.3.2 21
```

Configuring crossconnect with an incoming label outside the allocated range:

```
switch(config-mpls-static-lsp)# crossconnect input 20 output 11.0.3.2 99
 Failed to configure static LSP binding. Incoming label not in range allocated for
static LSP.
```

Configuring crossconnect with an interface that does not have routing enabled:

```
switch(config-mpls-static-lsp)# crossconnect input 20 output 1/1/8 11.0.3.2 21
The egress interface must have an IP address assigned.
```

Configuring crossconnect with a nexthop IP that is not in the same subnet as egress interface:

```
switch(config-if)# int 1/1/1
switch(config-if)# no shutdown
switch(config-if)# ip address 10.0.0.1/24
switch(config-if)# mpls enable
switch(config)# mpls
switch(config-mpls)# static-lsp
switch(config-mpls-static-lsp)# crossconnect input 35 output 1/1/1 60.0.0.20 40
The next hop IP address must be in the same subnet as interface 1/1/1.
```

Configuring crossconnect with a nexthop IP that is the same as the egress interface IP:

```
switch(config-if)# int 1/1/1
switch(config-if)# no shutdown
switch(config-if)# ip address 10.0.0.1/24
switch(config-if)# mpls enable
switch(config)# mpls
switch(config-mpls)# static-lsp
switch(config-mpls-static-lsp)# crossconnect input 35 output 1/1/1 10.0.0.1  40
The next hop IP address cannot be the same as any interface 1/1/1 primary or
secondary addresses.
```

Configuring crossconect with a nexthop IP and outgoing label of an already existing binding:

```
switch(config-if)# int 1/1/1
switch(config-if)# no shutdown
switch(config-if)# ip address 10.0.0.1/24
switch(config-if)# mpls enable
switch(config)# mpls
switch(config-mpls)# static-lsp
switch(config-mpls-static-lsp)# crossconnect input 35 output 1/1/1 10.0.0.2 40
switch(config-mpls-static-lsp)# crossconnect input 36 output 1/1/1 10.0.0.2 40
 A static LSP binding with the same nexthop and outgoing label already exists.
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-mpls-static-lsp`	Administrators or local user group members with execution rights for this command.

enable (mpls globally)

```
enable
no enable
```

Description

Enables MPLS forwarding of IPv4 traffic globally.

The `no` form of this command disables MPLS forwarding of IPv4 traffic globally.

Examples

Enabling MPLS forwarding of IPv4 traffic:

```
switch(config)# mpls
switch(config-mpls)# enable
```

Disabling MPLS forwarding of IPv4 traffic:

```
switch(config)# mpls
switch(config-mpls)# no enable
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-mpls`	Administrators or local user group members with execution rights for this command.

enable (mpls ldp)

```
enable
no enable
```

Description

Enables MPLS LDP.

The `no` form of this command disable MPLS LDP.

Usage

- The LDP back off timer cannot be configured. It is set to exponentially back off session retry attempts with initial value of 15 seconds and a maximum of 2 minutes.

Examples

Enabling MPLS LDP:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# enable
```

Disabling MPLS LDP:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# no enable
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-mpls-ldp`	Administrators or local user group members with execution rights for this command.

enable (mpls static lsp)

```
enable
no enable
```

Description

Enables MPLS static LSPs.

The `no` form of this command disables static LSPs.

Usage

A static LSP binding will be processed when MPLS is globally enabled, static LSP is enabled, and the ingress and egress interface has MPLS enabled.

Examples

Enabling MPLS static LSPs:

```
switch(config)# mpls
switch(config-mpls)# enable
switch(config-mpls)# static-lsp
switch(config-mpls-static-lsp)# enable
```

Disabling static LSPs:

```
switch(config)# mpls
switch(config-mpls)# static-lsp
switch(config-mpls-static-lsp)# no enable
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced.

Command Information

Platforms	Command context	Authority

6400
8360
	`config`
	`config-mpls`
	`config-mpls-static-lsp`
		Administrators or local user group members with execution rights for this command.

enable mpls (interface)

```
mpls enable
no mpls enable
```

Description

Enables MPLS forwarding of IP traffic for the interface.

The `no` form of this command disables MPLS forwarding of IP traffic for the interface.

Usage

- Routing must be configured before enabling MPLS on an interface.

Examples

Enabling MPLS forwarding:

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# mpls enable
```

Enabling MPLS on a layer 2 interface:

```
switch(config)# interface 1/1/2
switch(config-if)# mpls enable
Routing must be enabled on this interface to use MPLS
```

Disabling MPLS forwarding:

```
switch(config)# interface 1/1/2
switch(config-if)# no mpls enable
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-if`	Administrators or local user group members with execution rights for this command.

graceful-restart (mpls ldp)

`graceful-restart`

Description

Enables LDP graceful restart. Graceful restart is enabled by default. With graceful restart enabled, the MPLS forwarding state will be temporarily retained if the control plane restarts. The switch will wait after losing LDP neighbors before deleting bindings from that neighbor. See graceful-restart-timers (mpls ldp) for details. Graceful restart is enabled for LDP sessions only when the LDP setting and the overall router setting are enabled. If either is disabled, then graceful restart will not occur for LDP sessions.

Upon being disabled or enabled, any LDP sessions will be restarted, which may result in temporary traffic loss.

The `no` form of this command disables LDP graceful restart

Examples

Enabling MPLS LDP graceful restart:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# graceful-restart
Enabling graceful restart will restart any LDP sessions.
This may result in traffic loss.

Continue (y/n)? y
```

Disabling MPLS LDP graceful restart:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# no graceful-restart
Enabling graceful restart will restart any LDP sessions.
This may result in traffic loss.

Continue (y/n)? y
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Command introduced.

Command Information

Platforms	Command context	Authority
6400 8360	`config-mpls-ldp`	Administrators or local user group members with execution rights for this command.

graceful-restart-timers (mpls ldp)

```
graceful-restart-timers {forwarding-holding <SECONDS> | max-recovery <SECONDS> |
neighbor-liveness <SECONDS>}
no graceful-restart-timers {forwarding-holding <SECONDS> | max-recovery <SECONDS> |
neighbor-liveness <SECONDS>}
```

Description

Configures MPLS LDP discovery hold time for peers found via hello packets.

The `no` form of this command resets the discovery hello hold time to its default value of 15 seconds.

The BGP restart timer must be configured as 180 seconds or higher for graceful restart to work with MPLS.

It is recommended to configure the OSPF graceful restart timer as lower than the LDP forward-holding timer, which in turn should be configured as lower than the BGP graceful restart timer.

Parameter	Description
`forwarding-holding <SECONDS>`	Specifies the amount of time in seconds that the MPLS forwarding state should be preserved after the control plane restarts. Range: 30-600. Default: 150.
`max-recovery <SECONDS>`	Specifies the amount of time in seconds that the stale label bindings should be kept on the router after the LDP session has been reestablished. Range: 15-600. Default: 120.
`neighbor-liveness <SECONDS>`	Specifies the amount of time in seconds that the router will wait for the LDP session to be reestablished. If the router cannot reestablish the LDP session within that time, the router deletes all the stale LDP bindings received from that LDP neighbor. Range: 5-300. Default: 120.

Examples

Configuring the MPLS LDP graceful restart forwarding holding time for 30 seconds:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# graceful-restart-timers forwarding-holding 30
Changing the timer value will restart any LDP sessions.
This may result in traffic loss.
Continue (y/n)? y
```

Resetting the MPLS LDP graceful restart forwarding holding time to default:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# no graceful-restart-timers forwarding-holding
switch(config-mpls-ldp)# no graceful-restart-timers forwarding-holding 30
Changing the timer value will restart any LDP sessions.
This may result in traffic loss.
Continue (y/n)? y
```

Configuring the MPLS LDP graceful restart max recovery time for 30 seconds:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# graceful-restart-timers max-recovery 30
Changing the timer value will restart any LDP sessions.
This may result in traffic loss.

Continue (y/n)? y
```

Resetting the MPLS LDP graceful restart max recovery time to default:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# no graceful-restart-timers max-recovery
switch(config-mpls-ldp)# no graceful-restart-timers max-recovery 30
Changing the timer value will restart any LDP sessions.
This may result in traffic loss.

Continue (y/n)? y
```

Configuring the MPLS LDP graceful restart neighbor liveness timefor 30 seconds:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# graceful-restart-timers neighbor-liveness 30
Changing the timer value will restart any LDP sessions.
This may result in traffic loss.

Continue (y/n)? y
```

Resetting the MPLS LDP graceful restart neighbor liveness to default:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# no graceful-restart-timers neighbor-liveness
Changing the timer value will restart any LDP sessions.
This may result in traffic loss.

Continue (y/n)? y
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
6400 8360	`config-mpls-ldp`	Administrators or local user group members with execution rights for this command.

label-range (static lsp)

```
label-range <start-label-range> <end-label-range>
no label-range <start-label-range> <end-label-range>
```

Description

Allocates MPLS labels for use exclusively by static LSP.

The `no` form of this command removes the configured allocation, returning to the default state with no labels allocated for static LSP usage.

Parameter	Description
<start-label-range>	Selects the start of the static LSP label range. Range: 16-1048575.
<end-label-range>	Selects the end of the static LSP label range. Range: 16-1048575.

Usage

- The range arguments are inclusive. Configuring a range of 20-30 will allocate the labels 20, 21, ..., 29, 30.
- Static LSP labels must not overlap with labels used by any other protocol, i.e. LDP. This label range allocation command will fail if any labels are shared between protocols.
- Any change to the static LSP label allocation will fail if any static LSP bindings are configured. All bindings must be removed before the static LSP label range can be reallocated.
- Allocated label range affects only the ingress packets. Labels for the outgoing packets must be within the allocated label range of the next hop device.

Examples

Allocating a valid static LSP label range:

```
switch(config-mpls-static-lsp)# label-range 100 2000
```

Changing the static LSP label range while LSP bindings are configured:

```
switch(config-mpls-static-lsp)# label-range 100 2000
All static LSP bindings must first be deleted.
```

Deallocating static LSP label range; use either command:

```
switch(config-mpls-static-lsp)# no label-range 100 2000
switch(config-mpls-static-lsp)# no label-range
```

Configuring a static LSP range that intersects with LDP:

```
switch(config-mpls-static-lsp)# label-range 30 99
 The static LSP label range cannot overlap with any other MPLS range.
```

Deallocating static LSP range when bindings are still configured:

```
switch(config-mpls-static-lsp)# no label-range
 All static LSP bindings must be removed before removing the label range.
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-mpls-static-lsp`	Administrators or local user group members with execution rights for this command.

label-protocol ldp

```
label-protocol ldp
no label-protocol ldp
```

Description

Configures the Label Distribution Protocol (LDP).

The `no` form of this command removes all LDP-related configuration.

Examples

Configuring LDP:

```
switch(config-mpls)# label-protocol ldp
```

Removing all LDP-related configuration:

```
switch(config-mpls)# no label-protocol ldp
All MPLS LDP configuration will be deleted.

Continue (y/n)? y
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-mpls`	Administrators or local user group members with execution rights for this command.

mpls

```
mpls
no mpls
```

Description

Configures MPLS forwarding of IPv4 traffic globally.

The `no` form of the command removes all MPLS-related configuration.

Examples

Configuring MPLS forwarding for IPv4 traffic:

```
switch(config)# mpls
```

Removing MPLS configuration for IPv4 traffic:

```
switch(config)# no mpls
All MPLS configuration will be deleted.

Continue (y/n)? y
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config`	Administrators or local user group members with execution rights for this command.

mpls ldp enable

```
mpls ldp enable
no mpls ldp enable
```

Description

Enables LDP protocol in the interface level.

The `no` form of this command disables LDP.

Enabling/disabling interface level LDP will also enable/disable `php-mode-explicit-null` by default. `php-mode-explicit-null` is currently the only mode supported and there is no option to disable it when LDP is enabled on an interface.

Usage

- Routing must be configured before enabling LDP on an interface.
- MPLS must be enabled on the interface prior to enabling LDP.

Examples

Enabling the LDP protocol:

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# mpls enable
switch(config-if)# mpls ldp enable
```

Enabling LDP prior to enabling MPLS:

```
switch(config)# interface 1/1/2
switch(config-if)# routing
switch(config-if)# mpls ldp enable
MPLS must be enabled on this interface to use LDP.
```

Enabling MPLS on a layer 2 interface:

```
switch(config)# interface 1/1/2
switch(config-if)# mpls ldp enable
Routing must be enabled on this interface to use MPLS.
```

Disabling MPLS forwarding:

```
switch(config)# interface 1/1/2
switch(config-if)# no mpls ldp enable
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-if`	Administrators or local user group members with execution rights for this command.

mpls ldp entropy label insertion

```
entropy label
no entropy label
```

Description

Enables MPLS LDP entropy label insertion.

The `no` form of this command disables MPLS LDP entropy label insertion. Label insertion happens at the ingress LER where native packets are encapsulated with MPLS labels for forwarding. Entropy label insertion is disabled by default.

Examples

Enabling MPLS LDP entropy label insertion:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# entropy-label
```

Disabling MPLS LDP entropy label insertion:

```
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# no entropy-label
```

Command History

Release	Modification
10.09	Command introduced

Command Information

Platforms	Command context	Authority
8360	`config-mpls-ldp`	Administrators or local user group members with execution rights for this command.

mpls ldp discovery hello hold time (global)

```
discovery hello holdtime <SECONDS>
no discovery hello holdtime <SECONDS>
```

Description

Configures MPLS LDP discovery hold time for peers found via hello packets.

The `no` form of this command resets the discovery hello hold time to its default value of 15 seconds.

Parameter	Description
<SECONDS>	Specifies the discovery hold time in seconds. Range: 15-65535. Default: 15.

Usage

- The default value of discovery hello hold time is 15 seconds
- The discovery hello hold time configured on an interface supersedes the global configuration.
- The discovery hello interval time is auto-computed as one third of the hello hold time.

Examples

Configuring the MPLS LDP discovery hello hold time:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# discovery hello holdtime 30
```

Changing discovery hello hold time:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# discovery hello holdtime 50
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-mpls-ldp`	Administrators or local user group members with execution rights for this command.

mpls ldp discovery hello hold time (interface)

```
mpls ldp discovery hello holdtime <SECONDS>
no mpls ldp discovery hello holdtime <SECONDS>
```

Description

Overrides the global MPLS LDP discovery hold time for peers found via hello packets from the given interface.

The `no` form of this command resets the discovery hello hold time for the given interface to the global value (if configured) or default value of 15 seconds if global value is not specified.

Parameter	Description
<SECONDS>	Specifies the discover hello hold time on an interface. Range: 15-65535. Default: 15.

Usage

- The interface LDP discovery hello hold time overrides global hello hold time.
- Routing must be configured before changing the LDP discovery hold time on an interface.

Examples

Configuring the interface MPLS LDP discovery hello hold time:

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# mpls ldp discovery hello holdtime 30
```

Removing the interface MPLS LDP discovery hello hold time configuration:

```
switch(config)# interface 1/1/1
switch(config-if)# no mpls ldp discovery hello holdtime
```

Configuring the interface MPLS LDP discovery hello hold time on a Layer 2 interface:

```
switch(config)# interface 1/1/2
switch(config-if)# mpls ldp discovery hello holdtime 30
Routing must be enabled on this interface to use MPLS.
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-if`	Administrators or local user group members with execution rights for this command.

mpls ldp session holdtime (interface)

```
mpls ldp session holdtime <TIME>
no mpls ldp session holdtime <TIME>
```

Description

Configures MPLS LDP session hold time for an interface.

The `no` form of this command resets the session hold time to its default value of 15 seconds.

Parameter	Description
<TIME>	Specifies the session hold time for the interface in seconds. Range: 15-65535. Default: 40.

Usage

- The interface LDP session hold time overrides global hello hold time.
- Routing must be configured before changing LDP session holdtime on an interface.

Examples

Configuring the MPLS LDP session hold time for an interface:

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# mpls ldp session holdtime 30
```

Removing the MPLS LDP session hold time for the interface:

```
switch(config)# interface 1/1/1
switch(config-if)# no mpls ldp session holdtime 30
```

Configuring the MPLS LDP session hold time on a layer 2 interface:

```
switch(config)# interface 1/1/2
switch(config-if)# mpls ldp session holdtime 30
Routing must be enabled on this interface to use MPLS.
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-if`	Administrators or local user group members with execution rights for this command.

ping mpls

```
ping mpls ipv4 <IP-ADDR/MASK> [source <IP-ADDR> | destination <IP-ADDR> | ttl <HOPS> |
size <BYTES> | repeat <NUMBER> | timeout <TIME> | interval <TIME>]
```

Description

Ping MPLS is a command which sends LSP ping packets on the MPLS network and displays the responses from the remote target. It is used as a debugging and analytics tool to verify connectivity within MPLS networks.

Parameter	Description
`ipv4 <IP-ADDR/MASK>`	Specifies target IP address and mask of the remote subnet to ping.
`source <IP-ADDR>`	Specifies the source IPv4 address for the request packet.
`destination <IP-ADDR>`	Specifies the destination address for the request packet. Default: 127.0.0.1.
`ttl <HOPS>`	Specifies the max number of hops a packet can take en route to its destination. Range: 1-255. Default: 64.
`size <BYTES>`	Specifies the size of the packet to be sent in bytes. Range: 0-9600. Default: 0.
`repeat <NUMBER>`	Specifies the number of packets to be sent. Range 1-10000. Default: 5.
`timeout <TIME>`	Specifies the amount of time in seconds after which a packet is considered dropped. Range 1-60. Default: 2.
`interval <TIME>`	Specifies the interval time between packets in seconds. Range: 1-60 seconds. Default: 1.

Examples

Sending 5 successful pings to the destination to the 10.10.10.10/32 subnet with a source IP address 20.20.20.1, a destination IP of 127.0.0.1, a zero byte payload, 64 hop time to live, 3 second interval between packets, and a 5 second timeout:

```
switch# ping mpls ipv4 10.10.10.10/32 source 20.20.20.1 destination 127.0.0.1
repeat 5 size 0 ttl 64 interval 3 timeout 5
Sending 5 MPLS Echo packets of size 0 bytes to 10.10.10.0/32 from source
20.20.20.1,
timeout is 5 sec, send interval is 3 sec:
Codes: '!' - success, 'Q' - request not sent, '.' - timeout,
       'U' - unreachable, 'M' - malformed request, 'T' - unsupported TLV,
       'E' - malformed response, 'R' - transit router
Type escape sequence (Ctrl + C) to abort.
!!!!!
1908 Success rate is 100 percent (5/5), round-trip min/avg/max = 7/10/13 ms
```

Sending an unsuccessful ping that fails because the network is unreachable:

```
switch# ping mpls ipv4 10.10.10.10/32 source 20.20.20.1 destination 127.0.0.1
repeat 5 size 0
Sending 5 MPLS Echo packets of size 0 bytes to 10.10.10.0/32 from source
20.20.20.1,
timeout is 2 sec, send interval is 1 sec:
Codes: '!' - success, 'Q' - request not sent, '.' - timeout,
       'U' - unreachable, 'M' - malformed request, 'T' - unsupported TLV,
       'E' - malformed response, 'R' - transit router
Type escape sequence (Ctrl + C) to abort.
Network unreachable
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.10	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	Manager (#)	Administrators or local user group members with execution rights for this command.

router-id (mpls ldp)

```
router-id <IFNAME> [confirm]
no router-id <IFNAME> [confirm]
```

Description

Configures MPLS LDP router ID which is the IP address of a loopback interface.

The `no` form of this command removes the MPLS LDP router ID configuration.

Parameter	Description
<IFNAME>	Specifies the loopback interface for the MPLS LDP router ID.

Usage

- There is a possibility of MPLS traffic disruption whenever a router ID is deleted or updated to another loopback interface.
- The MPLS router ID interface must be a loopback interface with an IPv4 address configured.
- The confirmation prompt is skipped if the router ID is being configured for the first time by the user.
- Changing the IP address of the loopback interface may interrupt MPLS traffic.

Examples

Configuring an MPLS LDP router ID:

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 1.1.1.1/32

switch(config)# mpls
switch(config-mpls)# enable
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# enable
switch(config-mpls-ldp)# router-id loopback1
```

Changing the MPLS LDP router ID loopback interface:

```
switch(config)# interface loopback 2
switch(config-loopback-if)# ip address 2.2.2.2/32

switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# router-id loopback2
Changing the router ID interface may disrupt MPLS traffic.

Continue (y/n)?
```

Changing the MPLS LDP router ID interface without prompting for confirmation:

```
switch(config)# interface loopback 2
switch(config-loopback-if)# ip address 2.2.2.2/24

switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# router-id loopback2 confirm
```

Removing the MPLS LDP router ID configuration:

```
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# no router-id loopback2
Removing the router ID interface may disrupt MPLS traffic.
```

```
Continue (y/n)?
```

Removing the MPLS LDP router ID without providing loopback interface name:

```
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# no router-id
Removing the router ID interface may disrupt MPLS traffic.

Continue (y/n)?
```

Removing the configuration of an MPLS LDP router ID with an interface name which is different than the one configured as router ID:

```
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# router-id loopback1 confirm
switch(config-mpls-ldp)# no router-id loopback2
The value to disable does not match the currently configured value.
```

Removing the MPLS LDP router ID configuration prompting for confirmation:

```
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# no router-id loopback2 confirm
```

Configuring an MPLS LDP router ID with system interface:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# enable
switch(config-mpls-ldp)# router-id 1/1/1
The router ID must be a loopback interface with an IP address assigned.
```

Configuring MPLS LDP router ID with a loopback interface without an IPv4 address configured:

```
switch(config)# interface loopback 1

switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# enable
switch(config-mpls-ldp)# router-id loopback1
The router ID interface must have an IP address assigned.
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-loopback-if` `config-mpls-ldp`	Administrators or local user group members with execution rights for this command.

session hold time (mpls ldp globally)

```
session holdtime <SECONDS>
no session holdtime <SECONDS>
```

Description

Configures MPLS LDP session hold time.

The `no` form of this command resets the session hold time to its default value of 40 seconds.

Parameter	Description
<SECONDS>	Specifies the session hold time in seconds. Range: 15-65535. Default: 40.

Usage

- The default session hold time is 40 seconds
- The session hold time configured on an interface supersedes the global configuration.
- The session keepalive interval time is auto computed as one sixth of the hold time.

Examples

Configuring the MPLS LDP session hold time:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(config-mpls-ldp)# session holdtime 30
```

Changing the session hold time:

```
switch(config)# mpls
switch(config-mpls)# label-protocol ldp
switch(switch(config-mpls-ldp)# session holdtime 50)# session holdtime 50
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-mpls-ldp`	Administrators or local user group members with execution rights for this command.

show bgp vpnv4 unicast

show bgp VPNv4 unicast [[*<IP-ADDR>/<MASK>*] | community | extcommunity |
neighbors [*<IP-ADDR>*] | paths | summary | vsx-peer]

Description

Shows all vpnv4 entries in the BGP routing table .

Parameter	Description
<IP-ADDR>/<MASK>	Specifies the IP network and mask of a specific BGP route in IPv4 format (x.x.x.x/M), where x is a decimal number from 0 to 255 and M is the number of bits in CIDR format from 0 to 32.
`community`	Selects routes that belong to specified BGP communities.
`extcommunity`	Selects unicast routes with extended communities.
neighbors [*<IP-ADDR>*]	Selects BGP neighbor connection parameters for all neighbors or the IP address of a specific neighbor in IPv4 format (x.x.x.x) where x is a decimal number from 0 to 255.
`paths`	Selects AS Path information of the vpnv4 routes in BGP RIB.
`summary`	Selects a summary of BGP neighbor status.
`vsx-peer`	Selects VSX peer switch information.

Examples

Showing all VPNv4 entries in the BGP routing table:

```
switch# show bgp vpnv4 unicast
VRF : default
BGP Summary
-----------
 Local AS               : 100          BGP Router Identifier  : 4.4.4.4
 Peers                  : 0            Log Neighbor Changes   : No
 Cfg. Hold Time         : 180          Cfg. Keep Alive        : 60
 Confederation Id       : 0

PE2# show bgp vpnv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 4.4.4.4
```

```
    Network             Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 100:100              (Label 22)
*>i 11.1.1.0/30        1.1.1.1         0          100        0       ?

Route Distinguisher: 1.1.1.1:200          (Label 23)
*>i 11.1.2.0/30        1.1.1.1         0          100        0       ?

Route Distinguisher: 100:300              (Label 24)
*>i 11.1.3.0/30        1.1.1.1         0          100        0       ?

Route Distinguisher: 100:400              (Label 25)
*>i 11.1.4.0/30        1.1.1.1         0          100        0       ?
Total number of entries 4
```

Showing entries in the BGP routing table for the *11.1.3.0/30* network:

```
switch# show bgp vpnv4 unicast 11.1.3.0/30

VRF : default
BGP Local AS 100         BGP Router-id 4.4.4.4

   Network            : 11.1.3.0/30              Nexthop             : 1.1.1.1
   Peer               : 1.1.1.1                  Origin              : incomplete
   Metric             : 0                        Local Pref          : 100
   Weight             : 0                        Calc. Local Pref    : 100
   Best               : Yes                      Valid               : Yes
   Type               : internal                 Stale               : No
   Originator ID      : 0.0.0.0                  Path ID             : 0
   Aggregator ID      :
   Aggregator AS      :
   Atomic Aggregate   :

   AS-Path            :
   Cluster List       :
   Communities        :
   Ext-Communities    :
```

Showing entries in the BGP routing table for routes with extended communities:

```
switch# show bgp vpnv4 unicast extcommunity
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 4.4.4.4

    Network             Next Hop           Ecommunity
*>i 11.1.1.0/30         1.1.1.1            100:100

*>i 11.1.2.0/30         1.1.1.1            4.4.4.4:200

*>i 11.1.3.0/30         1.1.1.1            100:300

*>i 11.1.4.0/30         1.1.1.1            100:400

Total number of entries 4
```

Showing BGP neighbor connection parameters for all neighbors:

```
switch# show bgp vpnv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 1.1.1.1 (Internal)
    Description         : MPBGP Session to PE2
    Peer-group          :

    Remote Router Id    : 1.1.1.1            Local Router Id    : 4.4.4.4
    Remote AS           : 100                Local AS           : 100
    Remote Port         : 179                Local Port         : 38335
    State               : Established        Admin Status       : Up
    Conn. Established   : 1                  Conn. Dropped      : 0
    Passive             : No                 Update-Source      : loopback0
    Cfg. Hold Time      : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180                Neg. Keep Alive    : 60
    Up/Down Time        : 00h:56m:46s        Connect-Retry Time : 120
    Local-AS Prepend    : No                 Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled            Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300                Remove Private-AS  : No
    TTL                 : 255                Local Cluster-ID   :
    Weight              : 0                  Fall-over          : No
    Confederation-Peers : No

    Message statistics       Sent     Rcvd
    -------------------     -----    -----
    Open                        1        1
    Notification                0        0
    Updates                     7        7
    Keepalives                 64       65
    Route Refresh               0        0
    Total                      72       73

    Capability                           Advertised      Received
    -----------                          -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No

    Address Family : VPNv4 Unicast
    ------------------------------

    Rt. Reflect. Client : No                  Send Community    : both
    Allow-AS in         : 0                   Advt. Interval    : 30
    Max. Prefix         : 32500               Soft Reconfig In  :
    Nexthop-Self        :                     Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :
```

```
    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :
```

Showing BGP neighbor connection parameters for the neighbor with IP address *1.1.1.1*:

```
switch# show bgp vpnv4 unicast neighbors 1.1.1.1
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 1.1.1.1 (Internal)
    Description         : MPBGP Session to PE2
    Peer-group          :

    Remote Router Id    : 1.1.1.1            Local Router Id    : 4.4.4.4
    Remote AS           : 100                Local AS           : 100
    Remote Port         : 179                Local Port         : 38335
    State               : Established        Admin Status       : Up
    Conn. Established   : 1                  Conn. Dropped      : 0
    Passive             : No                 Update-Source      : loopback0
    Cfg. Hold Time      : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time      : 180                Neg. Keep Alive    : 60
    Up/Down Time        : 00h:58m:52s        Connect-Retry Time : 120
    Local-AS Prepend    : No                 Alt. Local-AS      : 0
    BFD                 : Disabled
    Password            :
    Last Err Sent       : No Error
    Last SubErr Sent    : No Error
    Last Err Rcvd       : No Error
    Last SubErr Rcvd    : No Error

    Graceful-Restart    : Enabled            Gr. Restart Time   : 120
    Gr. Stalepath Time  : 300                Remove Private-AS  : No
    TTL                 : 255                Local Cluster-ID   :
    Weight              : 0                  Fall-over          : No
    Confederation-Peers : No

    Message statistics        Sent     Rcvd
    -------------------      -----    -----
    Open                         1        1
    Notification                 0        0
    Updates                      7        7
    Keepalives                  67       67
    Route Refresh                0        0
    Total                       75       75

    Capability                           Advertised      Received
    -----------                          -----------     ----------
    Route Refresh                        Yes             Yes
    Graceful Restart                     Yes             Yes
    Add-Path                             No              No
    Four Octet ASN                       Yes             Yes
    Address family IPv4 Unicast          No              No
    Address family IPv6 Unicast          No              No
    Address family VPNv4 Unicast         Yes             Yes
    Address family L2VPN EVPN            No              No
    Address Family : VPNv4 Unicast
```

```
    ------------------------------

    Rt. Reflect. Client : No                    Send Community    : both
    Allow-AS in         : 0                     Advt. Interval    : 30
    Max. Prefix         : 32500                 Soft Reconfig In  :
    Nexthop-Self        :                       Default-Originate :
    Cfg. Add-Path       :
    Neg. Add-Path       :

    Routemap In         :
    Routemap Out        :
    ORF type            : Prefix-list
    ORF capability      :
```

Showing AS Path information of the vpnv4 routes in BGP RIB:

```
switch# show bgp vpnv4 unicast paths
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
VRF : default
Local Router-ID 4.4.4.4

    Network            Next Hop        PathID      Path

Route Distinguisher: 100:100              (Label 22)
* i 11.1.1.0/30        1.1.1.1         0             ?

Route Distinguisher: 1.1.1.1:200          (Label 23)
* i 11.1.2.0/30        1.1.1.1         0             ?

Route Distinguisher: 100:300              (Label 24)
* i 11.1.3.0/30        1.1.1.1         0             ?

Route Distinguisher: 100:400              (Label 25)
* i 11.1.4.0/30        1.1.1.1         0             ?
Total number of entries 4
```

Showing a summary of BGP neighbor status:

```
switch(config-bgp)# show bgp vpnv4 unicast summary
VRF : default
BGP Summary
Local AS               : 100         BGP Router Identifier  : 4.4.4.4
Peers                  : 0           Log Neighbor Changes   : No
Cfg. Hold Time         : 180         Cfg. Keep Alive        : 60
Confederation Id       : 0
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show capacities mpls

```
show capacities mpls
show capacities-status mpls
```

Description

For capacities command, shows the maximum number of label endpoints, label switch entries, and service label entries that can be configured on the device. For capacities-status command, shows the total number of label endpoints, label switch entries, and service label entries that are currently configured on the device.

Examples

Showing capacities of configurable MPLS options:

```
switch# show capacities mpls
System Capacities: Filter MPLS
Capacities Name
                           Value
----------------------------------------------------------------------------------
---------------------------------
Maximum number of MPLS Label Endpoints configurable in a system
                            8192
Maximum number of MPLS Label Switch entries configurable in a system
                            8192
Maximum number of MPLS Service Label entries configurable in a system
                            8192
```

Showing the configuration of currently configured MPLS options in relation to their capacities:

```
switch# show capacities-status mpls
System Capacities Status: Filter MPLS
Capacities Status Name
                          Value Maximum
----------------------------------------------------------------------------------
-------
Number of MPLS Label Endpoints currently configured
                              0    8192
Number of MPLS Label Switch entries currently configured
                              0    8192
Number of MPLS Service Label entries currently configured
                              0    8192
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	Manager (#)	Administrators or local user group members with execution rights for this command.

show mpls forwarding

```
show mpls forwarding [detail]
```

Description

Shows the MPLS forwarding table.

Usage

- Forwarding table filters will be implemented at a later date.
- When running this command on a huge-scale setup, showing the full tables might take a while.

Examples

Showing the MPLS forwarding table:

```
switch# show mpls forwarding
MPLS Bindings
Entry Bindings   : 2
Exit Bindings    : 2
Transit Bindings : 1
PHP Mode         : Explicit-Null
QoS Mode         : Uniform
TTL Propagation  : Uniform
Entry Bindings:
Origin   Prefix             Ingress          Nexthop             Outgoing
Egress        Egress           Status
                            VRF              Address             Label
Interface     VRF
----------------------------------------------------------------------------------
---------------------------------------
LDP      4.4.4.4/32         default          192.168.10.2        3002      1/1/6
        default           operational
BGP      20.20.20.0/24      vrf-blue         4.4.4.4             5001      1/1/6
        default           operational
Exit Bindings:
Origin   Prefix             Incoming  Service   Egress             Status
Label     Label     VRF
-------------------------------------------------------------------------------
static   n/a                exp-null  -         default            operational
BGP      n/a                imp-null  2001      vrf-blue           operational
Transit Bindings:
Origin  Prefix       Incoming  Egress     Egress   Nexthop        Outgoing  Status
```

```
                     Label     Interface  VRF      Address        Label
-----------------------------------------------------------------------------------
-----
LDP     4.4.4.4/32   2002      1/1/6      default  192.168.10.2   3002
operational
```

```
switch# show mpls forwarding detail
MPLS Bindings
Entry Bindings   : 2
Exit Bindings    : 2
Transit Bindings : 1
PHP Mode         : Explicit-Null
QoS Mode         : Uniform
TTL Propagation  : Uniform
Entry Bindings:
Origin   Prefix             Ingress           Nexthop             Outgoing
Egress        Egress            Status                 Tx Packets             Tx
Bytes
VRF                Address             Label     Interface     VRF
-----------------------------------------------------------------------------------
----------------------------------------------------------------------------------
LDP      4.4.4.4/32         default           192.168.10.2        3002      1/1/6
        default           operational                  99                  100
BGP      20.20.20.0/24      vrf-blue          4.4.4.4             5001      1/1/6
        default           operational                  66                   88
Exit Bindings:
Origin   Prefix             Incoming  Service   Egress            Status
       Rx Packets             Rx Bytes
Label     Label     VRF
-----------------------------------------------------------------------------------
---------------------------------------
static   n/a                exp-null  -         default           operational
                33                    44
BGP      n/a                imp-null  2001      vrf-blue          operational
                22                   33
Transit Bindings:
Origin   Prefix             Incoming  Egress        Egress            Nexthop
        Outgoing  Status                Rx Packets            Rx Bytes
Tx Packets             Tx Bytes
Label     Interface     VRF               Address             Label
-----------------------------------------------------------------------------------
-----------------------------------------------------------------------------------
---------------------------------
LDP      4.4.4.4/32         2002      1/1/6         default
192.168.10.2        3002      operational                   11
22                   22                    33
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show mpls label-range static-lsp

```
show mpls label-range static-lsp
```

Description

Shows the range of MPLS labels allocated for use in static LSP bindings and the range of labels currently used by static LSP bindings.

Examples

Showing the range and usage of static LSp labels on the switch:

```
switch# show mpls label-range static-lsp

     Static LSP Labels

       Allocated : 16-100
       In use    : 16-30,35
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show mpls ldp bindings

```
show mpls ldp bindings
```

Description

Shows information about all MPLS LDP bindings.

Examples

Showing information about MPLS LDP bindings:

```
switch# show mpls ldp bindings

10.10.2.0/24
        local binding: label: imp-null
        remote binding: lsr:10.255.255.255:0, label:16
        remote binding: lsr:10.256.256.256:0, label: exp-null
10.10.3.0/24
        local binding: label:20
        remote binding: lsr:10.256.256.256:0, label:22
5.43.9.98/32
        local binding: label:21
        No remote binding
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show mpls ldp discovery

```
show mpls ldp discovery [<IP-ADDR>]
no syntax
```

Description

Shows information about discovered LDP peers.

Parameter	Description
<IP-ADDR>	Specifies the peer MPLS LDP router ID in x.x.x.x format, where x is a decimal value from 0 to 255.

Examples

Showing information about discovered LDP peers:

```
switch# show mpls ldp discovery
Local LDP Identifier: 10.44.44.44:0
Discovery Sources:
     Interfaces:
         1/1/1 : recv
         LDP Id: 10.33.33.33:0, Transport address: 10.33.33.33
                 Path vector limit: 10
                 Distribution type: Downstream-on-demand
                 Adjacency type: Link
                 Hold time: 15 sec (local: 15 sec, peer: 15 sec, remaining: 10s)
                 BFD status: Activating
         1/1/2 : recv
         LDP Id: 10.33.33.34:0, Transport address: 10.33.33.33
                 Path vector limit: 10
                 Distribution type: Downstream-unsolicited
                 Adjacency type: Targeted
                 Hold time: 15 sec (local: 15 sec, peer: 15 sec, remaining: 10s)
                 BFD status: Active

Local LDP Identifier: 10.44.44.44:2
Discovery Sources:
    Interfaces:
        1/1/3 : recv
        LDP Id: 10.33.38.33:0, Transport address: 10.43.33.33
                Path vector limit: 10
                Distribution type: Downstream-unsolicited
                Adjacency type: Link
                Hold time: 15 sec (local: 15 sec, peer: 15 sec, remaining: 10s)
                BFD status: Active
```

Showingu information about a specific LDP peer:

```
switch# show mpls ldp discovery 10.33.33.34
Local LDP Identifier: 10.44.44.44:0
Discovery Sources:
    Interfaces:
        1/1/2 : recv
        LDP Id: 10.33.33.34:0, Transport address: 10.33.33.33
                Path vector limit: 10
                Distribution type: Downstream-unsolicited
                Adjacency type: Targeted
                Hold time: 15 sec (local: 15 sec, peer: 15 sec, remaining: 10s)
                BFD status: Active
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show mpls ldp graceful-restart

`show mpls ldp graceful-restart`

Description

Shows graceful restart parameters and status.

Examples

Showing graceful restart parameters and status when graceful restart is not configured:

```
switch# show mpls ldp graceful-restart
Max recovery time                       : 50 sec
Neighbor liveness time                  : 50 sec
Forwarding holding time                 : 70 sec
Number of graceful restart events       : 7
Graceful restart in progress            : true
Forwarding holding time remaining       : 300 sec
Current graceful restart status         : in-progress
Graceful restart exit history (last 5) : complete, complete, complete, cancelled,
cancelled
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.10	Command introduced.

Command Information

Platforms	Command context	Authority
6400 8360	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show mpls ldp neighbor

`show mpls ldp neighbor`

Description

Shows information about LDP neighbors in the current session(s). The reconnect and recovery time are the times advertised by the peer device.

Examples

Showing LDP neighbors:

```
switch# show mpls ldp neighbor
Local LDP Identifier: 10.44.44.44:0, Peer LDP Identifier: 10.22.22.22:0
    TCP connection: 10.22.22.22:646 - 10.33.33.33:65530
    Graceful Restart: No
    Session Holdtime: 180 sec
    State: Operational; Msgs sent/rcvd: 46/43
    Up time: 00:31:21
    LDP Discovery Sources: 1/1/1
    Addresses bound to this peer:
         10.22.22.22 10.10.2.1
```

Showing LDP neighbors when graceful restart has been configured:

```
switch# show mpls ldp neighbor
Local LDP Identifier: 1.1.1.1:0, Peer LDP Identifier: 11.1.1.2:0
    Graceful Restart: Yes
    Peer Reconnect Time: 120 sec
    Peer Recovery Time: 300 sec
    Session Holdtime: 40 sec
    Up time: 00:02:59
    State: operational
    LDP Discovery Sources: 1/1/32
    Addresses bound to this peer:
         11.1.1.2 12.1.1.1 2.2.2.2
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

static-lsp

```
static-lsp
no static-lsp
```

Description

Configures MPLS static Label Switched Paths (LSP).

The `no` form of this command removes all static LSP configurations including label range allocation and static LSP binding.

Examples

Configuring MPLS static LSP:

```
switch(config-mpls)# static-lsp
```

Removing MPLS static LSP configuration:

```
switch(config-mpls)# no static-lsp
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.09	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`config-mpls`	Administrators or local user group members with execution rights for this command.

traceroute mpls

```
traceroute mpls ipv4 <IP-ADDR/MASK> [source <IP-ADDR> | destination <IP-ADDR> | ttl
<HOPS> | timeout <TIME> | fec-type ldp]
```

Description

Send LSP ping packets on the MPLS network and display the responses all intermediate routers as well as the destination host. Use this command as a debugging and analytics tool to verify connectivity within the MPLS networks.

Parameter	Description
`ipv4 <IP-ADDR/MASK>`	Specifies the IP address and netmask of the remote subnet to traceroute.
`source <IP-ADDR>`	Specifies the source IPv4 address for the request packet.

Parameter	Description
`destination <IP-ADDR>`	Specifies the destination IPv4 address for the request packet
`ttl <HOPS>`	Specifies the max number of hops a packet can take en route to its destination. Range: 1-255. Default: 255.
`timeout <SECONDS>`	Specifies the number of seconds after which a packet is considered dropped. Range: 1-60 seconds. Default: 2.
`fec-type ldp`	Selects the target Forward Equivalence Class (FEC) type. The only supported option is the default value of ldp.

Example

Successfully tracing the route a target with IP address 1.1.4.1/32 with a maximum TTL of 3 hops and a 3 second timeout:

```
switch# traceroute mpls ipv4 1.1.4.1/32 ttl 3 timeout 3
Tracing MPLS Label Switched Path to 1.1.4.1/32 from source 10.0.0.2,
timeout is 3 seconds and ttl is 3

Codes: '!' - success, 'Q' - request not sent, '.' - timeout, 'N' - no label entry,
  'R' - transit router, 'D' - DS Map mismatch, 'F' - no FEC mapping,
  'M' - malformed request, ‘T' - unsupported tlvs, ‘Z' - return code 0

Type escape sequence to abort.
  0 10.0.0.2 MRU 1500 [Labels: 17]
R 1 10.0.0.1 MRU 1500 [Labels: explicit-null] 10 ms
! 2 10.0.1.2 1 ms
```

For more information on features that use this command, refer to the MPLS Guide for your switch model.

Command History

Release	Modification
10.11	Support for the Aruba 6400 Series Switch added.
10.10	Command introduced

Command Information

Platforms	Command context	Authority
6400 8360	`Manager (#)`	Administrators or local user group members with execution rights for this command.

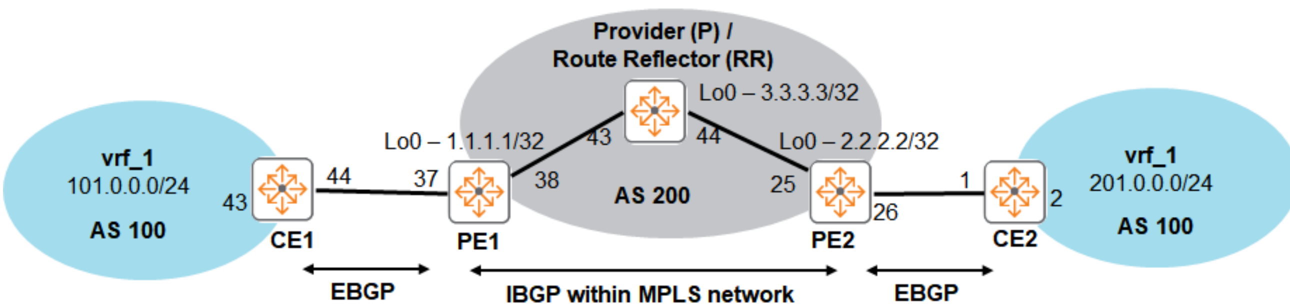
Chapter 8

Debugging and troubleshooting

The following is recommended prior to beginning the troubleshooting process:

- Have a topology diagram ready
- Ensure IPs, interface, and AS details are included
- Check physical cabling and generate `show tech` when opening a TAC case
- Check network:
    - Run **show LLDP neighbor**
    - Ensure unicast network works using ping and traceroute between loopbacks and interfaces
    - Fix any issues found

The figure below provides an example:



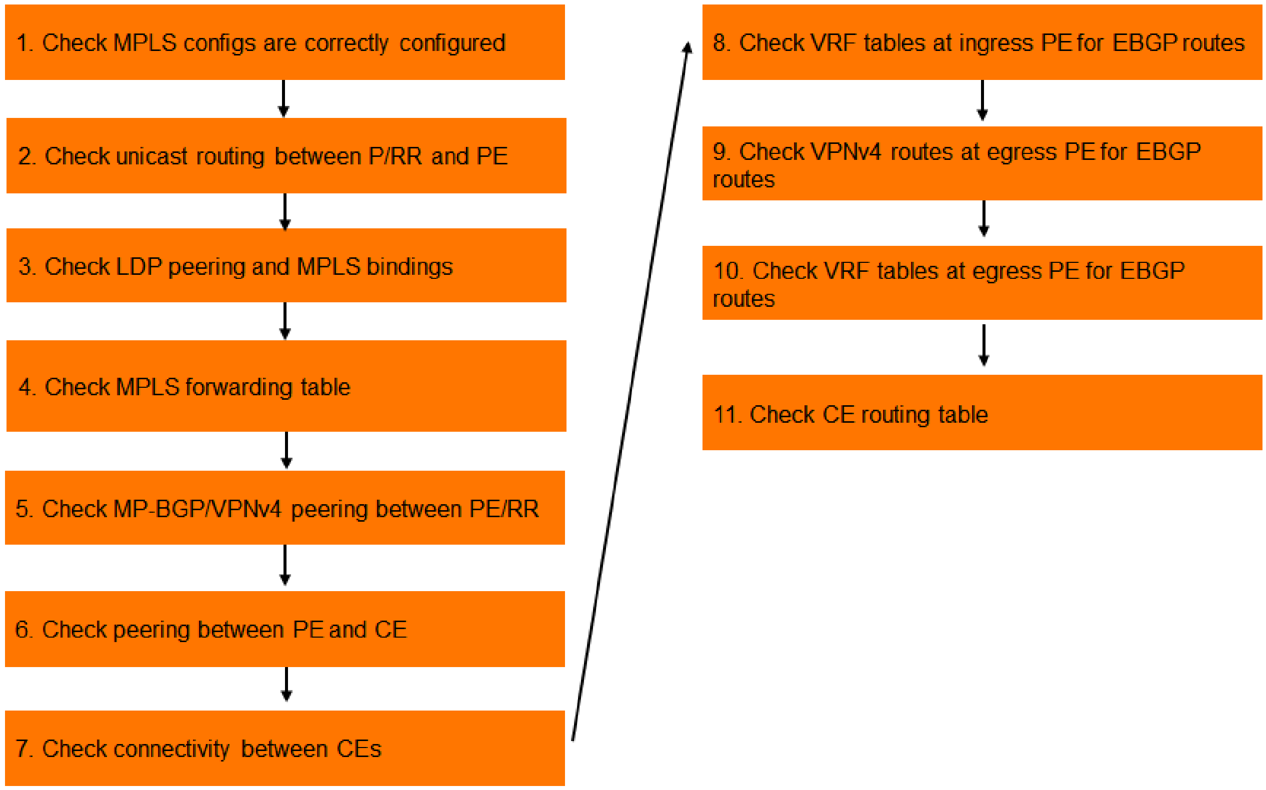
Mirror traffic and packet capture if required:

```
mirror session 1
    enable
    destination interface 1/1/40
    source interface 1/1/51 both
```

If requested by TAC engineer:

```
diagnostics
diag-dump mpls basic

debug mpls all
sh debug buffer
no debug all
```

The recommended MPLS troubleshooting flow is provided in the diagram below.



Troubleshooting steps

Follow the process outlines below to troubleshoot MPLS-related issues:

Step 1: Verify that MPLS configurations are correct

Refer to the Use cases section for CE, PE and P sample configurations.

Step 2: Check unicast routing between P/RR and PE

On PE, verify loopbacks from remote P and PE have been learned. If not seen, verify that OSPF is enabled and advertising loopbacks on remote P and PE:

```
PE1# show ip route

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix                Nexthop           Interface     VRF(egress)   Origin/
Distance/     Age
                                                                      Type      Metric
-----------------------------------------------------------------------------------
```

```
------------
1.1.1.1/32          -               loopback0      -             L         [0/0]
       -
2.2.2.2/32          30.0.0.2        1/1/38         -             O
[110/20]     00h:12m:32s
3.3.3.3/32          30.0.0.2        1/1/38         -             O
[110/10]     00h:12m:52s
30.0.0.0/24         -               1/1/38         -             C         [0/0]
       -
30.0.0.1/32         -               1/1/38         -             L         [0/0]
       -
30.1.0.0/24         30.0.0.2        1/1/38         -             O
[110/20]     00h:12m:52s

Total Route Count : 6
```

Step 3: Check LDP peering and MPLS bindings

Verify LDP neighbor is up:

```
PE1# show mpls ldp neighbor
Local LDP Identifier: 1.1.1.1:0, Peer LDP Identifier: 3.3.3.3:0
    Graceful Restart: No
    Session Holdtime: 40 sec
    Up time: 00:13:49
    State: operational
    LDP Discovery Sources: 1/1/38
    Addresses bound to this peer:
         3.3.3.3 30.0.0.2 30.1.0.1
```

Verify labels and MPLS bindings for remote P and PE exist:

```
PE1# show mpls ldp bindings
30.1.0.1/32
        No local binding
        remote binding: lsr:3.3.3.3:0 label: exp-null
1.1.1.1/32
        local binding: label: exp-null
        remote binding: lsr:3.3.3.3:0 label:16
30.0.0.1/32
        local binding: label: exp-null
        No remote binding
30.0.0.2/32
        No local binding
        remote binding: lsr:3.3.3.3:0 label: exp-null
3.3.3.3/32
        local binding: label: 17
        remote binding: lsr:3.3.3.3:0 label: exp-null
2.2.2.2/32
        local binding: label: 18
        remote binding: lsr:3.3.3.3:0 label:17
```

If the bolded examples are not seen, verify that remote PE has LDP established.

Step 4: Check MPLS forwarding table

Verify nexthop, labels, and routes exist for remote VPNv4 (outgoing label 18) and PE (outgoing label 17):

```
PE1# show mpls forwarding

MPLS Bindings

Entry Bindings   : 4
Exit Bindings    : 3
Transit Bindings : 2

Entry Bindings:
Origin  Prefix         Ingress    Nexthop    Outgoing  Egress      Egress    Status

                       VRF        Address    Label     Interface   VRF

---------------------------------------------------------------------------------
----
LDP   2.2.2.2/32    default   30.0.0.2   17        1/1/38      default
operational
LDP   3.3.3.3/32    default   30.0.0.2   exp-null  1/1/38      default
operational
BGP   20.2.0.0/24    vrf_1     2.2.2.2   18        1/1/38      default
operational
BGP   201.0.0.0/24   vrf_1     2.2.2.2   18        1/1/38      default
operational

Exit Bindings:

Origin   Prefix              Incoming  Service   Egress            Status
                             Label     Label     VRF
------------------------------------------------------------------------------
static   n/a                 exp-null  -         default           operational
BGP      n/a                 imp-null  16        vrf_1             operational
static   n/a                 7         -         default           operational
```

Step 5: Check MP-BGP VPNv4 peering between PE/RR

Verify that MP-BGP VPNv4 peering is established. If it's down, verify and fix network reachability between source and destination peering IPs.

```
PE1# show bgp vpnv4 un neighbors
Codes: ^ Inherited from peer-group

VRF : default

BGP Neighbor 3.3.3.3 (Internal)
    Description         :

    Peer-group          :

    Remote Router Id    : 3.3.3.3            Local Router Id    : 1.1.1.1
    Remote AS           : 200                Local AS           : 200
    Remote Port         : 43923              Local Port         : 179
    State               : Established        Admin Status       : Up
    Conn. Established   : 2                  Conn. Dropped      : 1
    Passive             : No                 Update-Source      : loopback0
```

Step 6: Check peering between PE and CE

Verify EBGP peering to CE on the PE

```
PE1# show bgp vrf vrf_1 ipv4 unicast neighbors
Codes: ^ Inherited from peer-group

VRF : vrf_1

BGP Neighbor 20.1.0.1 (External)
    Description          :
    Peer-group           :

    Remote Router Id     : 20.1.0.1           Local Router Id    : 20.1.0.2
    Remote AS            : 100                Local AS           : 200
    Remote Port          : 51588              Local Port         : 179
    State                : Established        Admin Status       : Up
    Conn. Established    : 1                  Conn. Dropped      : 0
    Passive              : No                 Update-Source      :
    Cfg. Hold Time       : 180                Cfg. Keep Alive    : 60
    Neg. Hold Time       : 180                Neg. Keep Alive    : 60
    Up/Down Time         : 00h:20m:28s        Connect-Retry Time : 120
    Local-AS Prepend     : No                 Alt. Local-AS      : 0
    BFD                  : Disabled
    Password             :
    Last Err Sent        : Cease
    Last SubErr Sent     : Administrative Shutdown
    Last Err Rcvd        : No Error
    Last SubErr Rcvd     : No Error
```

Step 7: Check connectivity between PEs and between CEs

Ping from CE to remote CE WAN and LAN to verify MPLS L3 VPN connectivity:

```
CE1# ping 20.2.0.2
PING 20.2.0.2 (20.2.0.2) 100(128) bytes of data.
108 bytes from 20.2.0.2: icmp_seq=1 ttl=61 time=0.307 ms
108 bytes from 20.2.0.2: icmp_seq=2 ttl=61 time=0.328 ms
108 bytes from 20.2.0.2: icmp_seq=3 ttl=61 time=0.340 ms
108 bytes from 20.2.0.2: icmp_seq=4 ttl=61 time=0.335 ms
108 bytes from 20.2.0.2: icmp_seq=5 ttl=61 time=0.354 ms
```

Traceroute from CE to remote CE to verify the hop information over MPLS L3 VPN:

```
CE1# traceroute 20.2.0.2
traceroute to 20.2.0.2 (20.2.0.2), 1 hops min, 30 hops max, 3 sec. timeout, 3
probes
  1   20.1.0.2  0.177ms  0.121ms  0.087ms
  2   30.0.0.2  0.233ms  0.120ms  0.122ms
  3   30.1.0.2  0.260ms  0.123ms  0.125ms
  4   20.2.0.2  0.189ms  0.149ms  0.144ms
```

Ping from PE to remote PE to verify MPLS L3 VPN connectivity:

```
PE1# ping 20.2.0.1 vrf vrf_1
PING 20.2.0.1 (20.2.0.1) 100(128) bytes of data.
108 bytes from 20.2.0.1: icmp_seq=1 ttl=60 time=0.334 ms
108 bytes from 20.2.0.1: icmp_seq=2 ttl=60 time=0.279 ms
108 bytes from 20.2.0.1: icmp_seq=3 ttl=60 time=0.280 ms
108 bytes from 20.2.0.1: icmp_seq=4 ttl=60 time=0.315 ms
108 bytes from 20.2.0.1: icmp_seq=5 ttl=60 time=0.291 ms
```

```
--- 20.2.0.1 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4091ms
rtt min/avg/max/mdev = 0.279/0.299/0.334/0.021 ms
```

Traceroute from PE to remote PE to verify hop information over MPLS L3 VPN:

```
PE1# traceroute 20.2.0.1 vrf vrf_1
traceroute to 20.2.0.1 (20.2.0.1), 1 hops min, 30 hops max, 3 sec. timeout, 3
probes
  1   30.0.0.2  0.224ms  0.156ms  0.158ms
  2   30.1.0.2  0.186ms  0.138ms  0.136ms
  3   20.2.0.1  0.160ms  0.147ms  0.127ms
```

Step 8: Check VRF tables at ingress PE for EBGP routes

On ingress PE, verify that routes from both the connected CE and remote PE are learned. If the expected route is not learned, check if the CE and remote PE are learning and advertising routes. In the example below, 20.2.0.0/24 is the route from the remote PE and 101.0.0.0/24 is the route from the directly connected CE:

```
PE1# show ip rib vrf vrf_1

Displaying ipv4 routes in RIB

Origin Codes: R - RIP, O - OSPFv2, B - BGP
              C - connected, S - static, L - local
Type Codes:   E - External BGP, I - Internal BGP
              IA - OSPF inter area, ia - OSPF intra area
              E1 - OSPF external type 1, E2 - OSPF external type 2
              EV - BGP EVPN, V - BGP VPN
* indicates selected for forwarding

VRF: vrf_1

Prefix               Nexthop            Interface     VRF(egress)   Origin/
Distance/    Age
                                                                    Type     Metric
-----------------------------------------------------------------------------------
------------
*20.1.0.0/24         -                  1/1/37        -             C        [0/0]
      -
*20.1.0.2/32         -                  1/1/37        -             L        [0/0]
      -
*20.2.0.0/24         2.2.2.2            -             -             B/V
[200/0]      00h:24m:27s
*101.0.0.0/24        20.1.0.1           -             -             B/E      [20/0]
      00h:14m:53s
*201.0.0.0/24        2.2.2.2            -             -             B/V
[200/0]      00h:14m:50s

Total Route Count : 5
```

```
PE1# show ip route vrf vrf_1

Displaying ipv4 routes selected for forwarding
```

```
Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: vrf_1

Prefix              Nexthop          Interface     VRF(egress)   Origin/
Distance/    Age
                                                                 Type     Metric
----------------------------------------------------------------------------------
------------
20.1.0.0/24         -                1/1/37        -             C         [0/0]
     -
20.1.0.2/32         -                1/1/37        -             L         [0/0]
     -
20.2.0.0/24         2.2.2.2          1/1/38        default       B/V       [200/0]
     00h:26m:51s
101.0.0.0/24        20.1.0.1         1/1/37        -             B/E       [20/0]
     00h:11m:42s
201.0.0.0/24        2.2.2.2          1/1/38        default       B/V       [200/0]
     00h:11m:42s

Total Route Count : 5
```

Step 9: Check VPNv4 routes at egress PE for EBGP routes

Verify CE LAN routes appear in the output:

```
PE2# show bgp vpnv4 unicast
Status codes: s suppressed, d damped, h history, * valid, > best, = multipath,
              i internal, e external S Stale, R Removed, a additional-paths
Origin codes: i - IGP, e - EGP, ? - incomplete

VRF : default
Local Router-ID 2.2.2.2

    Network            Nexthop         Metric     LocPrf     Weight Path

Route Distinguisher: 1:1                  (Label 16)
*>i 20.1.0.0/24        1.1.1.1         0          100        0       ?

Route Distinguisher: 1:2                  (Label 16)
*>  20.2.0.0/24        0.0.0.0         0          100        0       ?

Route Distinguisher: 1:1                  (Label 16)
*>i 101.0.0.0/24       1.1.1.1         0          100        0       100 i

Route Distinguisher: 1:2                  (Label 16)
*>  201.0.0.0/24       0.0.0.0         0          100        0       100 i
Total number of entries 4
```

Step 10: Check VRF tables at egress PE for EBGP routes

Verify CE LAN routes appear in the output:

```
PE2# show ip rib vrf vrf_1

Displaying ipv4 routes in RIB

Origin Codes: R - RIP, O - OSPFv2, B - BGP
              C - connected, S - static, L - local
Type Codes:   E - External BGP, I - Internal BGP
              IA - OSPF inter area, ia - OSPF intra area
              E1 - OSPF external type 1, E2 - OSPF external type 2
              EV - BGP EVPN, V - BGP VPN
* indicates selected for forwarding

VRF: vrf_1

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
----------------------------------------------------------------------------------
------------
*20.1.0.0/24         1.1.1.1           -             -             B/V
[200/0]      00h:23m:48s
*20.2.0.0/24         -                 1/1/26        -             C        [0/0]
      -
*20.2.0.1/32         -                 1/1/26        -             L        [0/0]
      -
*101.0.0.0/24        1.1.1.1           -             -             B/V
[200/0]      00h:14m:14s
*201.0.0.0/24        20.2.0.2          -             -             B/E      [20/0]
      00h:14m:12s

Total Route Count : 5
```

```
PE2# show ip route vrf vrf_1

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: vrf_1

Prefix               Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                   Type     Metric
----------------------------------------------------------------------------------
------------
20.1.0.0/24         1.1.1.1          1/1/25       default           B/V
[200/0]      00h:21m:20s
20.2.0.0/24         -                1/1/26       -                 C
[0/0]        -
20.2.0.1/32         -                1/1/26       -                 L
[0/0]        -
101.0.0.0/24        1.1.1.1          1/1/25       default           B/V
[200/0]      00h:11m:46s
201.0.0.0/24        20.2.0.2         1/1/26       -                 B/E
[20/0]       00h:11m:44s

Total Route Count : 5
```

Step 11: Check CE routing table

Check routes on CE and verify that remote CE routes have been learned. In the output below, 20.2.0.0/24 is the remote CE WAN subnet and 201.0.0.0/24 is the remote CE LAN subnet:

```
CE1# show ip route

Displaying ipv4 routes selected for forwarding

Origin Codes: C - connected, S - static, L - local
              R - RIP, B - BGP, O - OSPF
Type Codes:   E - External BGP, I - Internal BGP, V - VPN, EV - EVPN
              IA - OSPF internal area, E1 - OSPF external type 1
              E2 - OSPF external type 2

VRF: default

Prefix                Nexthop           Interface     VRF(egress)   Origin/
Distance/    Age
                                                                    Type     Metric
-----------------------------------------------------------------------------------
------------
20.0.0.0/24         -                1/1/43        -                 C
[0/0]        -
20.0.0.2/32         -                1/1/43        -                 L
[0/0]        -
20.1.0.0/24         -                1/1/44        -                 C
[0/0]        -
20.1.0.1/32         -                1/1/44        -                 L
[0/0]        -
20.2.0.0/24         20.1.0.2         1/1/44        -                 B/E
[20/0]       00h:13m:15s
101.0.0.0/24        20.0.0.1         1/1/43        -                 B/I
[200/0]      00h:01m:10s
101.0.1.0/24        20.0.0.1         1/1/43        -                 B/I
[200/0]      00h:01m:10s
101.0.2.0/24        20.0.0.1         1/1/43        -                 B/I
[200/0]      00h:01m:10s
101.0.3.0/24        20.0.0.1         1/1/43        -                 B/I
[200/0]      00h:01m:10s
101.0.4.0/24        20.0.0.1         1/1/43        -                 B/I
[200/0]      00h:01m:12s
201.0.0.0/24        20.1.0.2         1/1/44        -                 B/E
[20/0]       00h:01m:12s
201.0.1.0/24        20.1.0.2         1/1/44        -                 B/E
[20/0]       00h:01m:12s
201.0.2.0/24        20.1.0.2         1/1/44        -                 B/E
[20/0]       00h:01m:12s
201.0.3.0/24        20.1.0.2         1/1/44        -                 B/E
[20/0]       00h:01m:12s
201.0.4.0/24        20.1.0.2         1/1/44        -                 B/E
[20/0]       00h:01m:12s

Total Route Count : 15
```

Frequently asked questions

1. Can L2 subnets be extended across an MPLS L3 VPN network?

No, only L3 routing between CEs is supported across MPLS L3 VPN network. VXLAN can be used on CEs to provide L2 subnet stretching over the MPLS L3 VPN network.

2. Is multicast and IPv6 routing and supported across an MPLS L3 VPN network with 10.09?

No, 10.09 MPLS L3 VPN only supports IPv4 unicast routing. VXLAN can be used on CEs to provide multicast and IPv6 routing over the MPLS L3 VPN network.

3. Is VSX required on PEs for high availability?

VSX is not required on PEs as L3 routing across 2 standalone PEs (from a dual homed CE) provides PE redundancy.

4. How does the L3VPN solution enable site redundancy?

Site redundancy can be achieved using dual standalone PE devices connected to the site CE device that requires redundancy instead of using the PE devices in VSX mode. In order to ensure optimal best path selection, BGP configurations should implemented as explained in the examples above with same RD configurations on both PE devices.

5. Are non-L3VPN forwarding deployments supported?

Non-L3VPN forwarding deployments are supported with static LSP. Please refer to the lsp label imposition command for configuration details. LDP/BGP need to be configured on devices as explained in the examples above for all L3VPN deployments.

6. Why is BGP peer establishment failing with a non-Aruba device as PE and Aruba device as P router?

It is important to make sure that explicit-null is correctly configured on the non-Aruba PE device for forwarding between the Aruba P device and non-Aruba PE device. The Aruba P device requires proper explicit-null configuration for LDP label distribution. This enables BGP session establishment through the MPLS core.

7. Why is BGP peer establishment causing intermittent flaps in the MPLS domain?

Make sure that all interfaces that are enabled for MPLS are also configured to support jumbo frames. If MPLS-enabled interfaces do not support jumbo frames it's possible that the BGP route exchanges could be bigger than the configured MTU. Configuring jumbo frame support across the MPLS domain will prevent any traffic interruptions due to BGP session flaps.

8. Are IP/ICMP Ping IP options and dual homing supported with MPLS?

Packets over MPLS with IP options will get dropped. Ping or traceroute over MPLS paths on a dual-home network will fail and packets would be dropped.

Chapter 9
Support and Other Resources

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.arubanetworks.com/support-services/
AOS-CX Switch Software Documentation Portal	https://www.arubanetworks.com/techdocs/AOS-CX/help_portal/Content/home.htm
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	https://www.arubanetworks.com/support-services/contact-support/

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
HPE Aruba Networking Hardware	https://www.arubanetworks.com/techdocs/hardware/DocumentationPortal/Content/home.htm

Documentation and Translations Portal	
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://lms.arubanetworks.com/
End-of-Life information	https://www.arubanetworks.com/support-services/end-of-life/

Accessing Updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

Aruba Support Portal

https://asp.arubanetworks.com/downloads

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

https://www.hpe.com/networking/support

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

https://support.hpe.com/portal/site/hpsc/aae/home/

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://asp.arubanetworks.com/notifications/subscriptions (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty Information

To view warranty information for your product, go to https://www.arubanetworks.com/support-services/product-warranties/.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts

Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see https://www.arubanetworks.com/company/about-us/environmental-citizenship/.

Documentation Feedback

Aruba is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.